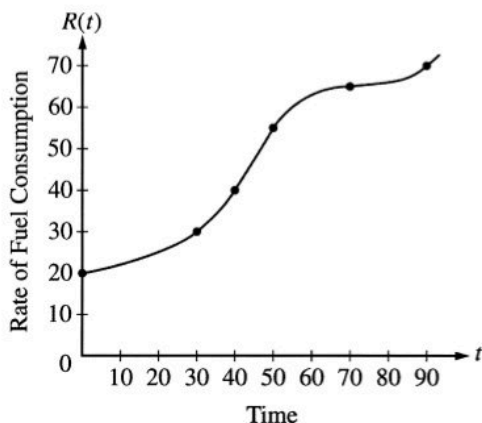


8-3b

1.  Oil is leaking from a tanker at the rate of $R(t) = 2,000e^{-0.2t}$ gallons per hour, where t is measured in hours. How much oil leaks out of the tanker from time $t = 0$ to $t = 10$?
- (A) 54 gallons
(B) 271 gallons
(C) 865 gallons
(D) 8,647 gallons ✓
(E) 14,778 gallons
2.  Water is pumped into a tank at a rate of $r(t) = 30(1 - e^{-0.16t})$ gallons per minute, where t is the number of minutes since the pump was turned on. If the tank contained 800 gallons of water when the pump was turned on, how much water, to the nearest gallon, is in the tank after 20 minutes?
- (A) 380 gallons
(B) 420 gallons
(C) 829 gallons
(D) 1220 gallons ✓
(E) 1376 gallons
3.  Insects destroyed a crop at the rate of $\frac{100e^{-0.1t}}{2 - e^{-3t}}$ tons per day, where time t is measured in days. To the nearest ton, how many tons did the insects destroy during the time interval $7 \leq t \leq 14$?
- (A) 125 ✓
(B) 100
(C) 88
(D) 50
(E) 12
4.  A pizza, heated to a temperature of 350 degrees Fahrenheit ($^{\circ}\text{F}$) is taken out of an oven and placed in a (75°F) room at time $t = 0$ minutes. The temperature of the pizza is changing at a rate of $-110e^{-0.4t}$ degrees Fahrenheit per minute. To the nearest degree, what is the temperature of the pizza at time $t = 5$ minutes?
- (A) 112°F ✓
(B) 119°F
(C) 147°F
(D) 238°F
(E) 335°F

8-3b

5.



t (minutes)	$R(t)$ (gallons per minute)
0	20
30	30
40	40
50	55
70	65
90	70

The rate of fuel consumption, in gallons per minute, recorded during an airplane flight is given by a twice-differentiable and strictly increasing function R of time t . The graph of R and a table of selected values of $R(t)$, for the time interval $0 \leq t \leq 90$ minutes, are shown above.

For $0 < b \leq 90$ minutes, explain the meaning of $\int_0^b R(t) dt$ in terms of fuel consumption for the plane.

Explain the meaning of $\frac{1}{b} \int_0^b R(t) dt$ in terms of fuel consumption for the plane. Indicate units of measure in both answers.

Part D

1 point is earned for the meaning of $\int_0^b R(t) dt$

$\int_0^b R(t) dt$ is the total amount of fuel in gallons consumed for the first b minutes.

< -1 > if no reference to time b

1 point is earned for the meaning of $\frac{1}{b} \int_0^b R(t) dt$

$\frac{1}{b} \int_0^b R(t) dt$ is the average value of the rate of fuel consumption in gallons/min during the first b minutes.

< -1 > if no reference to time b

1 point is earned for the correct units in both answers



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for the meaning of $\int_0^b R(t) dt$

$\int_0^b R(t) dt$ is the total amount of fuel in gallons consumed for the first b minutes.

< -1 > if no reference to time b

8-3b

1 point is earned for the meaning of $\frac{1}{b} \int_0^b R(t) dt$

$\frac{1}{b} \int_0^b R(t) dt$ is the average value of the rate of fuel consumption in gallons/min during the first b minutes.

< -1 > if no reference to time b

1 point is earned for the correct units in both answers

6. The tide removes sand from Sandy Point Beach at a rate modeled by the function R , given by

$$R(t) = 2 + 5 \sin\left(\frac{4\pi t}{25}\right).$$

A pumping station adds sand to the beach at a rate modeled by the function S , given by $S(t) = \frac{15t}{1+3t}$.

Both $R(t)$ and $S(t)$ have units of cubic yards per hour and t is measured in hours for $0 \leq t \leq 6$. At time $t = 0$, the beach contains 2500 cubic yards of sand.



How much sand will the tide remove from the beach during this 6-hour period? Indicate units of measure.

Part A

1 point is earned for integral

1 point is earned for the answer with units

$$\int_0^6 R(t) dt = 31.815 \text{ or } 31.816 \text{ yd}^3$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer with units

$$\int_0^6 R(t) dt = 31.815 \text{ or } 31.816 \text{ yd}^3$$

8-3b

7. A 12,000-liter tank of water is filled to capacity. At time $t = 0$, water begins to drain out of the tank at a rate modeled by $r(t)$, measured in liters per hour, where r is given by the piecewise-defined function

$$r(t) = \begin{cases} \frac{600t}{t+3} & \text{for } 0 \leq t \leq 5 \\ 1000e^{-0.2t} & \text{for } t > 5 \end{cases}$$

Write, but do not solve, an equation involving an integral to find the time A when the amount of water in the tank is 9000 liters.

Part D

1 point is earned for the integral

1 point is earned for the equation

$$12000 - \int_0^A r(t) dt = 9000$$



0	1	2
---	---	---

The student response earns all of the following points:

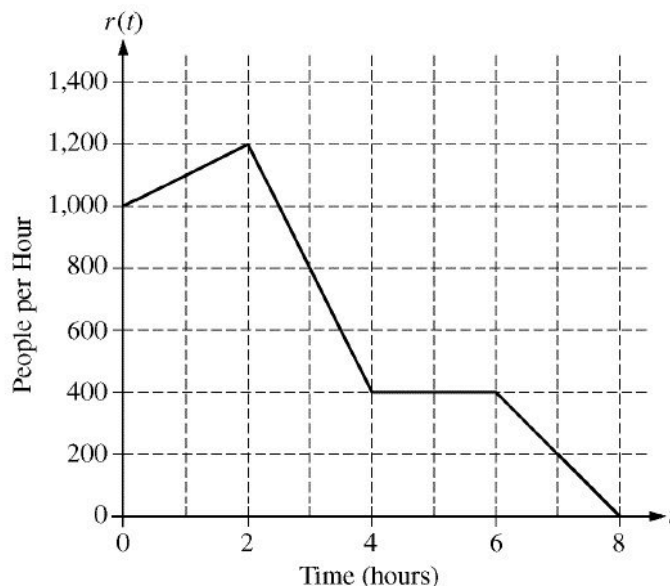
1 point is earned for the integral

1 point is earned for the equation

$$12000 - \int_0^A r(t) dt = 9000$$

8-3b

8.



There are 700 people in line for a popular amusement-park ride when the ride begins operation in the morning. Once it begins operation, the ride accepts passengers until the park closes 8 hours later. While there is a line, people move onto the ride at a rate of 800 people per hour. The graph above shows the rate, $r(t)$, at which people arrive at the ride throughout the day. Time t is measured in hours from the time the ride begins operation.

- How many people arrive at the ride between $t = 0$ and $t = 3$? Show the computations that lead to your answer.
- Is the number of people waiting in line to get on the ride increasing or decreasing between $t = 2$ and $t = 3$? Justify your answer.
- At what time t is the line for the ride the longest? How many people are in line at that time? Justify your answers.
- Write, but do not solve, an equation involving an integral expression of r whose solution gives the earliest time t at which there is no longer a line for the ride.

Part A

1 point is earned for integral

1 point is earned for the answer

$$\int_0^3 r(t) dt = 2 \cdot \frac{1000 + 1200}{2} + \frac{1200 + 800}{2} = 3200 \text{ people}$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for integral

1 point is earned for the answer

8-3b

$$\int_0^3 r(t) dt = 2 \cdot \frac{1000 + 1200}{2} + \frac{1200 + 800}{2} = 3200 \text{ people}$$

Part B

1 point is earned for answer with reason

The number of people waiting in line is increasing because people move onto the ride at a rate of 800 people per hour and for $2 < t < 3$, $r(t) > 800$.



0	1
---	---

The student response earns all of the following points:

1 point is earned for answer with reason

The number of people waiting in line is increasing because people move onto the ride at a rate of 800 people per hour and for $2 < t < 3$, $r(t) > 800$.

Part C

1 point is earned for identifying $t = 3$

1 point is earned for the number of people in line

1 point is earned for justification

$r(t) = 800$ only at $t = 3$

For $0 \leq t < 3$, $r(t) > 800$. For $3 < t \leq 8$, $r(t) < 800$.

Therefore, the line is longest at time $t = 3$.

There are $700 + 3200 - 800 \cdot 3 = 1500$ people waiting in line at time $t = 3$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for identifying $t = 3$

1 point is earned for the number of people in line

1 point is earned for justification

8-3b

$r(t) = 800$ only at $t = 3$

For $0 \leq t < 3$, $r(t) > 800$. For $3 < t \leq 8$, $r(t) < 800$.

Therefore, the line is longest at time $t = 3$.

There are $700 + 3200 - 800 \cdot 3 = 1500$ people waiting in line at time $t = 3$.

Part D

1 point is earned for $800t$

1 point is earned for integral

1 point is earned for the answer

$$0 = 700 + \int_0^t r(s)ds - 800t$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $800t$

1 point is earned for integral

1 point is earned for the answer

$$0 = 700 + \int_0^t r(s)ds - 800t$$

8-3b

The rate at which people enter an amusement park on a given day is modeled by the function E defined by

$$E(t) = \frac{15600}{(t^2 - 24t + 160)}$$

The rate at which people leave the same amusement park on the same day is modeled by the function L defined by

$$L(t) = \frac{9890}{(t^2 - 38t + 370)}$$

Both $E(t)$ and $L(t)$ are measured in people per hour and time t is measured in hours after midnight. These functions are valid for $9 \leq t \leq 23$, the hours during which the park is open. At time $t = 9$, there are no people in the park.

9.  The price of admission to the park is \$15 until 5:00 P.M. ($t = 17$). After 5:00 P.M., the price of admission to the park is \$11. How many dollars are collected from admissions to the park on the given day? Round your answer to the nearest whole number.

Part B

1 point is earned for the setup

$$15 \int_9^{17} E(t) dt + 11 \int_{17}^{23} E(t) dt = 104048.165$$

The amount collected was \$104,048.

or

$$\int_{17}^{23} E(t) dt = 1271.283$$

1271 people entered the park between 5 pm and 11 pm, so the amount collected was $15 * (6004) + 11 * (1271) = \$104,041$.



0	1
---	---

The student response earns all of the following points:

1 point is earned for the setup

$$15 \int_9^{17} E(t) dt + 11 \int_{17}^{23} E(t) dt = 104048.165$$

The amount collected was \$104,048.

or

8-3b

$$\int_{17}^{23} E(t) dt = 1271.283$$

1271 people entered the park between 5 pm and 11 pm, so the amount collected was $15 * (6004) + 11 * (1271) = \$104,041$.

10.  How many people have entered the park by 5:00 P.M. ($t = 17$)? Round your answers to the nearest whole number.

Part A

1 point is earned for limits

1 point is earned for integrand

1 point is earned for the answer

$$\int_9^{17} E(T) = 6004.270$$

6004 people enter the park by 5 pm.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for limits

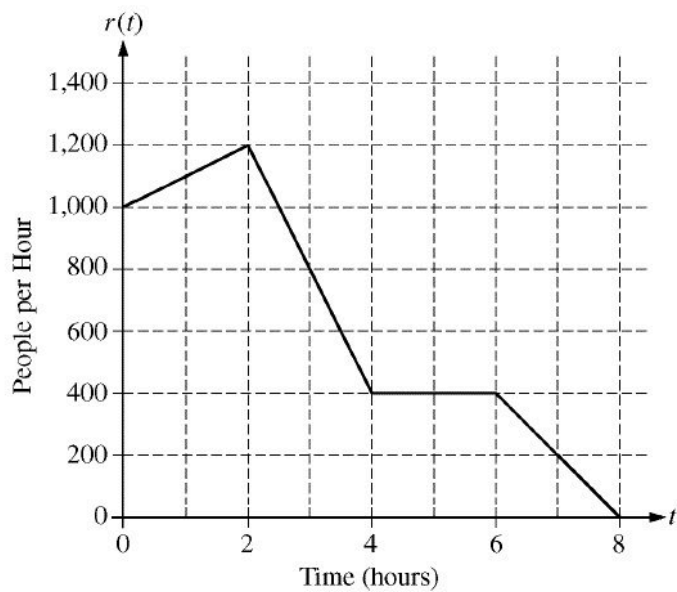
1 point is earned for integrand

1 point is earned for the answer

$$\int_9^{17} E(T) = 6004.270$$

6004 people enter the park by 5 pm.

8-3b



There are 700 people in line for a popular amusement-park ride when the ride begins operation in the morning. Once it begins operation, the ride accepts passengers until the park closes 8 hours later. While there is a line, people move onto the ride at a rate of 800 people per hour. The graph above shows the rate, $r(t)$, at which people arrive at the ride throughout the day. Time t is measured in hours from the time the ride begins operation.

11. How many people arrive at the ride between $t=0$ and $t=3$? Show the computations that lead to your answer.

Part A

1 point is earned for integral

1 point is earned for the answer

$$\int_0^3 r(t) dt = 2 \cdot \frac{1000 + 1200}{2} + \frac{1200 + 800}{2} = 3200 \text{ people}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

$$\int_0^3 r(t) dt = 2 \cdot \frac{1000 + 1200}{2} + \frac{1200 + 800}{2} = 3200 \text{ people}$$

8-3b

12. Write, but do not solve, an equation involving an integral expression of r whose solution gives the earliest time t at which there is no longer a line for the ride.

Part D

1 point is earned for $800t$

1 point is earned for integral

1 point is earned for the answer

$$0 = 700 + \int_0^t r(s) \, ds - 800t$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $800t$

1 point is earned for integral

1 point is earned for the answer

$$0 = 700 + \int_0^t r(s) \, ds - 800t$$

8-3b

The rate at which people enter an auditorium for a rock concert is modeled by the function R given by $R(t) = 1380t^2 - 675t^3$ for $0 \leq t \leq 2$ hours; $R(t)$ is measured in people per hour. No one is in the auditorium at time $t = 0$, when the doors open. The doors close and the concert begins at time $t = 2$.

13.  On average, how long does a person wait in the auditorium for the concert to begin? Consider all people who enter the auditorium after the doors open, and use the model for total wait time from part (c).

Part D

1 point is earned for integral

1 point is earned for answer

$$\frac{1}{980}w(2) = \frac{1}{980} \int_0^2 (2-t)R(t)dt = 0.77551$$

On average, a person waits 0.775 or 0.776 hour.



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for integral

1 point is earned for answer

$$\frac{1}{980}w(2) = \frac{1}{980} \int_0^2 (2-t)R(t)dt = 0.77551$$

On average, a person waits 0.775 or 0.776 hour.

8-3b

14.  The total wait time for all the people in the auditorium is found by adding the time each person waits, starting at the time the person enters the auditorium and ending when the concert begins. The function w models the total wait time for all the people who enter the auditorium before time t . The derivative of w is given by $w'(t) = (2-t)R(t)$. Find $w(2) - w(1)$, the total wait time for those who enter the auditorium after time $t = 1$.

Part C

1 point is earned for integral

1 point is earned for answer

$$w(2) - w(1) = \int_1^2 w'(t) dt = \int_1^2 (2-t)R(t) d(t) = 387.5$$

The total wait time for those who enter the auditorium after time $t = 1$ is 387.5 hours.



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for integral

1 point is earned for answer

$$w(2) - w(1) = \int_1^2 w'(t) dt = \int_1^2 (2-t)R(t) d(t) = 387.5$$

The total wait time for those who enter the auditorium after time $t = 1$ is 387.5 hours.

8-3b

15.  When a certain grocery store opens, it has 50 pounds of bananas on a display table. Customers remove bananas from the display table at a rate modeled by

$$f(t) = 10 + (0.8t) \sin\left(\frac{t^3}{100}\right) \text{ for } 0 < t \leq 12,$$

where $f(t)$ is measured in pounds per hour and t is the number of hours after the store opened. After the store has been open for three hours, store employees add bananas to the display table at a rate modeled by

$$g(t) = 3 + 2.4 \ln(t^2 + 2t) \text{ for } 3 < t \leq 12,$$

where $g(t)$ is measured in pounds per hour and t is the number of hours after the store opened.

- How many pounds of bananas are removed from the display table during the first 2 hours the store is open?
- Find $f'(7)$. Using correct units, explain the meaning of $f'(7)$ in the context of the problem.
- Is the number of pounds of bananas on the display table increasing or decreasing at time $t = 5$? Give a reason for your answer.
- How many pounds of bananas are on the display table at time $t = 8$?

Part A

1 point is earned for integral

1 point is earned for answer

$$\int_0^2 f(t) dt = 20.051175$$

20.051 pounds of bananas are removed from the display table during the first 2 hours the store is open.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for answer

$$\int_0^2 f(t) dt = 20.051175$$

20.051 pounds of bananas are removed from the display table during the first 2 hours the store is open.

Part B

1 point is earned for value

1 point is earned for meaning

$$f'(7) = -8.120 \text{ (or } -8.119)$$

8-3b

After the store has been open 7 hours, the rate at which bananas are being removed from the display table is decreasing by 8.120 (or 8.119) pounds per hour per hour.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for value

1 point is earned for meaning

$$f(7) = -8.120 \text{ (or } -8.119)$$

After the store has been open 7 hours, the rate at which bananas are being removed from the display table is decreasing by 8.120 (or 8.119) pounds per hour per hour.

Part C

1 point is earned for considers $f(5)$ and $g(5)$

1 point is earned for answer with reason

$$g(5) - f(5) = -2.263103 < 0$$

Because $g(5) - f(5) < 0$, the number of pounds of bananas on the display table is decreasing at time $t = 5$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for considers $f(5)$ and $g(5)$

1 point is earned for answer with reason

$$g(5) - f(5) = -2.263103 < 0$$

Because $g(5) - f(5) < 0$, the number of pounds of bananas on the display table is decreasing at time $t = 5$.

Part D

2 points are earned for integrals

1 point is earned for answer

$$50 + \int_3^8 g(t)dt - \int_0^8 f(t)dt = 23.347396$$

23.347 pounds of bananas are on the display table at time $t = 8$.

8-3b



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for integrals

1 point is earned for answer

$$50 + \int_3^8 g(t)dt - \int_0^8 f(t)dt = 23.347396$$

23.347 pounds of bananas are on the display table at time $t = 8$.

8-3b

16.  A water tank at Camp Newton holds 1200 gallons of water at time $t = 0$. During the time interval $0 \leq t \leq 18$ hours, water is pumped into the tank at the rate

$$W(t) = 95\sqrt{t} \sin^2\left(\frac{t}{6}\right) \text{ gallons per hour.}$$

During the same time interval, water is removed from the tank at the rate

$$R(t) = 275 \sin^2\left(\frac{t}{3}\right) \text{ gallons per hour.}$$

- (a) Is the amount of water in the tank increasing at time $t = 15$? Why or why not?
 (b) To the nearest whole number, how many gallons of water are in the tank at time $t = 18$?
 (c) At what time t , $0 \leq t \leq 18$, is the amount of water in the tank at an absolute minimum? Show the work that leads to your conclusion.
 (d) For $t > 18$, no water is pumped into the tank, but water continues to be removed at the rate $R(t)$ until the tank becomes empty. Let k be the time at which the tank becomes empty. Write, but do not solve, an equation involving an integral expression that can be used to find the value of k .

Part A

1 point is earned for the answer with reason

No; the amount of water is not increasing at $t = 15$ since $W(15) - R(15) = -121.09 < 0$.



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer with reason

No; the amount of water is not increasing at $t = 15$ since $W(15) - R(15) = -121.09 < 0$.

Part B

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$1200 + \int_0^{18} (W(t) - R(t)) dt = 1309.788$$

1310 gallons



0	1	2	3
---	---	---	---

The student response earns all of the following points:

8-3b

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$1200 + \int_0^{18} (W(t) - R(t))dt = 1309.788$$

1310 gallons

Part C

1 point is earned for the interior critical points

1 point is earned for the amount of water is least at $t = 6.494$ or 6.495

1 point is earned for the analysis for absolute minimum

$$W(t) - R(t) = 0$$

$$t = 0, 6.4948, 12.9748$$

t (hours)	gallons of water
0	1200
6.495	525
12.975	1697
18	1310

The values at the endpoints and the critical points show that the absolute minimum occurs when $t = 6.494$ or 6.495 .



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the interior critical points

1 point is earned for the amount of water is least at $t = 6.494$ or 6.495

1 point is earned for the analysis for absolute minimum

$$W(t) - R(t) = 0$$

$$t = 0, 6.4948, 12.9748$$

8-3b

t (hours)	gallons of water
0	1200
6.495	525
12.975	1697
18	1310

The values at the endpoints and the critical points show that the absolute minimum occurs when $t = 6.494$ or 6.495 .

Part D

1 point is earned for the limits

1 point is earned for the equation

$$\int_{18}^k R(t)dt = 1310$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the equation

$$\int_{18}^k R(t)dt = 1310$$

8-3b

17. A water tank at Camp Newton holds 1200 gallons of water at time $t = 0$. During the time interval $0 \leq t \leq 18$ hours, water is pumped into the tank at the rate

$$W(t) = 95\sqrt{t}\sin^2\left(\frac{t}{6}\right) \text{ gallons per hour.}$$

During the same time interval, water is removed from the tank at the rate

$$R(t) = 275\sin^2\left(\frac{t}{3}\right) \text{ gallons per hour.}$$

 For $t > 18$, no water is pumped into the tank, but water continues to be removed at the rate $R(t)$ until the tank becomes empty. Let k be the time at which the tank becomes empty. Write, but do not solve, an equation involving an integral expression that can be used to find the value of k .

Part D

1 point is earned for the limits

1 point is earned for the equation

$$\int_{18}^k R(t)dt = 1310$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the equation

$$\int_{18}^k R(t)dt = 1310$$

8-3b

18.

t (sec)	0	15	25	30	35	50	60
$v(t)$ (ft/sec)	-20	-30	-20	-14	-10	0	10
$a(t)$ (ft/sec ²)	1	5	2	1	2	4	2

A car travels on a straight track. During the time interval $0 \leq t \leq 60$ seconds, the car's velocity v , measured in feet per second, and acceleration a , measured in feet per second per second, are continuous functions. The table above shows selected values of these functions.

(a) Using appropriate units, explain the meaning of $\int_{30}^{60} |v(t)| dt$ in terms of the car's motion. Approximate

$\int_{30}^{60} |v(t)| dt$ using a trapezoidal approximation with the three subintervals determined by the table.

(b) Using appropriate units, explain the meaning of $\int_0^{30} a(t) dt$ in terms of the car's motion. Find the exact value of $\int_0^{30} a(t) dt$.

(c) For $0 < t < 60$, must there be a time t when $v(t) = -5$? Justify your answer.

(d) For $0 < t < 60$, must there be a time t when $a(t) = 0$? Justify your answer.

Part A

1 point is earned for the explanation

1 point is earned for the value

$\int_{30}^{60} |v(t)| dt$ is the distance in feet that the car travels from $t = 30$ sec to $t = 60$ sec.

Trapezoidal approximation for $\int_{30}^{60} |v(t)| dt$:

$$A = \frac{1}{2}(14 + 10)5 + \frac{1}{2}(10)(15) + \frac{1}{2}(10)(10) = 185 \text{ ft}$$

1 point is earned for units in both (a) and (b)

Units of ft in (a) and ft/sec in (b)



0	1	2	3
---	---	---	---

8-3b

The student response earns all of the following points:

1 point is earned for the explanation

1 point is earned for the value

$\int_{30}^{60} |v(t)| dt$ is the distance in feet that the car travels from $t = 30$ sec to $t = 60$ sec.

Trapezoidal approximation for $\int_{30}^{60} |v(t)| dt$:

$$A = \frac{1}{2}(14 + 10)5 + \frac{1}{2}(10)(15) + \frac{1}{2}(10)(10) = 185 \text{ ft}$$

1 point is earned for units in both (a) and (b)

Units of ft in (a) and ft/sec in (b)

Part B

1 point is earned for the explanation

1 point is earned for the value

$\int_0^{30} a(t) dt$ is the car's change in velocity in ft/sec from $t = 0$ sec to $t = 30$ sec.

$$\int_0^{30} a(t) dt = \int_0^{30} v'(t) dt = v(30) - v(0)$$

$$= -14 - (-20) = 6 \text{ ft/sec}$$

1 point is earned for units in both (a) and (b)

Units of ft in (a) and ft/sec in (b)



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the explanation

8-3b

1 point is earned for the value

$\int_0^{30} a(t) dt$ is the car's change in velocity in ft/sec from $t = 0$ sec to $t = 30$ sec.

$$\int_0^{30} a(t) dt = \int_0^{30} v'(t) dt = v(30) - v(0)$$

$$= -14 - (-20) = 6 \text{ ft/sec}$$

1 point is earned for units in both (a) and (b)

Units of ft in (a) and ft/sec in (b)

Part C

1 point is earned for $v(35) < -5 < v(50)$

1 point is earned for Yes; refers to IVT or hypotheses

Yes. Since $v(35) = -10 < -5 < 0 = v(50)$, the IVT guarantees a t in $(35, 50)$ so that $v(t) = -5$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $v(35) < -5 < v(50)$

1 point is earned for Yes; refers to IVT or hypotheses

Yes. Since $v(35) = -10 < -5 < 0 = v(50)$, the IVT guarantees a t in $(35, 50)$ so that $v(t) = -5$.

Part D

1 point is earned for $v(0) = v(25)$

1 point is earned for Yes; refers to MVT or hypotheses

Yes. Since $v(0) = v(25)$, the MVT guarantees a t in $(0, 25)$ so that $a(t) = v'(t) = 0$.



0	1	2
---	---	---

The student response earns all of the following points:

8-3b

1 point is earned for $v(0) = v(25)$

1 point is earned for Yes; refers to MVT or hypotheses

Yes. Since $v(0) = v(25)$, the MVT guarantees a t in $(0,25)$ so that $a(t) = v'(t) = 0$.

8-3b

19.

t (sec)	0	15	25	30	35	50	60
$v(t)$ (ft/sec)	-20	-30	-20	-14	-10	0	10
$a(t)$ (ft/sec ²)	1	5	2	1	2	4	2

A car travels on a straight track. During the time interval $0 \leq t \leq 60$ seconds, the car's velocity v , measured in feet per second, and acceleration a , measured in feet per second per second, are continuous functions. The table above shows selected values of these functions.

Using appropriate units, explain the meaning of $\int_0^{30} a(t) dt$ in terms of the car's motion. Find the exact value of

$$\int_0^{30} a(t) dt.$$

Part B

1 point is earned for the explanation

1 point is earned for the value

$\int_0^{30} a(t) dt$ is the car's change in velocity in ft/sec from $t=0$ sec to $t=30$ sec.

$$\begin{aligned} \int_0^{30} a(t) dt &= \int_0^{30} v'(t) dt = v(30) - v(0) \\ &= -14 - (-20) = 6 \text{ ft/sec} \end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the explanation

1 point is earned for the value

$\int_0^{30} a(t) dt$ is the car's change in velocity in ft/sec from $t=0$ sec to $t=30$ sec.

$$\begin{aligned} \int_0^{30} a(t) dt &= \int_0^{30} v'(t) dt = v(30) - v(0) \\ &= -14 - (-20) = 6 \text{ ft/sec} \end{aligned}$$

Extra Point

8-3b

1 point is earned for units in both (a) and (b)

Units of ft in (a) and ft/sec in (b)



0

1

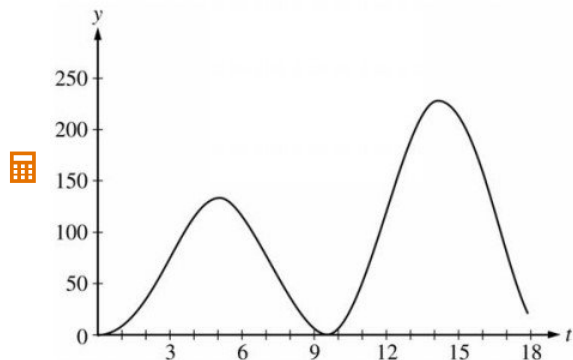
The student response earns one of the following points:

1 point is earned for units in both (a) and (b)

Units of ft in (a) and ft/sec in (b)

8-3b

20.



At an intersection in Thomasville, Oregon, cars turn left at the rate $L(t) = 60\sqrt{t} \sin^2\left(\frac{t}{3}\right)$ cars per hour over the time interval $0 \leq t \leq 18$ hours. The graph of $y = L(t)$ is shown above.

(a) To the nearest whole number, find the total number of cars turning left at the intersection over the time interval $0 \leq t \leq 18$ hours.

(b) Traffic engineers will consider turn restrictions when $L(t) \geq 150$ cars per hour. Find all values of t for which $L(t) \geq 150$ and compute the average value of L over this time interval. Indicate units of measure.

(c) Traffic engineers will install a signal if there is any two-hour time interval during which the product of the total number of cars turning left and the total number of oncoming cars traveling straight through the intersection is greater than 200,000. In every two-hour time interval, 500 oncoming cars travel straight through the intersection. Does this intersection require a traffic signal? Explain the reasoning that leads to your conclusion.

Part A

1 point is earned for the setup

1 point is earned for the answer

$$\int_0^{18} L(t) dt \approx 1658 \text{ cars}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the setup

1 point is earned for the answer

$$\int_0^{18} L(t) dt \approx 1658 \text{ cars}$$

Part B

1 point is earned for t -interval when $L(t) \geq 150$

8-3b

1 point is earned for the average value integral

1 point is earned for the answer with units

$L(t) = 150$ when $t = 12.42831, 16.121166$

Let $R = 12.42831$ and $S = 16.121166$

$L(t) \geq 150$ for t in the interval $[R, S]$

$$\frac{1}{S-R} \int_R^S L(t) dt = 199.426 \text{ cars per hour}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for t -interval when $L(t) \geq 150$

1 point is earned for the average value integral

1 point is earned for the answer with units

$L(t) = 150$ when $t = 12.42831, 16.121166$

Let $R = 12.42831$ and $S = 16.121166$

$L(t) \geq 150$ for t in the interval $[R, S]$

$$\frac{1}{S-R} \int_R^S L(t) dt = 199.426 \text{ cars per hour}$$

Part C

1 point is earned for considering 400 cars

1 point is earned for valid interval $[h, h + 2]$

1 point is earned for the value of $\int_h^{h+2} L(t) dt$

1 point is earned for the answer and explanation

OR

1 point is earned for considers 200 cars per hour

1 point is earned for solving $L(t) \geq 200$

1 point is earned if discusses 2 hour interval

1 point is earned for the answer and explanation

8-3b

For the product to exceed 200,000, the number of cars turning left in a two-hour interval must be greater than 400.

$$\int_{13}^{15} L(t) dt = 431.931 > 400$$

OR

The number of cars turning left will be greater than 400 on a two-hour interval if $L(t) \geq 200$ on that interval.

$L(t) \geq 200$ on any two-hour subinterval of $[13.25304, 15.32386]$.

Yes, a traffic signal is required.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for considering 400 cars

1 point is earned for valid interval $[h, h + 2]$

1 point is earned for the value of $\int_h^{h+2} L(t) dt$

1 point is earned for the answer and explanation

OR

1 point is earned for considers 200 cars per hour

1 point is earned for solving $L(t) \geq 200$

1 point is earned if discusses 2 hour interval

1 point is earned for the answer and explanation

For the product to exceed 200,000, the number of cars turning left in a two-hour interval must be greater than 400.

$$\int_{13}^{15} L(t) dt = 431.931 > 400$$

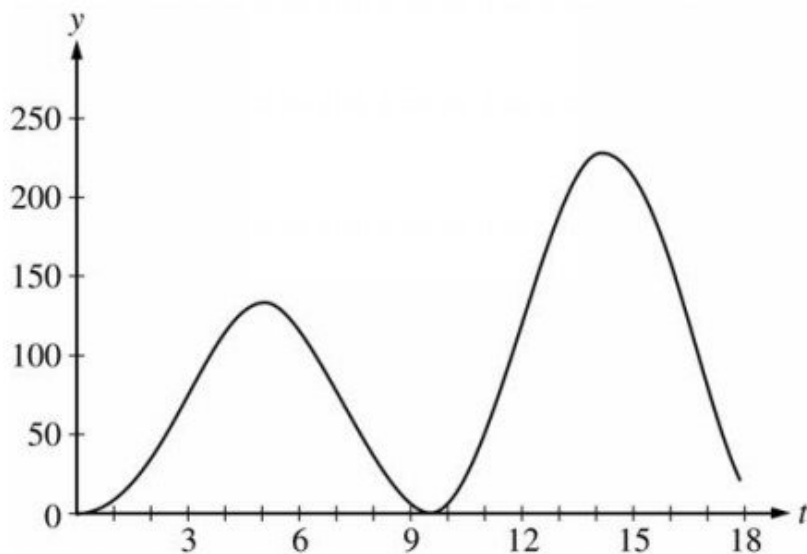
OR

The number of cars turning left will be greater than 400 on a two-hour interval if $L(t) \geq 200$ on that interval.

$L(t) \geq 200$ on any two-hour subinterval of $[13.25304, 15.32386]$

Yes, a traffic signal is required.

8-3b



At an intersection in Thomasville, Oregon, cars turn left at the rate $L(t) = 60\sqrt{t}\sin^2\left(\frac{t}{3}\right)$ cars per hour over the time interval $0 \leq t \leq 18$ hours. The graph of $y = L(t)$ is shown above.

21.  To the nearest whole number, find the total number of cars turning left at the intersection over the time interval $0 \leq t \leq 18$ hours.

Part A

1 point is earned for the setup

1 point is earned for the answer

$$\int_0^{18} L(t) dt \approx 1658 \text{ cars}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the setup

1 point is earned for the answer

$$\int_0^{18} L(t) dt \approx 1658 \text{ cars}$$

8-3b

22.  Traffic engineers will install a signal if there is any two-hour time interval during which the product of the total number of cars turning left and the total number of oncoming cars traveling straight through the intersection is greater than 200,000. In every two-hour interval, 500 oncoming cars travel straight through the intersection. Does this intersection require a traffic signal? Explain the reasoning that leads to your conclusion.

Part C

1 point is earned for considers 400 cars

1 point is earned for: $[h, h + 2]$

1 point is earned for the value of $\int_h^{h+2} L(t) dt$

1 point is earned for the answer and explanation

OR

1 point is earned for considers 200 cars

1 point is earned for solves $L(t) \geq 200$

1 point is earned for discusses 2 hour interval

1 point is earned for the answer and explanation

For the product to exceed 200,000, the number of cars turning in a two-hour interval must be greater than 400.

$$\int_{13}^{15} L(t) dt = 431.931 > 400$$

OR

The number of carts turning left will be greater than 400 on a two-hour interval if $L(t) \geq 200$ on that interval.

$L(t) \geq 200$ on any two-hour subinterval of $[13.25304, 15.32386]$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for considers 400 cars

1 point is earned for: $[h, h + 2]$

1 point is earned for the value of $\int_h^{h+2} L(t) dt$

1 point is earned for the answer and explanation

8-3b

OR

1 point is earned for considers 200 cars

1 point is earned for solves $L(t) \geq 200$

1 point is earned for discusses 2 hour interval

1 point is earned for the answer and explanation

For the product to exceed 200,000, the number of cars turning in a two-hour interval must be greater than 400.

$$\int_{13}^{15} L(t) dt = 431.931 > 400$$

OR

The number of carts turning left will be greater than 400 on a two-hour interval if $L(t) \geq 200$ on that interval. $L(t) \geq 200$ on any two-hour subinterval of $[13.25304, 15.32386]$

23. A particle moves along the x-axis so that its velocity v at any time t , for $0 \leq t \leq 16$ is given by $v(t) = e^{2\sin t} - 1$. At time $t = 0$, the particle is at the origin.

Is there any time t , $0 \leq t \leq 16$, at which the particle returns to the origin? Justify your answer.**Part D**

1 point is earned for no such time

1 point is earned for the reason

There is no such time because

$$\int_0^T v(t) dt > 0 \text{ for all } T > 0.$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for no such time

1 point is earned for the reason

There is no such time because

$$\int_0^T v(t) dt > 0 \text{ for all } T > 0.$$

8-3b

24. A particle with velocity at any time t given by $v(t) = e^t$ moves in a straight line. How far does the particle move from $t = 0$ to $t = 2$?

(A) $e^2 - 1$



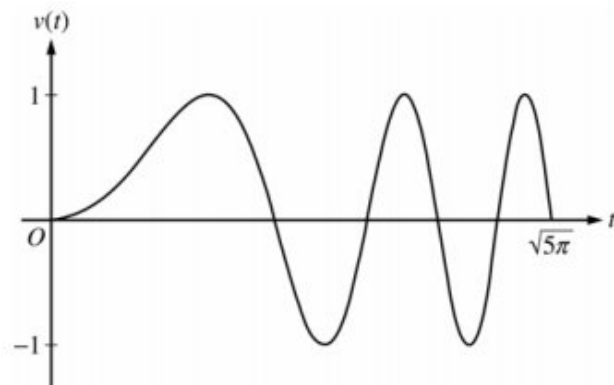
(B) $e - 1$

(C) $2e$

(D) e^2

(E) $\frac{e^3}{3}$

8-3b



A particle moves along the x -axis so that its velocity v at time $t \geq 0$ is given by $v(t) = \sin(t^2)$. The graph of v is shown above for $0 \leq t \leq \sqrt{5\pi}$. The position of the particle at time t is $x(t)$ and its position at time $t = 0$ is $x(0) = 5$.

25.  Find the total distance traveled by the particle from time $t=0$ to $t=3$.

Part B

1 point is earned for the setup

1 point is earned for the answer

$$\text{Distance} = \int_0^3 |v(t)| dt = 1.702$$

OR

For $0 < t < 3$, $v(t) = 0$ when $t = \sqrt{\pi} = 1.77245$ and

$$t = \sqrt{2\pi} = 2.50663$$

$$x(0) = 5$$

$$x(\sqrt{\pi}) = 5 + \int_0^{\sqrt{\pi}} v(t) dt = 5.89483$$

$$x(\sqrt{2\pi}) = 5 + \int_0^{\sqrt{2\pi}} v(t) dt = 5.43041$$

$$x(3) = 5 + \int_0^3 v(t) dt = 5.77356$$

$$|x(\sqrt{\pi}) - x(0)| + |x(\sqrt{2\pi}) - x(\sqrt{\pi})| + |x(3) - x(\sqrt{2\pi})| = 1.702$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the setup

8-3b

1 point is earned for the answer

$$\text{Distance} = \int_0^9 |v(t)| dt = 1.702$$

OR

For $0 < t < 3$, $v(t) = 0$ when $t = \sqrt{\pi} = 1.77245$ and

$$t = \sqrt{2\pi} = 2.50663$$

$$x(0) = 5$$

$$x(\sqrt{\pi}) = 5 + \int_0^{\sqrt{\pi}} v(t) dt = 5.89483$$

$$x(\sqrt{2\pi}) = 5 + \int_0^{\sqrt{2\pi}} v(t) dt = 5.43041$$

$$x(3) = 5 + \int_0^3 v(t) dt = 5.77356$$

$$|x(\sqrt{\pi}) - x(0)| + |x(\sqrt{2\pi}) - x(\sqrt{\pi})| + |x(3) - x(\sqrt{2\pi})| = 1.702$$

26.  Find the position of the particle at time $t=3$.

Part C

1 point is earned for integrand

1 point is earned if uses $x(0)=5$

1 point is earned for the answer

$$x(3) = 5 + \int_0^3 v(t) dt = 5.773 \text{ or } 5.774$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned if uses $x(0)=5$

1 point is earned for the answer

$$x(3) = 5 + \int_0^3 v(t) dt = 5.773 \text{ or } 5.774$$

8-3b

At a certain height, a tree trunk has a circular cross section. The radius $R(t)$ of that cross section grows at a rate modeled by the function

$$\frac{dR}{dt} = \frac{1}{16}(3 + \sin(t^2)) \text{ centimeters per year for}$$

$0 \leq t \leq 3$, where time t is measured in years. At time $t = 0$, the radius is 6 centimeters. The area of the cross section at time t is denoted by $A(t)$.

27.  Evaluate $\int_0^3 A'(t) dt$. Using appropriate units, interpret the meaning of that integral in terms of cross-sectional area.

Part C

1 point is earned for using Fundamental Theorem of Calculus

1 point is earned for value of $\int_0^3 A'(t) dt$.

1 point is earned for meaning of $\int_0^3 A'(t) dt$.

$$\int_0^3 A'(t) dt = A(3) - A(0) = 24.200 \text{ or } 24.201$$

From time $t=0$ to $t=3$ years, the cross-sectional area grows by 24.201 square centimeters.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for using Fundamental Theorem of Calculus

1 point is earned for value of $\int_0^3 A'(t) dt$.

8-3b

1 point is earned for meaning of $\int_0^3 A'(t) dt$.

$$\int_0^3 A'(t) dt = A(3) - A(0) = 24.200 \text{ or } 24.201$$

From time $t=0$ to $t=3$ years, the cross-sectional area grows by 24.201 square centimeters.

28.  Write an expression, involving an integral, for the radius $R(t)$ for $0 \leq t \leq 3$. Use your expression to find $R(3)$.

Part A

1 point is earned for the integral

1 point is earned for the expression for $R(t)$

1 point is earned for $R(3)$

$$R(t) = 6 + \int_0^t \frac{1}{16}(3 + \sin(x^2)) dx$$

$$R(3) = 6.610 \text{ or } 6.611$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the expression for $R(t)$

1 point is earned for $R(3)$

$$R(t) = 6 + \int_0^t \frac{1}{16}(3 + \sin(x^2)) dx$$

$$R(3) = 6.610 \text{ or } 6.611$$

8-3b

29. Mighty Cable Company manufactures cable that sells for \$120 per meter. For a cable of fixed length, the cost of producing a portion of the cable varies with its distance from the beginning of the cable. Mighty reports that the cost to produce a portion of a cable that is x meters from the beginning of the cable is $6\sqrt{x}$ dollars per meter. (Note: Profit is defined to be the difference between the amount of money received by the company for selling the cable and the company's cost of producing the cable.)

Write an expression, involving an integral, that represents Mighty's profit on the sale of a cable that is k meters long.

Part C

1 point is earned for the integral

1 point is earned for the answer

$$\text{Profit} = 120k - \int_0^k 6\sqrt{x} \, dx \text{ dollars}$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the integral

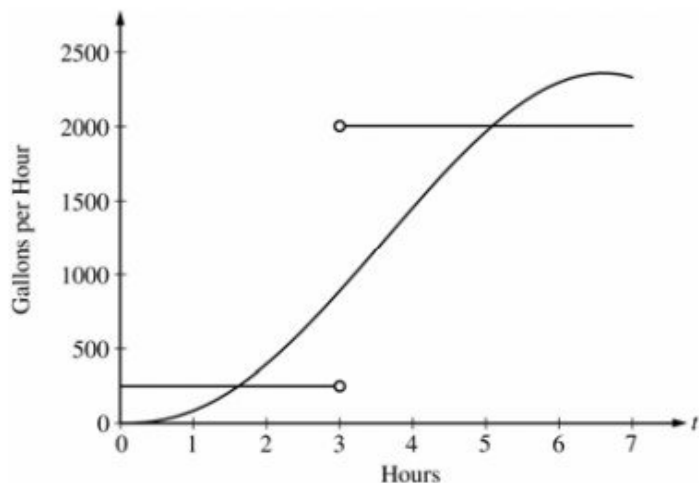
1 point is earned for the answer

$$\text{Profit} = 120k - \int_0^k 6\sqrt{x} \, dx \text{ dollars}$$

8-3b

The amount of water in a storage tank, in gallons, is modeled by a continuous function on the time interval $0 \leq t \leq 7$, where t is measured in hours. In this model, rates are given as follows:

- i. The rate at which water enters the tank is $f(t) = 100t^2 \sin(\sqrt{t})$ gallons per hour for $0 \leq t \leq 7$.
- ii. The rate at which water enters the tank is $g(t) = \begin{cases} 250 & \text{for } 0 \leq t < 3 \\ 2000 & \text{for } 3 < t \leq 7 \end{cases}$ gallons per hour.



The graphs of f and g , which intersect at $t = 1.617$ and $t = 5.076$, are shown in the figure above. At time $t = 0$, the amount of water in the tank is 5000 gallons.

30.  For $0 \leq t \leq 7$, at what time t is the amount of water in the tank greatest? To the nearest gallon, compute the amount of water at this time. Justify your answer.

Part C

1 point is earned for identifies $t = 3$ as a candidate

1 point is earned for the integrand

1 point is earned for the amount of water at $t = 3$

1 point is earned for the amount of water at $t = 7$

1 point is earned for the conclusion

Since $f(t) - g(t)$ changes sign from positive to negative only at $t = 3$, the candidates for the absolute maximum are $t = 0$, 3 , and 7 .

8-3b

t (hours)	gallons of water
0	5000
3	$5000 + \int_0^3 f(t) dt - 250(3) = 5126.591$
7	$5126.591 + \int_3^7 f(t) dt - 2000(4) = 4513.807$

The amount of water in the tank is greatest at 3 hours. At that time, the amount of water in the tank, rounded to the nearest gallon, is 5127 gallons.



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned for identifies $t = 3$ as a candidate

1 point is earned for the integrand

1 point is earned for the amount of water at $t = 3$

1 point is earned for the amount of water at $t = 7$

1 point is earned for the conclusion

Since $f(t) - g(t)$ changes sign from positive to negative only at $t = 3$, the candidates for the absolute maximum are $t = 0, 3$, and 7.

t (hours)	gallons of water
0	5000
3	$5000 + \int_0^3 f(t) dt - 250(3) = 5126.591$
7	$5126.591 + \int_3^7 f(t) dt - 2000(4) = 4513.807$

The amount of water in the tank is greatest at 3 hours. At that time, the amount of water in the tank, rounded to the nearest gallon, is 5127 gallons.

8-3b

31.  How many gallons of water enter the tank during the time interval $0 \leq t \leq 7$? Round your answer to the nearest gallon.
-

Part A

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^7 f(t) dt \approx 8264 \text{ gallons}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^7 f(t) dt \approx 8264 \text{ gallons}$$

8-3b

32. A cylindrical can of radius 10 millimeters is used to measure rainfall in Stormville. The can is initially empty, and rain enters the can during a 60-day period. The height of water in the can is modeled by the function S , where $S(t)$ is measured in millimeters and t is measured in days for $0 \leq t \leq 60$. The rate at which the height of the water is rising in the can is given by $S'(t) = 2\sin(0.03t) + 1.5$.



According to the model, what is the height of the water in the can at the end of the 60-day period?

Part A

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$S(60) = \int_0^{60} S'(t) dt = 171.813 \text{ mm}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$S(60) = \int_0^{60} S'(t) dt = 171.813 \text{ mm}$$

8-3b

33. Traffic flow is defined as the rate at which cars pass through an intersection, measured in cars per minute. The traffic flow at a particular intersection is modeled by the function F defined by $F(t) = 82 + 4 \sin\left(\frac{t}{2}\right)$ for $0 \leq t \leq 30$, where $F(t)$ is measured in cars per minute and t is measured in minutes.



To the nearest whole number, how many cars pass through the intersection over the 30-minute period?

Part A

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\int_0^{30} F(t) dt = 2474 \text{ cars}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

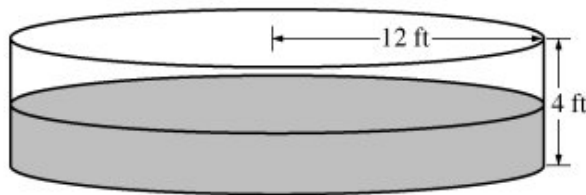
1 point is earned for the integrand

1 point is earned for the answer

$$\int_0^{30} F(t) dt = 2474 \text{ cars}$$

8-3b

t	0	2	4	6	8	10	12
$P(t)$	0	46	53	57	60	62	63



The figure above shows an above ground swimming pool in the shape of a cylinder with a radius of 12 feet and a height of 4 feet. The pool contains 1000 cubic feet of water at time $t=0$. During the time interval $0 \leq t \leq 12$ hours, water is pumped into the pool at the rate $P(t)$ cubic feet per hour. The table above gives values of $P(t)$ for selected values of t . During the same time interval, water is leaking from the pool at the rate $R(t) = 25e^{-0.05t}$ cubic feet per hour.

(Note: The volume V of a cylinder with radius r and height h is given by $V = \pi r^2 h$.)

34. Use the results from parts (a) and (b) to approximate the volume of water in the pool at time $t=12$ hours. Round your answer to the nearest cubic foot.

Part C

1 point is earned for the answer

$$1000 + \int_0^{12} p(t)dt - \int_0^{12} R(t)dt = 1434.406$$

At time $t=12$ hours, the volume of water in the pool is approximately 1434 ft³.



0

1

The student response earns one of the following points:

1 point is earned for the answer

$$1000 + \int_0^{12} p(t)dt - \int_0^{12} R(t)dt = 1434.406$$

At time $t=12$ hours, the volume of water in the pool is approximately 1434 ft³.

8-3b

35.  Calculate the total amount of water that leaked out of the pool during the time interval $0 \leq t \leq 12$ hours.

Part B

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^{12} R(t) dt = 225.594 ft^3$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^{12} R(t) dt = 225.594 ft^3$$

8-3b

36. Use a midpoint Riemann sum with three subintervals of equal length to approximate the total amount of water that was pumped into the pool during the time interval $0 \leq t \leq 12$ hours. Show the computations that lead to your answer.
-

Part A

1 point is earned for midpoint sum

1 point is earned for the answer

$$\int_0^{12} p(t) dt \approx 46.4 + 57.4 + 62.4 = 660 ft^3$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for midpoint sum

1 point is earned for the answer

$$\int_0^{12} p(t) dt \approx 46.4 + 57.4 + 62.4 = 660 ft^3$$

8-3b

There is no snow on Janet's driveway when snow begins to fall at midnight. From midnight to 9 A.M., snow accumulates on the driveway at a rate modeled by $f(t) = 7te^{\cos t}$ cubic feet per hour, where t is measured in hours since midnight. Janet starts removing snow at 6 A.M. ($t = 6$). The rate $g(t)$, in cubic feet per hour, at which Janet removes snow from the driveway at time t hours after midnight is modeled by

$$g(t) = \begin{cases} 0 & \text{for } 0 \leq t < 6 \\ 125 & \text{for } 6 \leq t < 7 \\ 108 & \text{for } 7 \leq t \leq 9. \end{cases}$$

37.  How many cubic feet of snow have accumulated on the driveway by 6 A.M?

Part A

1 point is earned for integral

1 point is earned for the answer

$$\text{Area} = \int_0^6 f(t) dt = 142.274 \text{ or } 142.275 \text{ cubic feet}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

$$\text{Area} = \int_0^6 f(t) dt = 142.274 \text{ or } 142.275 \text{ cubic feet}$$

8-3b

38.  How many cubic feet of snow are on the driveway at 9 A.M?
-

Part D

1 point is earned for integral

1 point is earned for $h(9)$

1 point is earned for the answer

Amount of snow is $\int_0^9 f(t) dt - h(9) = 26.334 \text{ or } 26.335 \text{ cubic feet}$



0	1	2	3
---	---	---	---

The student response earns all of the following points

1 point is earned for integral

1 point is earned for $h(9)$

1 point is earned for the answer

Amount of snow is

$\int_0^9 f(t) dt - h(9) = 26.334 \text{ or } 26.335 \text{ cubic feet}$

8-3b

39.

t (hours)	0	2	5	7	8
$E(t)$ (hundreds of entries)	0	4	13	21	23

A zoo sponsored a one-day contest to name a new baby elephant. Zoo visitors deposited entries in a special box between noon ($t=0$) and 8 P.M. ($t=8$). The number of entries in the box t hours after noon is modeled by a differentiable function E for $0 \leq t \leq 8$. Values of $E(t)$, in hundreds of entries, at various times t are shown in the table above.

 At 8 P.M., volunteers began to process the entries. They processed the entries at a rate modeled by the function P , where $P(t)=t^3-30t^2+298t-976$ hundreds of entries per hour for $8 \leq t \leq 12$. According to the model, how many entries had not yet been processed by midnight ($t=12$)?

Part C

1 point is earned for integral

1 point is earned for the answer

$$23 - \int_8^{12} P(t) dt = 23 - 16 = 7 \text{ hundred entries}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

$$23 - \int_8^{12} P(t) dt = 23 - 16 = 7 \text{ hundred entries}$$

8-3b

40. A storm washed away sand from a beach, causing the edge of the water to get closer to a nearby road. The rate at which the distance between the road and the edge of the water was changing during the storm is modeled by $f(t) = \sqrt{t} + \cos t - 3$ meters per hour, t hours after the storm began. The edge of the water was 35 meters from the road when the storm began, and the storm lasted 5 hours. The derivative of $f(t)$ is $f'(t) = \frac{1}{2\sqrt{t}} - \sin t$.

After the storm, a machine pumped sand back onto the beach so that the distance between the road and the edge of the water was growing at a rate of $g(p)$ meters per day, where p is the number of days since pumping began. Write an equation involving an integral expression whose solution would give the number of days that sand must be pumped to restore the original distance between the road and the edge of the water.

Part D

1 point is earned for the integral of g

1 point is earned for the answer

$$-\int_0^5 f(t)dt = \int_0^x g(p)dp$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral of g

1 point is earned for the answer

$$-\int_0^5 f(t)dt = \int_0^x g(p)dp$$

8-3b

41.

t (minutes)	0	2	5	9	10
$H(t)$ (degrees Celsius)	66	60	52	44	43

As a pot of tea cools, the temperature of the tea is modeled by a differentiable function H for $0 \leq t \leq 10$, where time t is measured in minutes and temperature $H(t)$ is measured in degrees Celsius. Values of $H(t)$ at selected values of time t are shown in the table above.

 At time $t = 0$, biscuits with temperature 100°C were removed from an oven. The temperature of the biscuits at time t is modeled by a differentiable function B for which it is known that $B'(t) = -13.84e^{-0.173t}$. Using the given models, at time $t = 10$, how much cooler are the biscuits than the tea?

Part D

1 point is earned for integrand

1 point is earned for using $B(0)=100$

1 point is earned for answer

$$B(10)=100+\int_0^{10} B'(t)dt = 34.18275, H(10)-B(10)=8.817$$

The biscuits are 8.817 degrees Celsius cooler than the tea



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned for using $B(0)=100$

1 point is earned for answer

$$B(10)=100+\int_0^{10} B'(t)dt = 34.18275, H(10)-B(10)=8.817$$

The biscuits are 8.817 degrees Celsius cooler than the tea

8-3b

t (minutes)	0	4	9	15	20
$W(t)$ (degrees Fahrenheit)	55.0	57.1	61.8	67.9	71.0

The temperature of water in a tub at time t is modeled by a strictly increasing, twice-differentiable function W , where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. At time $t = 0$, the temperature of the water is 55°F . The water is heated for 30 minutes, beginning at time $t = 0$. Values of $W(t)$ at selected times t for the first 20 minutes are given in the table above.

42.  For $20 \leq t \leq 25$, the function W that models the water temperature has first derivative given by $w'(t) = 0.4\sqrt{t} \cos(0.06t)$. Based on the model, what is the temperature of the water at time $t=25$?

part A

1 point is earned for the integral

1 point is earned for the answer

$$w(25) = 71.0 + \int_0^{25} w'(t) dt$$

$$= 71.0 + 2.043155 = 73.043$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$w(25) = 71.0 + \int_0^{25} w'(t) dt$$

$$= 71.0 + 2.043155 = 73.043$$

8-3b

43.  Use the data in the table to evaluate $\int_0^{20} W'(t) dt$. Using correct units, interpret the meaning of $\int_0^{20} W'(t) dt$ in the context of this problem.

Part B

1 point is earned for the value

$$\int_0^{20} w'(t) dt = w(20) - w(0) = 71.0 - 55.0 = 16$$

1 point is earned for the interpretation with units

The water has warmed by 16°F over the interval from $t=0$ to $t=20$ minutes.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the value

$$\int_0^{20} w'(t) dt = w(20) - w(0) = 71.0 - 55.0 = 16$$

1 point is earned for the interpretation with units

The water has warmed by 16°F over the interval from $t=0$ to $t=20$ minutes.

8-3b

On a certain workday, the rate, in tons per hour, at which unprocessed gravel arrives at a gravel processing plant is modeled by $G(t) = 90 + 45\cos(t^2/18)$, where t is measured in hours and $0 \leq t \leq 8$. At the beginning of the workday ($t = 0$), the plant has 500 tons of unprocessed gravel. During the hours of operation, $0 \leq t \leq 8$, the plant processes gravel at a constant rate of 100 tons per hour.

44.  Find the total amount of unprocessed gravel that arrives at the plant during the hours of operation on this workday.

Part B

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^8 G(t) dt = 825.551 \text{ tons}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^8 G(t) dt = 825.551 \text{ tons}$$

8-3b

45.  What is the maximum amount of unprocessed gravel at the plant during the hours of operation on this workday? Justify your answer.

Part D

1 point is earned for considering $A'(t)=0$

1 point is earned for the answer

1 point is earned for the justification

The amount of unprocessed gravel at time t is given by

$$A(t) = 500 + \int_0^t (G(s) - 100) ds.$$

$$A'(t) = G(t) - 100 = 0 \Rightarrow t = 4.923480$$

t	$A(t)$
0	500
4.92348	635.376123
8	525.551089

The maximum amount of unprocessed gravel at the plant during this workday is 635.376 tons.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considering $A'(t)=0$

1 point is earned for the answer

1 point is earned for the justification

The amount of unprocessed gravel at time t is given by

8-3b

$$A(t) = 500 + \int_0^t (G(s) - 100) ds.$$

$$A'(t) = G(t) - 100 = 0 \Rightarrow t = 4.923480$$

t	$A(t)$
0	500
4.92348	635.376123
8	525.551089

The maximum amount of unprocessed gravel at the plant during this workday is 635.376 tons.

8-3b

46. For time $t \geq 0$ hours, let $r(t) = 120(1 - e^{-10t^2})$ represent the speed, in kilometers per hour, at which a car travels along a straight road. The number of liters of gasoline used by the car to travel x kilometers is modeled by $g(x) = 0.05x(1 - e^{-x/2})$.



How many liters of gasoline have been used by the car when it reaches a speed of 80 kilometers per hour?

Part C

1 point is earned for $r(t) = 80$

2 points are earned for distance integral

1 point is earned for answer

Let T be the time at which the car's speed reaches 80 kilometers per hour.

Then, $r(T) = 80$ or $T = 0.331453$ hours.

At time T , the car has gone

$$x(T) = \int_0^T r(t) dt = 10.794097 \text{ kilometers}$$

and has consumed $g(x(T)) = 0.537$ liters of gasoline.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $r(t) = 80$

2 points are earned for distance integral

1 point is earned for answer

Let T be the time at which the car's speed reaches 80 kilometers per hour.

Then, $r(T) = 80$ or $T = 0.331453$ hours.

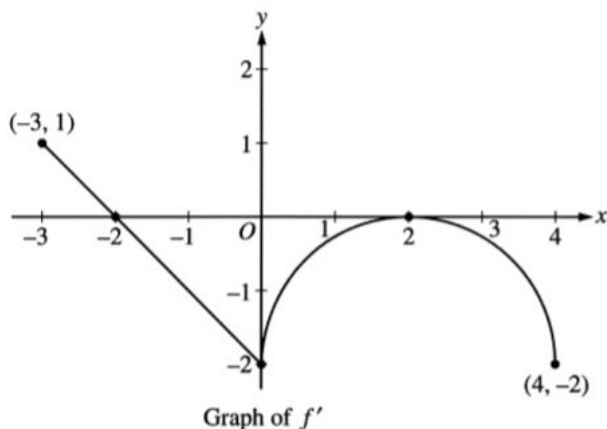
At time T , the car has gone

$$x(T) = \int_0^T r(t) dt = 10.794097 \text{ kilometers}$$

and has consumed $g(x(T)) = 0.537$ liters of gasoline.

8-3b

47.



Let f be a function defined on the closed interval $-3 \leq x \leq 4$ with $f(0) = 3$. The graph of f' , the derivative of f , consists of one line segment and a semicircle, as shown above.

Find $f(-3)$ and $f(4)$. Show the work that leads to your answers.

Part D

1 point is earned for the difference of areas of triangles, $\pm (\frac{1}{2} - 2)$

$$f(0) - f(-3) = \int_{-3}^0 f'(t) dt = \frac{1}{2}(1)(1) - \frac{1}{2}(2)(2) = -\frac{3}{2}$$

1 point is earned for the correct answer for $f(-3)$ using FTC

$$f(-3) = f(0) + \frac{3}{2} = \frac{9}{2}$$

1 point is earned for the area of rectangle - area of semicircle, $\pm (8 - \frac{1}{2}(2)^2\pi)$

$$f(4) - f(0) = \int_0^4 f'(t) dt = -(8 - \frac{1}{2}(2)^2\pi) = -8 + 2\pi$$

1 point is earned for the correct answer of $f(4)$ using FTC

$$f(4) = f(0) - 8 + 2\pi = -5 + 2\pi$$



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

1 point is earned for the difference of areas of triangles, $\pm (\frac{1}{2} - 2)$

$$f(0) - f(-3) = \int_{-3}^0 f'(t) dt = \frac{1}{2}(1)(1) - \frac{1}{2}(2)(2) = -\frac{3}{2}$$

1 point is earned for the correct answer for $f(-3)$ using FTC

$$f(-3) = f(0) + \frac{3}{2} = \frac{9}{2}$$

1 point is earned for the area of rectangle - area of semicircle, $\pm (8 - \frac{1}{2}(2)^2\pi)$

8-3b

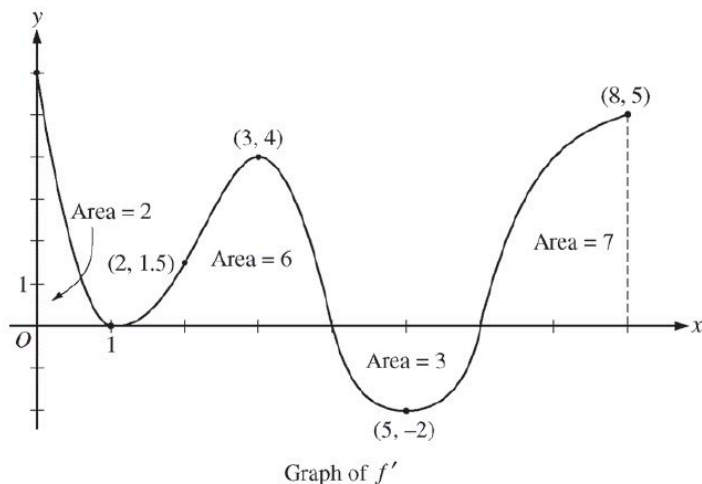
$$f(4) - f(0) = \int_0^4 f'(t) dt = -\left(8 - \frac{1}{2}(2)^2\pi\right) = -8 + 2\pi$$

1 point is earned for the correct answer of $f(4)$ using FTC

$$f(4) = f(0) - 8 + 2\pi = -5 + 2\pi$$

8-3b

48.



The figure above shows the graph of f' the derivative of a twice-differentiable function f , on the closed interval $0 \leq x \leq 8$. The graph of f' has horizontal tangent lines at $x=1$, $x=3$, and $x=5$. The areas of the regions between the graph of f' and the x -axis are labeled in the figure. The function f is defined for all real numbers and satisfied $f(8)=4$.

Determine the absolute minimum value of f on the closed interval $0 \leq x \leq 8$. Justify your answer.

Part B

1 point is earned for considering $x=0$ and $x=6$

1 point is earned for the answer

1 point is earned for justification

From part (a), the absolute minimum occurs either at end point

$$f(0) = f(8) + \int_0^8 f'(x) dx$$

$$= f(8) - \int_0^8 f'(x) dx = 4 - 12 = -8$$

$$f(6) = f(8) + \int_8^6 f'(x) dx$$

$$= f(8) - \int_6^8 f'(x) dx = 4 - 7 = -3$$

$$f(8) = 4$$

The absolute minimum value of f on the closed interval $[0, 8]$ is -8 .

8-3b



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considering $x=0$ and $x=6$

1 point is earned for the answer

1 point is earned for justification

From part (a), the absolute minimum occurs either at $x=6$ or at an endpoint.

$$f(0) = f(8) + \int_0^8 f'(x) dx$$

$$= f(8) - \int_0^8 f'(x) dx = 4 - 12 = -8$$

$$f(6) = f(8) + \int_8^6 f'(x) dx$$

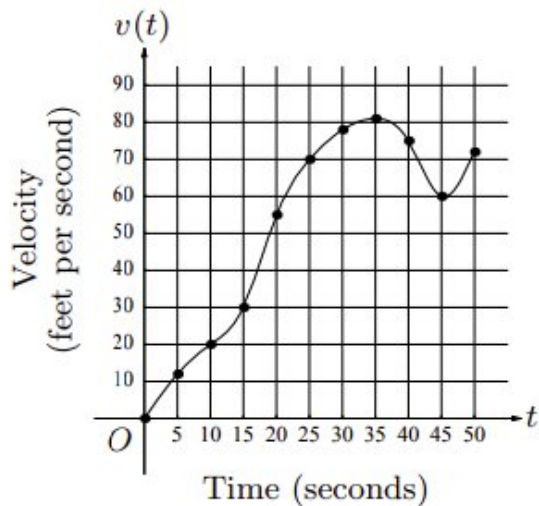
$$= f(8) - \int_6^8 f'(x) dx = 4 - 7 = -3$$

$$f(8) = 4$$

The absolute minimum value of f on the closed interval $[0, 8]$ is -8 .

8-3b

49.



t (seconds)	$v(t)$ (feet per second)
0	0
5	12
10	20
15	30
20	55
25	70
30	78
35	81
40	75
45	60
50	72

The graph of the velocity $v(t)$, in ft/sec, of a car traveling on a straight road, for $0 \leq t \leq 50$, is shown above. A table of values for $v(t)$, at 5 second intervals of time t , is shown to the right of the graph

Approximate $\int_0^{50} v(t) dt$ with a Riemann sum, using the midpoints of five subintervals of equal length. Using correct units, explain the meaning of this integral.

Part D

1 point is earned for midpoint Riemann sum

1 point is earned for answer

1 point is earned for meaning of integral

$$\int_0^{50} v(t) dt$$

$$\approx 10[v(5) + v(15) + v(25) + v(35) + v(45)]$$

$$= 10(12 + 30 + 70 + 81 + 60)$$

8-3b

=2530 feet

This integral is the total distance traveled in feet over the time 0 to 50 seconds.



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for midpoint Riemann sum

1 point is earned for answer

1 point is earned for meaning of integral

$$\int_0^{50} v(t) dt$$

$$\approx 10[v(5) + v(15) + v(25) + v(35) + v(45)]$$

$$= 10(12 + 30 + 70 + 81 + 60)$$

$$= 2530 \text{ feet}$$

This integral is the total distance traveled in feet over the time 0 to 50 seconds.

8-3b

50.

Distance x (cm)	0	1	5	6	8
Temperature $T(x)$ ($^{\circ}\text{C}$)	100	93	70	62	55

A metal wire of length 8 centimeters (cm) is heated at one end. The table above gives selected values of the temperature $T(x)$, in degrees Celsius ($^{\circ}\text{C}$), of the wire x cm from the heated end. The function T is decreasing and twice differentiable.

Find $\int_0^8 T'(x)dx$, and indicate units of measure. Explain the meaning of $\int_0^8 T'(x)dx$ in terms of the temperature of the wire.

Part C

1 point is earned for the value

1 point is earned for the meaning

$$\int_0^8 T'(x)dx = T(8) - T(0) = 55 - 100 = -45^{\circ}\text{C}$$

The temperature drops 45°C from the heated end of the wire of the other end of the wire.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the value

1 point is earned for the meaning

$$\int_0^8 T'(x)dx = T(8) - T(0) = 55 - 100 = -45^{\circ}\text{C}$$

The temperature drops 45°C from the heated end of the wire of the other end of the wire.

8-3b

51.

t (minutes)	0	2	5	7	11	12
$r'(t)$ (feet per minute)	5.7	4.0	2.0	1.2	0.6	0.5

The volume of a spherical hot air balloon expands as the air inside the balloon is heated. The radius of the balloon, in feet, is modeled by a twice-differentiable function r of time t , where t is measured in minutes. For $0 < t < 12$, the graph of r is concave down. The table above gives selected values of the rate of change, $r'(t)$, of radius of the balloon over the time interval $0 \leq t \leq 12$. The radius of the balloon is 30 feet when $t = 5$.

(Note: The volume of a sphere of radius r is given by $V = \frac{4}{3}\pi r^3$.)

Use a right Riemann sum with the five subintervals indicated by the data in the table to approximate $\int_0^{12} r'(t) dt$.

Using correct units, explain the meaning of $\int_0^{12} r'(t) dt$ in terms of the radius of the balloon.

Part C

1 point is earned for the approximation

1 point is earned for the explanation

$$\int_0^{12} r'(t) dt \approx 2(4.0) + 3(2.0) + 2(1.2) + 4(0.6) + 1(0.5)$$

$$= 19.3 \text{ ft}$$

$$\int_0^{12} r'(t) dt \text{ is the change in the radius, in feet, from}$$

$t = 0$ to $t = 12$ minutes.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the approximation

1 point is earned for the explanation

$$\int_0^{12} r'(t) dt \approx 2(4.0) + 3(2.0) + 2(1.2) + 4(0.6) + 1(0.5)$$

$$= 19.3 \text{ ft}$$

$$\int_0^{12} r'(t) dt \text{ is the change in the radius, in feet, from}$$

$t = 0$ to $t = 12$ minutes.

8-3b

52.

t (min)	0	5	10	15	20	25	30	35	40
$v(t)$ (mpm)	7.0	9.2	9.5	7.0	4.5	2.4	2.4	4.3	7.3

A test plane flies in a straight line with positive velocity $v(t)$, in miles per minute at time t minutes, where v is a differentiable function of t . Selected values of $v(t)$ for $0 \leq t \leq 40$ are shown in the table above.

Use a midpoint Riemann sum with four subintervals of equal length and values from the table to approximate

$\int_0^{40} v(t) dt$. Show the computations that lead to your answer. Using correct units, explain the meaning of in $\int_0^{40} v(t) dt$ terms of the plane's flight.

Part A

1 point is earned for $v(5)+v(15)+v(25)+v(35)$

1 point is earned for the answer

1 point is earned for the meaning with units

Midpoint Riemann sum is $10*[v(5)+v(15)+v(25)+v(35)] = 10*[9.2+7.0+2.4+4.3]=229$ The integral gives the total distance in miles that the plane flies during the 40 minutes.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $v(5)+v(15)+v(25)+v(35)$

1 point is earned for the answer

1 point is earned for the meaning with units

Midpoint Riemann sum is $10*[v(5)+v(15)+v(25)+v(35)] = 10*[9.2+7.0+2.4+4.3]=229$

The integral gives the total distance in miles that the plane flies during the 40 minutes.

8-3b

53.

t (hours)	$R(t)$ (gallons per hour)
0	9.6
3	10.4
6	10.8
9	11.2
12	11.4
15	11.3
18	10.7
21	10.2
24	9.6

The rate at which water flows out of a pipe, gallons per hour, is given by a differentiable function R of time t . The table above shows the rate as measured every 3 hours for a 24-hour period.

Use a midpoint Riemann sum with 4 subdivisions of equal length to approximate $\int_0^{24} R(t)dt$. Using correct units, explaining the meaning of your answer in terms of water flow.

Part A

1 point is earned for: $R(3) + R(9) + R(15) + R(21)$

1 point is earned for the answer

1 point is earned for the explanation

$$\int_0^{24} R(t)dt \approx 6 [R(3) + R(9) + R(15) + R(21)]$$

$$6 [10.4 + 11.2 + 11.3 + 10.2]$$

$$258.6 \text{ gallons}$$

This is an approximation to the total flow in gallons of water from the pipe in the 24-hour period.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $R(3) + R(9) + R(15) + R(21)$

1 point is earned for the answer

1 point is earned for the explanation

8-3b

$$\int_0^{24} R(t) dt \approx 6 [R(3) + R(9) + R(15) + R(21)]$$
$$6 [10.4 + 11.2 + 11.3 + 10.2]$$
$$258.6 \text{ gallons}$$

This is an approximation to the total flow in gallons of water from the pipe in the 24-hour period.

54. A city is built around a circular lake that has a radius of 1 mile. The population density of the city is $f(r)$ people per square mile, where r is the distance from the center of the lake, in miles. Which of the following expressions gives the number of people who live within 1 mile of the lake?

(A) $2\pi \int_0^1 r f(r) dr$

(B) $2\pi \int_0^1 r (1 + f(r)) dr$

(C) $2\pi \int_0^2 r (1 + f(r)) dr$

(D) $2\pi \int_1^2 r f(r) dr$



(E) $2\pi \int_1^2 r (1 + f(r)) dr$

55.  A particle moves along the x -axis so that at any time $t > 0$, its acceleration is given by $a(t) = \ln(1 + 2^t)$. If the velocity of the particle is 2 at time $t = 1$, then the velocity of the particle at time $t = 2$ is

(A) 0.462

(B) 1.609

(C) 2.555

(D) 2.886

(E) 3.346



8-3b

56. Oil is leaking from a pipeline on the surface of a lake and forms an oil slick whose volume increases at a constant rate of 2000 cubic centimeters per minute. The oil slick takes the form of a right circular cylinder with both its radius and height changing with time. (Note: The volume V of a right circular cylinder with radius r and height h is given by $V = \pi r^2 h$.)

By the time the recovery device began removing oil, 60,000 cubic centimeters of oil had already leaked. Write, but do not evaluate, an expression involving an integral that gives the volume of oil at the time found previously.

General

The response can earn up to 2 points:

1 point: For the correct limits and initial condition

1 point: For the integrand

The volume of oil, in cm³, in the slick at time $t = 25$ minutes is given by $60,000 + \int_0^{25} (2000 - R(t))dt$.



0	1	2
---	---	---

The response earns both of the following points:

1 point: For the correct limits and initial condition

1 point: For the integrand

The volume of oil, in cm³, in the slick at time $t = 25$ minutes is given by $60,000 + \int_0^{25} (2000 - R(t))dt$.

8-3b

The rate at which rainwater flows into a drainpipe is modeled by the function R , where $R(t) = 20 \sin\left(\frac{t^2}{35}\right)$ cubic feet per hour, t is measured in hours, and $0 \leq t \leq 8$. The pipe is partially blocked, allowing water to drain out the other end of the pipe at a rate modeled by $D(t) = -0.04t^3 + 0.4t^2 + 0.96t$ or $0 \leq t \leq 8$. There are 30 cubic feet of water in the pipe at time $t=0$.

57.  How many cubic feet of rainwater flow into the pipe during the 8-hour time interval $0 \leq t \leq 8$?

Part A

1 point is earned for the Integrand

1 point is earned for the answer

$$\int_8^0 R(t) dt = 76.570$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the Integrand

1 point is earned for the answer

$$\int_8^0 R(t) dt = 76.570$$

8-3b

58. The pipe can hold 50 cubic feet of water before overflowing. For $t > 8$, water continues to flow into and out of the pipe at the given rates until the pipe begins to overflow. Write, but do not solve, an equation involving one or more integrals that gives the time w when the pipe will begin to overflow.

Part D

1 point is earned for the integral

1 point is earned for the equation

$$30 + \int_0^w [R(t) - D(t)] dt = 50$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the equation

$$30 + \int_0^w [R(t) - D(t)] dt = 50$$

8-3b

59.

t (hours)	0	1	3	4	7	8	9
$L(t)$ (people)	120	156	176	126	150	80	0

Concert tickets went on sale at noon ($t = 0$) and were sold out within 9 hours. The number of people waiting in line to purchase tickets at time t is modeled by a twice-differentiable function L for $0 \leq t \leq 9$. Values of $L(t)$ at various times t are shown in the table above.



The rate at which tickets were sold for $0 \leq t \leq 9$ is modeled by $r(t) = 550te^{-t/2}$ tickets per hour. Based on the model, how many tickets were sold by 3 P.M. ($t = 3$), to the nearest whole number?

Part D

The response can earn up to 2 points:

1 point: For the correct integrand

1 point: For the correct limits and correct answer

$$\int_0^{3r} r(t) dt = 972.784$$

There were approximately 973 tickets sold by 3 P.M.



0	1	2
---	---	---

The response can earn up to 2 points:

1 point: For the correct integrand

1 point: For the correct limits and correct answer

$$\int_0^{3r} r(t) dt = 972.784$$

There were approximately 973 tickets sold by 3 P.M.

8-3b

60. The rate of consumption of cola in the United States is given by $S(t) = Ce^{kt}$, where S is measured in billions of gallons per year and t is measured in years from the beginning of 1980.

Using correct units, explain the meaning of $\int_5^7 S(t) dt$ in terms of cola consumption.

Part D

1 point is earned for the correct total consumption in a time period

1 point is earned for the correct time period and liquid measure

This gives the total consumption, in billions of gallons, during the years 1985 and 1986.

0/2 "rate of consumption"



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the correct total consumption in a time period

1 point is earned for the correct time period and liquid measure

This gives the total consumption, in billions of gallons, during the years 1985 and 1986.

0/2 "rate of consumption"

8-3b

61. Water is pumped into an underground tank at a constant rate of 8 gallons per minute. Water leaks out of the tank at the rate of $\sqrt{t+1}$ gallons per minute, for $0 \leq t \leq 120$ minutes. At time $t = 0$, the tank contains 30 gallons of water.

How many gallons of water leak out of the tank from $t = 0$ to $t = 3$ minutes?

Part A

Method 1:

2 points are earned for the definite integral

1: limits

1: integrand

1 point is earned for the answer

- or -

Method 2:

1 point is earned for antiderivative with C

1 point is earned for solves for C using $L(0) = 0$

1 point is earned for the answer

Method 1: $\int_0^3 \sqrt{t+1} dt = \frac{2}{3}(t+1)^{3/2} \Big|_0^3 = \frac{14}{3}$
 -or-

Method 2: $L(t)$ = gallons leaked in first t minutes

$$\frac{dL}{dt} = \sqrt{t+1}; L(t) = \frac{2}{3}(t+1)^{3/2} + C$$

$$L(0) = 0; C = -\frac{2}{3}$$

$$L(t) = \frac{2}{3}(t+1)^{3/2} - \frac{2}{3}; L(3) = \frac{14}{3}$$



0	1	2	3	4	5	6
---	---	---	---	---	---	---

The student response earns all of the following points:

Method 1:

2 points are earned for the definite integral

1: limits

1: integrand

8-3b

1 point is earned for the answer

- or -

Method 2:

1 point is earned for antiderivative with C

1 point is earned for solves for C using $L(0) = 0$

1 point is earned for the answer

Method 1: $\int_0^3 \sqrt{t+1} dt = \frac{2}{3}(t+1)^{3/2} \Big|_0^3 = \frac{14}{3}$
— or —

Method 2: $L(t)$ = gallons leaked in first t minutes

$$\frac{dL}{dt} = \sqrt{t+1}; L(t) = \frac{2}{3}(t+1)^{3/2} + C$$

$$L(0) = 0; C = -\frac{2}{3}$$

$$L(t) = \frac{2}{3}(t+1)^{3/2} - \frac{2}{3}; L(3) = \frac{14}{3}$$

8-3b

62. The temperature outside a house during a 24-hour period is given by

$$F(t) = 80 - 10 \cos\left(\frac{\pi t}{12}\right), \quad 0 \leq t \leq 24,$$

where $F(t)$ is measured in degrees Fahrenheit and t is measured in hours.



The cost of cooling the house accumulates at the rate of \$0.05 per hour for each degree the outside temperature exceeds 78 degrees Fahrenheit. What was the total cost, to the nearest cent, to cool the house for this 24-hour period?

Part D

1 point is earned for limits and 0.05

1 point is earned for integrand

1 point is earned for answer

$$C = 0.05 \int_{5.231}^{18.770} \left(\left[80 - 10 \cos\left(\frac{\pi t}{12}\right) \right] - 78 \right) dt$$

or
18.769
or
5.230

$$= 0.05(101.92741) = 5.096 \approx \$5.10$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for limits and 0.05

1 point is earned for integrand

1 point is earned for answer

$$C = 0.05 \int_{5.231}^{18.770} \left(\left[80 - 10 \cos\left(\frac{\pi t}{12}\right) \right] - 78 \right) dt$$

or
18.769
or
5.230

$$= 0.05(101.92741) = 5.096 \approx \$5.10$$

8-3b

63.  The rate of change of the altitude of a hot-air balloon is given by $r(t) = t^3 - 4t^2 + 6$ for $0 \leq t \leq 8$. Which of the following expressions gives the change in altitude of the balloon during the time the altitude is decreasing?

(A) $\int_{1.572}^{3.514} r(t) dt$ ✓

(B) $\int_0^8 r(t) dt$

(C) $\int_0^{2.667} r(t) dt$

(D) $\int_{1.572}^{3.514} r'(t) dt$

(E) $\int_0^{2.667} r'(t) dt$

8-3b

64.



t (hours)	0	1	3	4	7	8	9
$L(t)$ (people)	120	156	176	126	150	80	0

Concert tickets went on sale at noon ($t = 0$) and were sold out within 9 hours. The number of people waiting in line to purchase tickets at time t is modeled by a twice-differentiable function L for $0 \leq t \leq 9$. Values of $L(t)$ at various times t are shown in the table above.

- (a) Use the data in the table to estimate the rate at which the number of people waiting in line was changing at 5:30 P.M. ($t = 5.5$). Show the computations that lead to your answer. Indicate units of measure.
- (b) Use a trapezoidal sum with three subintervals to estimate the average number of people waiting in line during the first 4 hours that tickets were on sale.
- (c) For $0 \leq t \leq 9$, what is the fewest number of times at which $L'(t)$ must equal 0? Give a reason for your answer.
- (d) The rate at which tickets were sold for $0 \leq t \leq 9$ is modeled by $r(t) = 550te^{-t/2}$ tickets per hour. Based on the model, how many tickets were sold by 3 P.M. ($t = 3$), to the nearest whole number?

Part A

1 point is earned for estimation

1 point is earned for units

$$L'(5.5) = \frac{L(7) - L(4)}{7 - 4} = \frac{150 - 126}{3} = 8 \text{ people per hour}$$



0	1	2
---	---	---

The student earns all of the following points:

1 point is earned for estimation

1 point is earned for units

$$L'(5.5) = \frac{L(7) - L(4)}{7 - 4} = \frac{150 - 126}{3} = 8 \text{ people per hour}$$

Part B

1 point is earned for trapezoidal sum

1 point is earned for answer

The average number of people waiting in line during the first 4 hours is approximately

8-3b

$$\frac{1}{4} \left(\frac{L(0) + L(1)}{2}(1 - 0) + \frac{L(1) + L(3)}{2}(3 - 1) + \frac{L(3) + L(4)}{2}(4 - 3) \right)$$

= 155.25 people



0	1	2
---	---	---

The student earns all of the following points:

1 point is earned for trapezoidal sum

1 point is earned for answer

The average number of people waiting in line during the first 4 hours is approximately

$$\frac{1}{4} \left(\frac{L(0) + L(1)}{2}(1 - 0) + \frac{L(1) + L(3)}{2}(3 - 1) + \frac{L(3) + L(4)}{2}(4 - 3) \right)$$

= 155.25 people

Part C

1 point is earned for considering change in sign of L'

1 point is earned for the analysis

1 point is earned for the correct conclusion

L is differentiable $[0, 9]$ so the Mean Value Theorem implies $L'(t) > 0$ for some t in $(1, 3)$ and some t in $(4, 7)$. Similarly, $L'(t) < 0$ for some t in $(3, 4)$ and some t in $(7, 8)$. Then, since L' is continuous on $[0, 9]$, the Intermediate Value Theorem implies that $L'(t) = 0$ for at least three values of t in $[0, 9]$.

OR

1 point is earned for considering relative extrema of L on $(0, 9)$

1 point is earned for the analysis

1 point is earned for the correct conclusion

The continuity of L on $[1, 4]$ implies that L attains a maximum value there. Since $L(3) > L(1)$ and $L(3) > L(4)$, this maximum occurs on $(1, 4)$. Similarly, L attains a minimum on $(3, 7)$ and a maximum on $(4, 8)$. L is differentiable, so $L'(t) = 0$ at each relative extreme point on $(0, 9)$. Therefore $L'(t) = 0$ for at least three values of t in $[0, 9]$.

[Note: There is a function L that satisfies the given conditions with $L'(t) = 0$ for exactly three values of t .]

8-3b



0	1	2	3
---	---	---	---

The student earns all of the following points:

1 point is earned for considering change in sign of L'

1 point is earned for the analysis

1 point is earned for the correct conclusion

L is differentiable $[0, 9]$ so the Mean Value Theorem implies $L'(t) > 0$ for some t in $(1, 3)$ and some t in $(4, 7)$. Similarly, $L'(t) < 0$ for some t in $(3, 4)$ and some t in $(7, 8)$. Then, since L' is continuous on $[0, 9]$, the Intermediate Value Theorem implies that $L'(t) = 0$ for at least three values of t in $[0, 9]$.

OR

1 point is earned for considering relative extrema of L on $(0, 9)$

1 point is earned for the analysis

1 point is earned for the correct conclusion

The continuity of L on $[1, 4]$ implies that L attains a maximum value there. Since $L(3) > L(1)$ and $L(3) > L(4)$, this maximum occurs on $(1, 4)$. Similarly, L attains a minimum on $(3, 7)$ and a maximum on $(4, 8)$. L is differentiable, so $L'(t) = 0$ at each relative extreme point on $(0, 9)$. Therefore $L'(t) = 0$ for at least three values of t in $[0, 9]$.

[Note: There is a function L that satisfies the given conditions with $L'(t) = 0$ for exactly three values of t .]

Part D

1 point is earned for the correct integrand

1 point is earned for the correct limits and answer

$$\int_0^3 r(t) dt = 972.784$$

There were approximately 973 tickets sold by 3 P.M.



0	1	2
---	---	---

The student earns all of the following points:

1 point is earned for the correct integrand

1 point is earned for the correct limits and answer

8-3b

$$\int_0^3 r(t) \, dt = 972.784$$

There were approximately 973 tickets sold by 3 P.M.

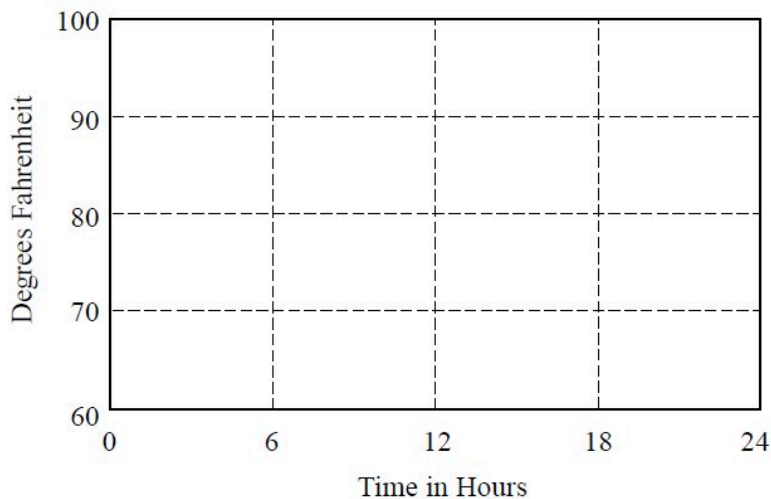
8-3b

65.  The temperature outside a house during a 24-hour period is given by

$$F(t) = 80 - 10 \cos\left(\frac{\pi t}{12}\right), \quad 0 \leq t \leq 24,$$

where $F(t)$ is measured in degrees Fahrenheit and t is measured in hours.

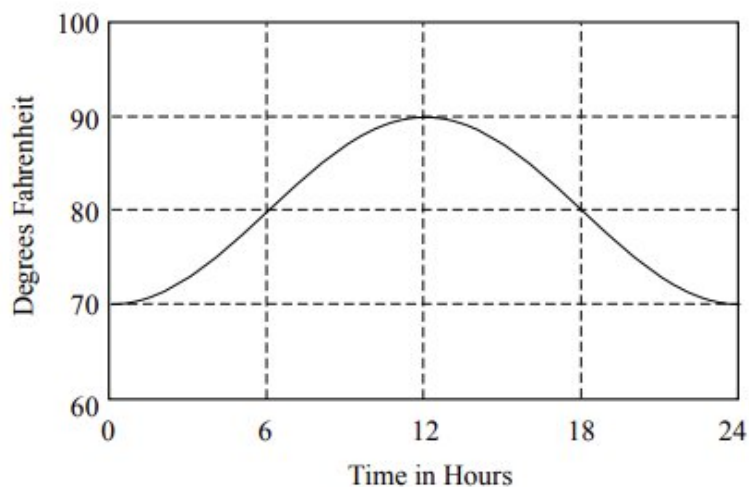
- (a) Sketch the graph of F on the grid below.



- (b) Find the average temperature, to the nearest degree Fahrenheit, between $t = 6$ and $t = 14$.
- (c) An air conditioner cooled the house whenever the outside temperature was at or above 78 degrees Fahrenheit. For what values of t was the air conditioner cooling the house?
- (d) The cost of cooling the house accumulates at the rate of \$0.05 per hour for each degree the outside temperature exceeds 78 degrees Fahrenheit. What was the total cost, to the nearest cent, to cool the house for this 24-hour period?

Part A

1 point is earned for bell-shaped graph minimum 70 at $t = 0$, $t = 24$ only maximum 90 at $t = 12$ only



8-3b

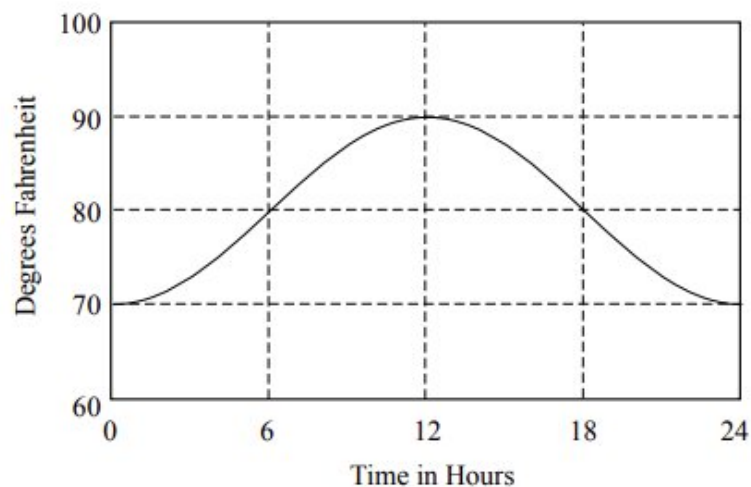
✓

0

1

The student response earns all of the following points:

1 point is earned for bell-shaped graph minimum 70 at $t = 0$, $t = 24$ only maximum 90 at $t = 12$ only

**Part B**

1 point is earned for limits and $1/(14 - 6)$

1 point is earned for integrand

1 point is earned for answer

$$\begin{aligned}
 \text{Avg.} &= \frac{1}{14-6} \int_6^{14} \left[80 - 10 \cos \left(\frac{\pi t}{12} \right) \right] dt \\
 &= \frac{1}{8} (697.2957795) \\
 &= 87.162 \text{ or } 87.161 \\
 &= 87^\circ \text{F}
 \end{aligned}$$

✓

0

1

2

3

The student response earns all of the following points:

1 point is earned for limits and $1/(14 - 6)$

1 point is earned for integrand

8-3b

1 point is earned for answer

$$\begin{aligned}\text{Avg.} &= \frac{1}{14-6} \int_6^{14} \left[80 - 10 \cos \left(\frac{\pi t}{12} \right) \right] dt \\ &= \frac{1}{8} (697.2957795) \\ &= 87.162 \text{ or } 87.161 \\ &= 87^\circ \text{F}\end{aligned}$$

Part C

1 point is earned for inequality or equation

1 point is earned for solutions with interval

$$\begin{aligned}[80 - 10 \cos \left(\frac{\pi t}{12} \right)] - 78 &\geq 0 \\ 2 - 10 \cos \left(\frac{\pi t}{12} \right) &\geq 0 \\ \left. \begin{array}{c} 5.230 \\ \text{or} \\ 5.231 \end{array} \right\} \leq t \leq \left\{ \begin{array}{c} 18.769 \\ \text{or} \\ 18.770 \end{array} \right.\end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for inequality or equation

1 point is earned for solutions with interval

$$\begin{aligned}[80 - 10 \cos \left(\frac{\pi t}{12} \right)] - 78 &\geq 0 \\ 2 - 10 \cos \left(\frac{\pi t}{12} \right) &\geq 0 \\ \left. \begin{array}{c} 5.230 \\ \text{or} \\ 5.231 \end{array} \right\} \leq t \leq \left\{ \begin{array}{c} 18.769 \\ \text{or} \\ 18.770 \end{array} \right.\end{aligned}$$

Part D

1 point is earned for limits and 0.05

1 point is earned for integrand

8-3b

1 point is earned for answer

$$\begin{array}{c}
 18.770 \\
 \text{or} \\
 18.769 \\
 C = 0.05 \int_{5.231}^{} \left(\left[80 - 10 \cos \left(\frac{\pi t}{12} \right) \right] - 78 \right) dt \\
 5.231 \\
 \text{or} \\
 5.230 \\
 = 0.05 (101.92741) = 5.096 \approx \$5.10
 \end{array}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for limits and 0.05

1 point is earned for integrand

1 point is earned for answer

$$\begin{array}{c}
 18.770 \\
 \text{or} \\
 18.769 \\
 C = 0.05 \int_{5.231}^{} \left(\left[80 - 10 \cos \left(\frac{\pi t}{12} \right) \right] - 78 \right) dt \\
 5.231 \\
 \text{or} \\
 5.230 \\
 = 0.05 (101.92741) = 5.096 \approx \$5.10
 \end{array}$$

8-3b

66.

t (minutes)	0	2	5	9	10
$H(t)$ (degrees Celsius)	66	60	52	44	43

 As a pot of tea cools, the temperature of the tea is modeled by a differentiable function H for $0 \leq t \leq 10$, where time t is measured in minutes and temperature $H(t)$ is measured in degrees Celsius. Values of $H(t)$ at selected values of time t are shown in the table above.

(a) Use the data in the table to approximate the rate at which the temperature of the tea is changing at time $t = 3.5$. Show the computations that lead to your answer.

(b) Using correct units, explain the meaning of $\frac{1}{10} \int_0^{10} H(t) dt$ in the context of this problem. Use a trapezoidal sum with the four subintervals indicated by the table to estimate $\frac{1}{10} \int_0^{10} H(t) dt$.

(c) Evaluate $\int_0^{10} H'(t) dt$. Using correct units, explain the meaning of the expression in the context of this problem.

(d) At time $t = 0$, biscuits with temperature 100°C were removed from an oven. The temperature of the biscuits at time t is modeled by a differentiable function B for which it is known that $B'(t) = -13.84e^{-0.173t}$. Using the given models, at time $t = 10$, how much cooler are the biscuits than the tea?

Part A

1 point is earned for the answer

$$H'(3.5) \approx \frac{H(5) - H(2)}{5 - 2}$$

$$= \frac{52 - 60}{3} = -2.666 \text{ or } -2.667 \text{ degrees Celsius per minute}$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

$$H'(3.5) \approx \frac{H(5) - H(2)}{5 - 2}$$

$$= \frac{52 - 60}{3} = -2.666 \text{ or } -2.667 \text{ degrees Celsius per minute}$$

Part B

8-3b

1 point is earned for meaning of the expression

1 point is earned for trapezoidal sum

1 point is earned for estimate

$\frac{1}{10} \int_0^{10} H(t) dt$ is the average temperature of the tea, in degrees Celsius, over the 10 minutes.

$$\begin{aligned} \frac{1}{10} \int_0^{10} H(t) dt &\approx \frac{1}{10} \left(2 \cdot \frac{66+60}{2} + 3 \cdot \frac{60+52}{2} + 4 \cdot \frac{52+44}{2} + 1 \cdot \frac{44+33}{2} \right) \\ &= 52.95 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for meaning of the expression

1 point is earned for trapezoidal sum

1 point is earned for estimate

$\frac{1}{10} \int_0^{10} H(t) dt$ is the average temperature of the tea, in degrees Celsius, over the 10 minutes.

$$\begin{aligned} \frac{1}{10} \int_0^{10} H(t) dt &\approx \frac{1}{10} \left(2 \cdot \frac{66+60}{2} + 3 \cdot \frac{60+52}{2} + 4 \cdot \frac{52+44}{2} + 1 \cdot \frac{44+33}{2} \right) \\ &= 52.95 \end{aligned}$$

Part C

1 point is earned for the value of integral

1 point is earned for meaning of the expression

$$\int_0^{10} H'(t) dt = H(10) - H(0) = 43 - 66 = -23$$

The temperature of the tea drops 23 degrees Celsius from time $t = 0$ to time $t = 10$ minutes.

8-3b



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the value of integral

1 point is earned for meaning of the expression

$$\int_0^{10} H'(t) dt = H(10) - H(0) = 43 - 66 = -23$$

The temperature of the tea drops 23 degrees Celsius from time $t = 0$ to time $t = 10$ minutes.

Part D

1 point is earned for integrand

1 point is earned for using $B(0) = 100$

1 point is earned for answer $B(10) = 100 + \int_0^{10} B'(t) dt = 34.18275$; $H(10) - H(0) = 8.817$

The biscuits are 8.817 degrees Celsius cooler than the tea.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned for using $B(0) = 100$

1 point is earned for answer

$$B(10) = 100 + \int_0^{10} B'(t) dt = 34.18275; H(10) - H(0) = 8.817$$

The biscuits are 8.817 degrees Celsius cooler than the tea.

8-3b

67.  The rate at which rainwater flows into a drainpipe is modeled by the function R , where $R(t) = 20 \sin\left(\frac{t^2}{35}\right)$ cubic feet per hour, t is measured in hours, and $0 \leq t \leq 8$. The pipe is partially blocked, allowing water to drain out the other end of the pipe at a rate modeled by $D(t) = -0.04t^3 + 0.4t^2 + 0.96t$ cubic feet per hour, for $0 \leq t \leq 8$. There are 30 cubic feet of water in the pipe at time $t = 0$.
- How many cubic feet of rainwater flow into the pipe during the 8-hour time interval $0 \leq t \leq 8$?
 - Is the amount of water in the pipe increasing or decreasing at time $t = 3$ hours? Give a reason for your answer.
 - At what time t , $0 \leq t \leq 8$, is the amount of water in the pipe at a minimum? Justify your answer.
 - The pipe can hold 50 cubic feet of water before overflowing. For $t > 8$, water continues to flow into and out of the pipe at the given rates until the pipe begins to overflow. Write, but do not solve, an equation involving one or more integrals that gives the time w when the pipe will begin to overflow.

Part A

1 point is earned for the Integrand

1 point is earned for the answer

$$\int_0^8 R(t) dt = 76.570$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the Integrand

1 point is earned for the answer

$$\int_0^8 R(t) dt = 76.570$$

Part B

1 point is earned for considering $R(3)$ and $D(3)$

1 point is earned for the answer and reason

$$R(3) - D(3) = -0.313632 < 0$$

Since $R(3) < D(3)$, the amount of water in the pipe is decreasing at time $t = 3$ hours.

8-3b



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for considering $R(3)$ and $D(3)$

1 point is earned for the answer and reason

$$R(3) - D(3) = -0.313632 < 0$$

Since $R(3) < D(3)$, the amount of water in the pipe is decreasing at time $t = 3$ hours.

Part C

1 point is earned for considering $R(t) - D(t) = 0$

1 point is earned for the answer

1 point is earned for the justification

The amount of water in the pipe at time t , $0 \leq t \leq 8$, is $30 + \int_0^t [R(x) - D(x)] dx$.

$$R(t) - D(t) = 0 \Rightarrow t = 0, 3.271658$$

t	Amount of water in the pipe
0	30
3.271658	27.964561
8	48.543686

The amount of water in the pipe is a minimum at time $t = 3.272$ (or 3.271) hours.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considering $R(t) - D(t) = 0$

1 point is earned for the answer

1 point is earned for the justification

The amount of water in the pipe at time t , $0 \leq t \leq 8$, is $30 + \int_0^t [R(x) - D(x)] dx$.

$$R(t) - D(t) = 0 \Rightarrow t = 0, 3.271658$$

8-3b

t	Amount of water in the pipe
0	30
3.271658	27.964561
8	48.543686

The amount of water in the pipe is a minimum at time $t = 3.272$ (or 3.271) hours.

Part D

1 point is earned for the integral

1 point is earned for the equation

$$30 + \int_0^w [R(t) - D(t)]dt = 50$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the equation

$$30 + \int_0^w [R(t) - D(t)]dt = 50$$

8-3b

68.

t (hours)	0	2	5	7	8
$E(t)$ (hundreds of entries)	0	4	13	21	23

A zoo sponsored a one-day contest to name a new baby elephant. Zoo visitors deposited entries in a special box between noon ($t = 0$) and 8 P.M. ($t = 8$). The number of entries in the box t hours after noon is modeled by a differentiable function E for $0 \leq t \leq 8$. Values of $E(t)$, in hundreds of entries, at various times t are shown in the table above.

(a) Use the data in the table to approximate the rate, in hundreds of entries per hour, at which entries were being deposited at time $t = 6$. Show the computations that lead to your answer.

(b) Use a trapezoidal sum with the four subintervals given by the table to approximate the value of $18 \int_0^8 E(t) dt$.

. Using correct units, explain the meaning of $18 \int_0^8 E(t) dt$ in terms of the number of entries.

(c) At 8 P.M., volunteers began to process the entries. They processed the entries at a rate modeled by the function P , where $P(t) = t^3 - 30t^2 + 298t - 976$ hundreds of entries per hour for $8 \leq t \leq 12$. According to the model, how many entries had not yet been processed by midnight ($t = 12$)?

(d) According to the model from part (c), at what time were the entries being processed most quickly? Justify your answer.

Part A

1 point is earned for answer

$$E'(6) \approx \frac{E(7) - E(5)}{7 - 5} = 4 \text{ hundred entries per hour}$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for answer

$$E'(6) \approx \frac{E(7) - E(5)}{7 - 5} = 4 \text{ hundred entries per hour}$$

Part B

1 point is earned for trapezoidal sum

1 point is earned for approximation

1 point is earned for meaning

8-3b

$$\begin{aligned}\frac{1}{8} \int_0^8 E(t) dt &\approx \\ \frac{1}{8} \left(2 \cdot \frac{E(0)+E(2)}{2} + 3 \cdot \frac{E(2)+E(5)}{2} + 2 \cdot \frac{E(5)+E(7)}{2} + 1 \cdot \frac{E(7)+E(8)}{2} \right) \\ &= 10.687 \text{ or } 10.688\end{aligned}$$

$$\frac{1}{8} \int_0^8 E(t) dt \text{ is the average number of hundreds of entries in the box between noon and 8 P.M.}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for trapezoidal sum

1 point is earned for approximation

1 point is earned for meaning

$$\begin{aligned}\frac{1}{8} \int_0^8 E(t) dt &\approx \\ \frac{1}{8} \left(2 \cdot \frac{E(0)+E(2)}{2} + 3 \cdot \frac{E(2)+E(5)}{2} + 2 \cdot \frac{E(5)+E(7)}{2} + 1 \cdot \frac{E(7)+E(8)}{2} \right) \\ &= 10.687 \text{ or } 10.688\end{aligned}$$

$$\frac{1}{8} \int_0^8 E(t) dt \text{ is the average number of hundreds of entries in the box between noon and 8 P.M.}$$

Part C

1 point is earned for integral

1 point is earned for the answer

$$23 - \int_8^{12} P(t) dt = 23 - 16 = 7 \text{ hundred entries}$$



0	1	2
---	---	---

8-3b

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

$$23 - \int_8^{12} P(t) dt = 23 - 16 = 7 \text{ hundred entries}$$

Part D

1 point is earned for considering $P'(t) = 0$

1 point is earned for identifying candidates

1 point is earned for answering with justification

$P'(t) = 0$ when $t = 9.183503$ and $t = 10.816497$.

t	$P(t)$
8	0
9.183503	5.088662
10.816497	2.911338
12	8

Entries are being processed most quickly at time $t = 12$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considering $P'(t) = 0$

1 point is earned for identifying candidates

1 point is earned for answering with justification

$P'(t) = 0$ when $t = 9.183503$ and $t = 10.816497$.

t	$P(t)$
8	0
9.183503	5.088662
10.816497	2.911338
12	8

Entries are being processed most quickly at time $t = 12$.

8-3b

69.  The rate at which people enter an amusement park on a given day is modeled by the function E defined by

$$E(t) = \frac{15600}{(t^2 - 24t + 160)}.$$

The rate at which people leave the same amusement park on the same day is modeled by the function L defined by

$$L(t) = \frac{9890}{(t^2 - 38t + 370)}.$$

Both $E(t)$ and $L(t)$ are measured in people per hour and time t is measured in hours after midnight. These functions are valid for $9 \leq t \leq 23$, the hours during which the park is open. At time $t = 9$, there are no people in the park.

- (a) How many people have entered the park by 5:00 P.M. ($t = 17$)? Round your answer to the nearest whole number.
- (b) The price of admission to the park is \$15 until 5:00 P.M. ($t = 17$). After 5:00 P.M., the price of admission to the park is \$11. How many dollars are collected from admissions to the park on the given day? Round your answer to the nearest whole number.
- (c) Let $H(t) = \int_9^t (E(x) - L(x))dx$ for $9 \leq t \leq 23$. The value of $H(17)$ to the nearest whole number is 3725. Find the value of $H'(17)$, and explain the meaning of $H(17)$ and $H'(17)$ in the context of the amusement park.
- (d) At what time t , for $9 \leq t \leq 23$, does the model predict that the number of people in the park is a maximum?

Part A

1 point is earned for limits

1 point is earned for integrand

1 point is earned for the answer

$$\int_9^{17} E(t)dt = 6004.270$$

6004 people entered the park by 5 pm.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for limits

1 point is earned for integrand

1 point is earned for the answer

$$\int_9^{17} E(t)dt = 6004.270$$

8-3b

6004 people entered the park by 5 pm.

Part B

1 point is earned for the setup

$$15 \int_9^{17} E(t) dt + 11 \int_{17}^{23} E(t) dt = 104048.165$$

The amount collected was \$104,048.

or

$$\int_{17}^{23} E(t) dt = 1271.283$$

1271 people entered the park between 5 pm and 11 pm, so the amount collected was

$$\$15 \cdot (6004) + \$11 \cdot (1271) = \$104,041.$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the setup

$$15 \int_9^{17} E(t) dt + 11 \int_{17}^{23} E(t) dt = 104048.165$$

The amount collected was \$104,048.

or

$$\int_{17}^{23} E(t) dt = 1271.283$$

1271 people entered the park between 5 pm and 11 pm, so the amount collected was

$$\$15 \cdot (6004) + \$11 \cdot (1271) = \$104,041.$$

Part C

8-3b

1 point is earned for value $H'(17)$

2 points are earned for meanings

1: meaning of $H(17)$

1: meaning of $H'(17)$

<-1> if no reference to $t = 17$

$$H'(17) = E(17) - L(17) = -380.281$$

There were 3725 people in the park at $t = 17$. The number of people in the park was decreasing at the rate of approximately 380 people/hr at time $t = 17$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for value $H'(17)$

2 points are earned for meanings

1: meaning of $H(17)$

1: meaning of $H'(17)$

<-1> if no reference to $t = 17$

$$H'(17) = E(17) - L(17) = -380.281$$

There were 3725 people in the park at $t = 17$. The number of people in the park was decreasing at the rate of approximately 380 people/hr at time $t = 17$.

Part D

1 point is earned for $E(t) - L(t) = 0$

1 point is earned for the answer

$$H'(t) = E(t) - L(t) = 0$$

$$t = 15.794 \text{ or } 15.795$$



0	1	2
---	---	---

The student response earns all of the following points:

8-3b

1 point is earned for $E(t) - L(t) = 0$

1 point is earned for the answer

$$H'(t) = E(t) - L(t) = 0$$

$$t = 15.794 \text{ or } 15.795$$

8-3b

70.  The rate at which people enter an auditorium for a rock concert is modeled by the function R given by $R(t) = 1380t^2 - 675t^3$ for $0 \leq t \leq 2$ hours; $R(t)$ is measured in people per hour. No one is in the auditorium at time $t = 0$, when the doors open. The doors close and the concert begins at time $t = 2$.
- (a) How many people are in the auditorium when the concert begins?
- (b) Find the time when the rate at which people enter the auditorium is a maximum. Justify your answer.
- (c) The total wait time for all the people in the auditorium is found by adding the time each person waits, starting at the time the person enters the auditorium and ending when the concert begins. The function w models the total wait time for all the people who enter the auditorium before time t . The derivative of w is given by $w'(t) = (2 - t)R(t)$. Find $w(2) - w(1)$, the total wait time for those who enter the auditorium after time $t = 1$.
- (d) On average, how long does a person wait in the auditorium for the concert to begin? Consider all people who enter the auditorium after the doors open, and use the model for total wait time from part (c).

Part A

1 point is earned for integral

1 point is earned for answer

$$\int_0^2 R(t) dt = 980 \text{ people}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for answer

$$\int_0^2 R(t) dt = 980 \text{ people}$$

Part B

1 point is earned if considers $R'(t) = 0$

1 point is earned for interior critical point

1 point is earned for the answer plus justification

$$R'(t) = 0 \text{ when } t = 0 \text{ and } t = 1.36296$$

The maximum rate may occur at 0, $a = 1.36296$, or 2.

$$R(0) = 0$$

8-3b

$$R(a) = 854.527$$

$$R(2) = 120$$

The maximum rate occurs when $t = 1.362$ or 1.363 .



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if considers $R'(t) = 0$

1 point is earned for interior critical point

1 point is earned for the answer plus justification

$$R'(t) = 0 \text{ when } t = 0 \text{ and } t = 1.36296$$

The maximum rate may occur at 0, $a = 1.36296$, or 2.

$$R(0) = 0$$

$$R(a) = 854.527$$

$$R(2) = 120$$

The maximum rate occurs when $t = 1.362$ or 1.363 .

Part C

1 point is earned for integral

1 point is earned for answer

$$w(2) - w(1) = \int_1^2 w'(t) dt = \int_1^2 (2 - t)R(t) dt = 387.5$$

The total wait time for those who enter the auditorium after time $t = 1$ is 387.5 hours.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for answer

8-3b

$$w(2) - w(1) = \int_1^2 w'(t) dt = \int_1^2 (2 - t)R(t) dt = 387.5$$

The total wait time for those who enter the auditorium after time $t = 1$ is 387.5 hours.

Part D

1 point is earned for integral

1 point is earned for answer

$$\frac{1}{980}w(2) = \frac{1}{980} \int_0^2 (2 - t)R(t) dt = 0.77551$$

On average, a person waits 0.775 or 0.776 hour.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for answer

$$\frac{1}{980}w(2) = \frac{1}{980} \int_0^2 (2 - t)R(t) dt = 0.77551$$

On average, a person waits 0.775 or 0.776 hour.

8-3b

71.  For time $t \geq 0$ hours, let $r(t) = 120(1 - e^{-10t^2})$ represent the speed, in kilometers per hour, at which a car travels along a straight road. The number of liters of gasoline used by the car to travel x kilometers is modeled by $g(x) = 0.05x(1 - e^{-x/2})$.

- (a) How many kilometers does the car travel during the first 2 hours?
 (b) Find the rate of change with respect to time of the number of liters of gasoline used by the car when $t = 2$ hours. Indicate units of measure.
 (c) How many liters of gasoline have been used by the car when it reaches a speed of 80 kilometers per hour?

Part A

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^2 r(t) dt = 206.370 \text{ kilometers}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^2 r(t) dt = 206.370 \text{ kilometers}$$

Part B

2 points are earned for using chain rule

1 point is earned for the answer with units

$$\begin{aligned} \frac{dg}{dt} &= \frac{dg}{dx} \cdot \frac{dx}{dt}; \quad \frac{dx}{dt} = r(t) \\ \frac{dg}{dt} \Big|_{t=2} &= \frac{dg}{dx} \Big|_{x=206.370} \cdot r(2) \end{aligned}$$

$$= (0.050)(120) = 6 \text{ liters/hour}$$

8-3b



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for using chain rule

1 point is earned for the answer with units

$$\begin{aligned}\frac{dg}{dt} &= \frac{dg}{dx} \cdot \frac{dx}{dt}; \quad \frac{dx}{dt} = r(t) \\ \frac{dg}{dt} \Big|_{t=2} &= \frac{dg}{dx} \Big|_{x=206.370} \cdot r(2) \\ &= (0.050)(120) = 6 \text{ liters/hour}\end{aligned}$$

Part C

1 point is earned for equation $r(t) = 80$

2 points are earned for distance integral

1 point is earned for answer

Let T be the time at which the car's speed reaches 80 kilometers per hour.

Then, $r(T) = 80$ or $T = 0.331453$ hours.

At time T , the car has gone

$$x(T) = \int_0^T r(t) \, dt = 10.794097 \text{ kilometers}$$

and has consumed $g(x(T)) = 0.537$ liters of gasoline.



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for equation $r(t) = 80$

2 points are earned for distance integral

1 point is earned for answer

Let T be the time at which the car's speed reaches 80 kilometers per hour.

Then, $r(T) = 80$ or $T = 0.331453$ hours.

8-3b

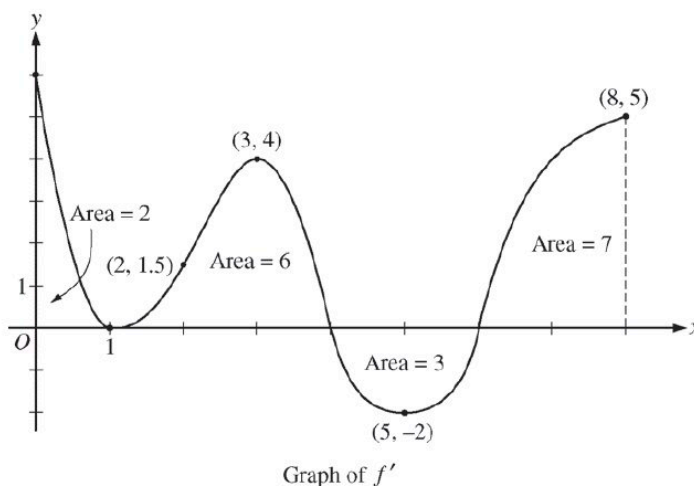
At time T , the car has gone

$$x(T) = \int_0^T r(t) \, dt = 10.794097 \text{ kilometers}$$

and has consumed $g(x(T)) = 0.537$ liters of gasoline.

8-3b

72.



The figure above shows the graph of f' , the derivative of a twice-differentiable function f , on the closed interval $0 \leq x \leq 8$. The graph of f' has horizontal tangent lines at $x = 1$, $x = 3$, and $x = 5$. The areas of the regions between the graph of f' and the x -axis are labeled in the figure. The function f is defined for all real numbers and satisfies $f(8) = 4$.

- Find all values of x on the open interval $0 < x < 8$ for which the function f has a local minimum. Justify your answer.
- Determine the absolute minimum value of f on the closed interval $0 \leq x \leq 8$. Justify your answer.
- On what open intervals contained in $0 < x < 8$ is the graph of f both concave down and increasing? Explain your reasoning.
- The function g is defined by $g(x) = (f(x))^3$. If $f(3) = -5/2$, find the slope of the line tangent to the graph of g at $x = 3$.

Part A

1 point is earned for the answer with justification

$x = 6$ is the only critical point at which f' changes sign from negative to positive. Therefore, f has a local minimum at $x = 6$.



0

1

The student response earns all of the following points:

1 point is earned for the answer with justification

$x = 6$ is the only critical point at which f' changes sign from negative to positive. Therefore, f has a local minimum at $x = 6$.

Part B

1 point is earned for considering $x = 0$ and $x = 6$

1 point is earned for the answer

1 point is earned for justification

8-3b

From part (a), the absolute minimum occurs either at $x = 6$ or at an endpoint.

$$\begin{aligned} f(0) &= f(8) + \int_8^0 f'(x) dx \\ &= f(8) - \int_0^8 f'(x) dx = 4 - 12 = -8 \\ f(6) &= f(8) + \int_8^6 f'(x) dx \\ &= f(8) - \int_6^8 f'(x) dx = 4 - 7 = -3 \end{aligned}$$

$$f(8) = 4$$

The absolute minimum value of f on the closed interval $[0, 8]$ is -8.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considering $x = 0$ and $x = 6$

1 point is earned for the answer

1 point is earned for justification

From part (a), the absolute minimum occurs either at $x = 6$ or at an endpoint.

$$\begin{aligned} f(0) &= f(8) + \int_8^0 f'(x) dx \\ &= f(8) - \int_0^8 f'(x) dx = 4 - 12 = -8 \\ f(6) &= f(8) + \int_8^6 f'(x) dx \\ &= f(8) - \int_6^8 f'(x) dx = 4 - 7 = -3 \end{aligned}$$

$$f(8) = 4$$

The absolute minimum value of f on the closed interval $[0, 8]$ is -8.

Part C

1 point is earned for the answer

1 point is earned for the explanation

8-3b

The graph of f is concave down and increasing on $0 < x < 1$ and $3 < x < 4$, because f' is decreasing and positive on these intervals.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the explanation

The graph of f is concave down and increasing on $0 < x < 1$ and $3 < x < 4$, because f' is decreasing and positive on these intervals.

Part D

2 points are earned for $g'(x)$

1 point is earned for the answer

$$g'(x) = 3[f(x)]^2 \cdot f'(x)$$

$$g'(3) = 3[f(3)]^2 \cdot f'(3) = 3\left(-\frac{5}{2}\right)^2 \cdot 4 = 75$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for $g'(x)$

1 point is earned for the answer

$$g'(x) = 3[f(x)]^2 \cdot f'(x)$$

$$g'(3) = 3[f(3)]^2 \cdot f'(3) = 3\left(-\frac{5}{2}\right)^2 \cdot 4 = 75$$

8-3b

73.

Distance x (cm)	0	1	5	6	8
Temperature $T(x)$ ($^{\circ}\text{C}$)	100	93	70	62	55

A metal wire of length 8 centimeters (cm) is heated at one end. The table above gives selected values of the temperature $T(x)$, in degrees Celsius ($^{\circ}\text{C}$), of the wire x cm from the heated end. The function T is decreasing and twice differentiable.

- (a) Estimate $T'(7)$. Show the work that leads to your answer. Indicate units of measure.
- (b) Write an integral expression in terms of $T(x)$ for the average temperature of the wire. Estimate the average temperature of the wire using a trapezoidal sum with the four subintervals indicated by the data in the table. Indicate units of measure.
- (c) Find $\int_0^8 T'(x)dx$, and indicate units of measure. Explain the meaning of $\int_0^8 T'(x)dx$ in terms of the temperature of the wire.
- (d) Are the data in the table consistent with the assertion that $T''(x) > 0$ for every x in the interval $0 < x < 8$? Explain your answer.

Part A

1 point is earned for the answer

$$\frac{T(8) - T(6)}{8 - 6} = \frac{55 - 62}{2} = -\frac{7}{2}^{\circ}\text{C/cm}$$

1 point is earned for: units in (a), (b), and (c)

Units of $^{\circ}\text{C/cm}$ in (a), and $^{\circ}\text{C}$ in (b) and (c)



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

$$\frac{T(8) - T(6)}{8 - 6} = \frac{55 - 62}{2} = -\frac{7}{2}^{\circ}\text{C/cm}$$

1 point is earned for: units in (a), (b), and (c)

Units of $^{\circ}\text{C/cm}$ in (a), and $^{\circ}\text{C}$ in (b) and (c)

Part B

8-3b

1 point is earned for: $\frac{1}{8} \int_0^8 T(x) dx$

1 point is earned for the trapezoidal sum

1 point is earned for the answer

$$\frac{1}{8} \int_0^8 T(x) dx$$

Trapezoidal approximation for $\int_0^8 T(x) dx$:

$$A = \frac{100 + 93}{2} \cdot 1 + \frac{93 + 70}{2} \cdot 4 + \frac{70 + 62}{2} \cdot 1 + \frac{62 + 55}{2} \cdot 2$$

$$\text{Average temperature} \approx \frac{1}{8} A = 75.6875^\circ \text{C}$$

1 point is earned for: units in (a), (b), and (c)

Units of $^\circ\text{C}/\text{cm}$ in (a), and $^\circ\text{C}$ in (b) and (c)



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for: $\frac{1}{8} \int_0^8 T(x) dx$

1 point is earned for the trapezoidal sum

1 point is earned for the answer

$$\frac{1}{8} \int_0^8 T(x) dx$$

Trapezoidal approximation for $\int_0^8 T(x) dx$:

$$A = \frac{100 + 93}{2} \cdot 1 + \frac{93 + 70}{2} \cdot 4 + \frac{70 + 62}{2} \cdot 1 + \frac{62 + 55}{2} \cdot 2$$

8-3b

$$\text{Average temperature} \approx \frac{1}{8}A = 75.6875^{\circ}\text{C}$$

1 point is earned for: units in (a), (b), and (c)

Units of $^{\circ}\text{C}/\text{cm}$ in (a), and $^{\circ}\text{C}$ in (b) and (c)

Part C

1 point is earned for the value

1 point is earned for the meaning

$$\int_0^8 T'(x)dx = T(8) - T(0) = 55 - 100 = -45^{\circ}\text{C}$$

The temperature drops 45°C from the heated end of the wire to the other end of the wire.

1 point is earned for: units in (a), (b), and (c)

Units of $^{\circ}\text{C}/\text{cm}$ in (a), and $^{\circ}\text{C}$ in (b) and (c)



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the value

1 point is earned for the meaning

$$\int_0^8 T'(x)dx = T(8) - T(0) = 55 - 100 = -45^{\circ}\text{C}$$

The temperature drops 45°C from the heated end of the wire to the other end of the wire.

1 point is earned for: units in (a), (b), and (c)

Units of $^{\circ}\text{C}/\text{cm}$ in (a), and $^{\circ}\text{C}$ in (b) and (c)

Part D

1 point is earned for the two slopes of secant lines

8-3b

1 point is earned for the answer with explanation

Average rate of change of temperature on $[1, 5]$ is $\frac{70 - 93}{5 - 1} = -5.75$.

Average rate of change of temperature on $[5, 6]$ is $\frac{62 - 70}{6 - 5} = -8$.

No. By the MVT, $T'(c_1) = -5.75$ for some c_1 in the interval $(1, 5)$ and $T'(c_2) = -8$ for some c_2 in the interval $(5, 6)$. It

follows that T' must decrease somewhere in the interval (c_1, c_2) . Therefore T'' is not positive for every x in $[0, 8]$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the two slopes of secant lines

1 point is earned for the answer with explanation

Average rate of change of temperature on $[1, 5]$ is $\frac{70 - 93}{5 - 1} = -5.75$.

Average rate of change of temperature on $[5, 6]$ is $\frac{62 - 70}{6 - 5} = -8$.

No. By the MVT, $T'(c_1) = -5.75$ for some c_1 in the interval $(1, 5)$ and $T'(c_2) = -8$ for some c_2 in the interval $(5, 6)$. It

follows that T' must decrease somewhere in the interval (c_1, c_2) . Therefore T'' is not positive for every x in $[0, 8]$.

8-3b

74.  A cylindrical can of radius 10 millimeters is used to measure rainfall in Stormville. The can is initially empty, and rain enters the can during a 60-day period. The height of water in the can is modeled by the function S , where $S(t)$ is measured in millimeters and t is measured in days for $0 \leq t \leq 60$. The rate at which the height of the water is rising in the can is given by $S'(t) = 2\sin(0.03t) + 1.5$.
- (a) According to the model, what is the height of the water in the can at the end of the 60-day period?
- (b) According to the model, what is the average rate of change in the height of water in the can over the 60-day period? Show the computations that lead to your answer. Indicate units of measure.
- (c) Assuming no evaporation occurs, at what rate is the volume of water in the can changing at time $t = 7$? Indicate units of measure.
- (d) During the same 60-day period, rain on Monsoon Mountain accumulates in a can identical to the one in Stormville. The height of the water in the can on Monsoon Mountain is modeled by the function M , where $M(t) = \frac{1}{400}(3t^3 - 30t^2 + 330t)$. The height $M(t)$ is measured in millimeters, and t is measured in days for $0 \leq t \leq 60$. Let $D(t) = M(t) - S(t)$. Apply the Intermediate Value Theorem to the function D on the interval $0 \leq t \leq 60$ to justify that there exists a time t , $0 < t < 60$, at which the heights of water in the two cans are changing at the same rate.

Part A

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$S(60) = \int_0^{60} S'(t) dt = 171.813 \text{ mm}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$S(60) = \int_0^{60} S'(t) dt = 171.813 \text{ mm}$$

Part B

1 point is earned for the answer

1 point earned for units in (b) or (c)

8-3b

$$\frac{S(60) - S(0)}{60} = 2.863 \text{ or } 2.864 \text{ mm/day}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point earned for units in (b) or (c)

$$\frac{S(60) - S(0)}{60} = 2.863 \text{ or } 2.864 \text{ mm/day}$$

Part C

1 point is earned for relationship between V and S

1 point is earned for the answer

1 point earned for units in (b) or (c)

$$V(t) = 100\pi S(t)$$

$$V'(7) = 100\pi S'(7) = 602.218$$

The volume of water in the can is increasing at a rate of $602.218 \text{ mm}^3/\text{day}$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for relationship between V and S

1 point is earned for the answer

1 point earned for units in (b) or (c)

$$V(t) = 100\pi S(t)$$

$$V'(7) = 100\pi S'(7) = 602.218$$

The volume of water in the can is increasing at a rate of $602.218 \text{ mm}^3/\text{day}$.

8-3b

Part D

1 point is earned if considers $D(0)$ and $D(60)$

1 point is earned for justification

$$D(0) = -0.675 < 0 \text{ and } D(60) = 69.37730 > 0$$

Because D is continuous, the Intermediate Value Theorem implies that there is a time t , $0 < t < 60$, at which $D(t) = 0$. At this time, the heights of water in the two cans are changing at the same rate.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned if considers $D(0)$ and $D(60)$

1 point is earned for justification

$$D(0) = -0.675 < 0 \text{ and } D(60) = 69.37730 > 0$$

Because D is continuous, the Intermediate Value Theorem implies that there is a time t , $0 < t < 60$, at which $D(t) = 0$. At this time, the heights of water in the two cans are changing at the same rate.

8-3b

75.

t (minutes)	0	2	5	7	11	12
$r'(t)$ (feet per minute)	5.7	4.0	2.0	1.2	0.6	0.5

The volume of a spherical hot air balloon expands as the air inside the balloon is heated. The radius of the balloon, in feet, is modeled by a twice-differentiable function r of time t , where t is measured in minutes. For $0 < t < 12$, the graph of r is concave down. The table above gives selected values of the rate of change, $r'(t)$, of the radius of the balloon over the time interval $0 \leq t \leq 12$. The radius of the balloon is 30 feet when $t = 5$.

(Note: The volume of a sphere of radius r is given by $V = \frac{4}{3}\pi r^3$.)

- (a) Estimate the radius of the balloon when $t = 5.4$ using the tangent line approximation at $t = 5$. Is your estimate greater than or less than the true value? Give a reason for your answer.
- (b) Find the rate of change of the volume of the balloon with respect to time when $t = 5$. Indicate units of measure.
- (c) Use a right Riemann sum with the five subintervals indicated by the data in the table to approximate $\int_0^{12} r'(t) dt$. Using correct units, explain the meaning of $\int_0^{12} r'(t) dt$ in terms of the radius of the balloon.
- (d) Is your approximation in part (c) greater than or less than $\int_0^{12} r'(t) dt$? Give a reason for your answer.

Part A

1 point is earned for the estimate

1 point is earned for the conclusion with reason

$$r(5.4) \approx r(5) + r'(5)\Delta t = 30 + 2(0.4) = 30.8 \text{ ft}$$

Since the graph of r is concave down on the interval $5 < t < 5.4$, this estimate is greater than $r(5.4)$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the estimate

1 point is earned for the conclusion with reason

$$r(5.4) \approx r(5) + r'(5)\Delta t = 30 + 2(0.4) = 30.8 \text{ ft}$$

Since the graph of r is concave down on the interval $5 < t < 5.4$, this estimate is greater than $r(5.4)$.

Part B

2 points are earned for : $\frac{dV}{dt}$

8-3b

1 point is earned for the answer

$$\frac{dV}{dt} = 3 \left(\frac{4}{3} \right) \pi r^2 \frac{dr}{dt}$$

$$\frac{dV}{dt} \Big|_{t=5} = 4\pi(30)^2 2 = 7200\pi \text{ ft}^3/\text{min}$$

1 point is earned for units in (b) and (c)

Units of ft^3/min in part (b) and ft in part (c)



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for : $\frac{dV}{dt}$

1 point is earned for the answer

$$\frac{dV}{dt} = 3 \left(\frac{4}{3} \right) \pi r^2 \frac{dr}{dt}$$

$$\frac{dV}{dt} \Big|_{t=5} = 4\pi(30)^2 2 = 7200\pi \text{ ft}^3/\text{min}$$

1 point is earned for units in (b) and (c)

Units of ft^3/min in part (b) and ft in part (c)

Part C

1 point is earned for the approximation

1 point is earned for the explanation

$$\int_0^{12} r'(t) dt \approx 2(4.0) + 3(2.0) + 2(1.2) + 4(0.6) + 1(0.5)$$

$$= 19.3 \text{ ft}$$

$$\int_0^{12} r'(t) dt \text{ is the change in the radius, in feet, from}$$

$t = 0$ to $t = 12$ minutes.

8-3b

1 point is earned for units in (b) and (c)

Units of ft^3/min in part (b) and ft in part (c)



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the approximation

1 point is earned for the explanation

$$\int_0^{12} r'(t) dt \approx 2(4.0) + 3(2.0) + 2(1.2) + 4(0.6) + 1(0.5) \\ = 19.3 \text{ ft}$$

$$\int_0^{12} r'(t) dt \text{ is the change in the radius, in feet, from}$$

$t = 0$ to $t = 12$ minutes.

1 point is earned for units in (b) and (c)

Units of ft^3/min in part (b) and ft in part (c)

Part D

1 point is earned for the conclusion with reason

Since r is concave down, r' is decreasing on $0 < t < 12$.

Therefore, this approximation, 19.3 ft, is less than

$$\int_0^{12} r'(t) dt.$$



0	1
---	---

The student response earns all of the following points:

8-3b

1 point is earned for the conclusion with reason

Since r is concave down, r' is decreasing on $0 < t < 12$.

Therefore, this approximation, 19.3 ft, is less than

$$\int_0^{12} r'(t) dt.$$

76.  Honey is poured through a funnel at a rate of $r(t) = 4e^{-0.35t}$ ounces per minute, where t is measured in minutes. How many ounces of honey are poured through the funnel from time $t = 0$ to time $t = 3$?

(A) 0.910

(B) 1.400

(C) 2.600

(D) 7.429

**Answer D**

$$\int_0^3 r(t) dt$$

8-3b

77. Mighty Cable Company manufactures cable that sells for \$120 per meter. For a cable of fixed length, the cost of producing a portion of the cable varies with its distance from the beginning of the cable. Mighty reports that the cost to produce a portion of a cable that is x meters from the beginning of the cable is $6\sqrt{x}$ dollars per meter. (Note: Profit is defined to be the difference between the amount of money received by the company for selling the cable and the company's cost of producing the cable.)
- Find Mighty's profit on the sale of a 25-meter cable.
 - Using correct units, explain the meaning of $\int_{25}^{30} 6\sqrt{x} \, dx$ in the context of this problem.
 - Write an expression, involving an integral, that represents Mighty's profit on the sale of a cable that is k meters long.
 - Find the maximum profit that Mighty could earn on the sale of one cable. Justify your answer.

Part A

1 point is earned for the integral

1 point is earned for the answer

$$\text{Profit} = 120 \cdot 25 - \int_0^{25} 6\sqrt{x} \, dx = 2500 \text{ dollars}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\text{Profit} = 120 \cdot 25 - \int_0^{25} 6\sqrt{x} \, dx = 2500 \text{ dollars}$$

Part B

1 point is earned for the answer with units

$\int_{25}^{30} 6\sqrt{x} \, dx$ is the difference in cost, in dollars, of producing a cable of length 30 meters and a cable of length 25 meters.



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer with units

8-3b

$\int_{25}^{30} 6\sqrt{x} \, dx$ is the difference in cost, in dollars, of producing a cable of length 30 meters and a cable of length 25 meters.

Part C

1 point is earned for the integral

1 point is earned for the expression

$$\text{Profit} = 120k - \int_0^k 6\sqrt{x} \, dx \text{ dollars}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the expression

$$\text{Profit} = 120k - \int_0^k 6\sqrt{x} \, dx \text{ dollars}$$

Part D

1 point is earned for $P'(k) = 0$

1 point is earned for $k = 400$

1 point is earned for answer

1 point is earned for justification

Let $P(k)$ be the profit for a cable of length k .

$$P'(k) = 120 - 6\sqrt{k} = 0 \text{ when } k = 400.$$

This is the only critical point for P , and P' changes from positive to negative at $k = 400$.

Therefore, the maximum profit is $P(400) = 16,000$ dollars.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

8-3b

1 point is earned for $P'(k) = 0$

1 point is earned for $k = 400$

1 point is earned for answer

1 point is earned for justification

Let $P(k)$ be the profit for a cable of length k .

$$P'(k) = 120 - 6\sqrt{k} = 0 \text{ when } k = 400.$$

This is the only critical point for P , and P' changes from positive to negative at $k = 400$.

Therefore, the maximum profit is $P(400) = 16,000$ dollars.

8-3b

78. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Fish enter a lake at a rate modeled by the function E given by $E(t) = 20 + 15 \sin\left(\frac{\pi t}{6}\right)$. Fish leave the lake at a rate modeled by the function L given by $L(t) = 4 + 2^{0.1t^2}$. Both $E(t)$ and $L(t)$ are measured in fish per hour, and t is measured in hours since midnight ($t = 0$).

- How many fish enter the lake over the 5-hour period from midnight ($t = 0$) to 5 A.M. ($t = 5$)? Give your answer to the nearest whole number.
- What is the average number of fish that leave the lake per hour over the 5-hour period from midnight ($t = 0$) to 5 A.M. ($t = 5$)?
- At what time t , for $0 \leq t \leq 8$, is the greatest number of fish in the lake? Justify your answer.
- Is the rate of change in the number of fish in the lake increasing or decreasing at 5 A.M. ($t = 5$)? Explain your reasoning.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point requires a correct integrand and limits.

The second point requires the first point to be earned. The second point is earned with an answer of 153, 154, or any noninteger answer that is accurate to the number of places presented after the decimal point (up to the first 3). An answer between 101 and 102, inclusive, might indicate that the response uses degree mode rather than radian mode. In this case, the second point is not earned, but the remaining points for the question may be earned for consistent answers using degree mode.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

8-3b

- ☐ integral
- ☐ answer

Solution:

$$\int_0^5 E(t) dt = 153.457690$$

To the nearest whole number, 153 fish enter the lake from midnight to 5 A.M.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point requires a correct integrand and limits. The $\frac{1}{5}$ is considered for the second point. If this integral is computed in Part A of the question, that number can be used in place of the integral to earn the first point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ integral
- ☐ answer

Solution:

$$\frac{1}{5-0} \int_0^5 L(t) dt = 6.059038$$

The average number of fish that leave the lake per hour from midnight to 5 A.M. is 6.059 fish per hour.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned through analytical, graphical, or verbal expressions to show where $E(t) - L(t) = 0$, where $E(t) - L(t)$ changes sign, or where $E(t)$ and $L(t)$ intersect.

The second point is earned with the correct value of t .

The third point requires a global argument for the sign of $E(t) - L(t)$ rather than a local argument.

Another approach uses the Candidates Test. Let $A(t)$ be the change in the number of fish in the lake from midnight to t hours after midnight.

$$A(t) = \int_0^t (E(s) - L(s)) ds$$

8-3b

$$A'(t) = E(t) - L(t) = 0 \Rightarrow t = C = 6.20356$$

t	$A(t)$
0	0
C	135.01492
8	80.91908

Therefore the greatest number of fish in the lake is at time $t = 6.204$ (or 6.203).



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ sets $E(t) - L(t) = 0$
- ☐ answer
- ☐ justification

Solution:

The rate of change in the number of fish in the lake at time t is given by $E(t) - L(t)$.

$$E(t) - L(t) = 0 \Rightarrow t = 6.20356$$

$E(t) - L(t) > 0$ for $0 \leq t < 6.20356$, and $E(t) - L(t) < 0$ for $6.20356 < t \leq 8$. Therefore the greatest number of fish in the lake is at time $t = 6.204$ (or 6.203).

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point requires an attempt to use $E'(5)$ and $L'(5)$. Incorrect symbolic differentiation for both evaluated at 5 earns the point. If values are reported without symbolic work, at least one must be correct to earn the point. These values need not be reported to three places after the decimal point.

The second point requires a direct connection between $E'(5)$ and $L'(5)$. The point is not earned if reported values are incorrect. The numerical value of $E'(5) - L'(5)$ is not required, but the response must indicate that this value is negative.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ considers $E'(5)$ and $L'(5)$

8-3b☐ answer with explanation**Solution:**

$$E'(5) - L'(5) = -10.7228 < 0$$

Because $E'(5) - L'(5) < 0$, the rate of change in the number of fish is decreasing at time $t = 5$.

8-3b

79. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

From 5 A.M. to 10 A.M., the rate at which vehicles arrive at a certain toll plaza is given by

$A(t) = 450\sqrt{\sin(0.62t)}$, where t is the number of hours after 5 A.M. and $A(t)$ is measured in vehicles per hour. Traffic is flowing smoothly at 5 A.M. with no vehicles waiting in line.

(a) Write, but do not evaluate, an integral expression that gives the total number of vehicles that arrive at the toll plaza from 6 A.M. ($t = 1$) to 10 A.M. ($t = 5$).

(b) Find the average value of the rate, in vehicles per hour, at which vehicles arrive at the toll plaza from 6 A.M. ($t = 1$) to 10 A.M. ($t = 5$).

(c) Is the rate at which vehicles arrive at the toll plaza at 6 A.M. ($t = 1$) increasing or decreasing? Give a reason for your answer.

(d) A line forms whenever $A(t) \geq 400$. The number of vehicles in line at time t , for $a \leq t \leq 4$, is given by

$N(t) = \int_a^t (A(x) - 400) dx$, where a is the time when a line first begins to form. To the nearest whole number, find the greatest number of vehicles in line at the toll plaza in the time interval $a \leq t \leq 4$. Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response must be a definite integral with correct lower and upper limits to earn this point.

Because $|A(t)| = A(t)$ for $1 \leq t \leq 5$, a response of $\int_1^5 450\sqrt{\sin(0.62t)} dt$ or $\int_1^5 |A(t)| dt$ earns the point.

A response missing dt or using dx is eligible to earn the point.

A response with a copy error in the expression for $A(t)$ will earn the point only in the presence of $\int_1^5 A(t) dt$.

8-3b



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

The total number of vehicles that arrive at the toll plaza from 6 a.m. to 10 a.m. is given by $\int_1^5 A(t) dt$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The use of the average value formula, indicating that $a = 1$ and $b = 5$, can be presented in single or multiple steps to earn the first point. For example, the following response earns both points: $\int_1^5 A(t) dt = 1502.147865$, so the average value is 375.536966 .

A response that presents a correct integral along with the correct average value, but provides incorrect or incomplete communication, earns 1 out of 2 points. For example, the following response earns 1 out of 2 points:

$\int_1^5 A(t) dt = 1502.147865 = 375.536966$. so the average value is 375.536966 .

The answer must be correct to three decimal places. For example, $\frac{1}{5-1} \int_1^5 A(t) dt = 375.536966 \approx 376$ earns only the first point.

Degree mode: A response that presents answers obtained by using a calculator in degree mode does not earn the first point it would have otherwise earned. The response is generally eligible for all subsequent points (unless no answer is possible in degree mode or the question is made simpler by using degree mode). In degree mode, $\frac{1}{4} \int_1^5 A(t) dt = 79.416068$.

Special case: $\frac{1}{5} \int_1^5 A(t) dt = 300.429573$ earns 1 out of 2 points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Uses average value formula: $\frac{1}{b-a} \int_a^b A(t) dt$

☐ Answer

Solution:

$$\text{Average} = \frac{1}{5-1} \int_1^5 A(t) dt = 375.536966$$

8-3b

The average rate at which vehicles arrive at the toll plaza from 6 a.m. to 10 a.m. is 375.537 (or 375.536) vehicles per hour.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response need not present the value of $A'(1)$. The second line of the model solution earns both points.

An incorrect value assigned to $A'(1)$ earns the first point (but will not earn the second point).

Without a reference to $t = 1$, the first point is earned by any of the following:

· 148.947 accurate to the number of decimals presented, with zero up to three decimal places (i.e., 149, 148, 148.9, 148.95, or 148.94)

· $A'(t) = 148.947$ by itself

To be eligible for the second point, the first point must be earned.

To earn the second point, there must be a reference to $t = 1$.

Degree mode: $A'(1) = 23.404311$



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Considers $A'(1)$
- ☐ Answer with reason

Solution:

$$A'(1) = 148.947272$$

Because $A'(1) > 0$, the rate at which the vehicles arrive at the toll plaza is increasing.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

It is not necessary to indicate that $A(t) = 400$ to earn the first point, although this statement alone would earn the first point.

A response of “ $A(t) \geq 400$ when $1.469372 \leq t \leq 3.597713$ ” will earn the first 2 points. A response of “ $A(t) \geq 400$ ” along with the presence of exactly one of the two numbers above will earn the first point, but not the second. A response of “ $A(t) \geq 400$ ” by itself will not earn either of the first 2 points.

To earn the second point the values for a and b must be accurate to the number of decimals presented, with at least one and up to three decimal places. These may appear only in a candidates table, as limits of integration, or on a number line.

8-3b

A response with incorrect notation involving t or x is eligible to earn all 4 points.

A response that does not earn the first point is still eligible for the remaining 3 points.

To earn the third point, a response must present the greatest number of vehicles. This point is earned for answers of either 71 or 71.254*** only.

A correct justification earns the fourth point, even if the third point is not earned because of a decimal presentation error.

When using a Candidates Test, the response must include the values for $N(a)$, $N(b)$, and $N(4)$ to earn the fourth point. These values must be correct to the number of decimals presented, with up to three decimal places. (Correctly rounded integer values are acceptable.)

Alternate solution for the third and fourth points:

For $a \leq t \leq b$, $A(t) \geq 400$. For $b \leq t \leq 4$, $A(t) \leq 400$.

Thus, $N(t) = \int_a^t (A(x) - 400) dx$ is greatest at $t = b$.

$N(b) = 71.254129$, and the greatest number of vehicles in line is 71.

Degree mode: The response is only eligible to earn the first point because in degree mode $A(t) < 400$.



0	1	2	3	4
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The student response accurately includes **all four** of the criteria below.

- ☐ Considers $N'(t) = 0$
- ☐ $t = a$ and $t = b$
- ☐ Answer
- ☐ Justification

Solution:

$$N'(t) = A(t) - 400 = 0$$

$$\Rightarrow A(t) = 400 \Rightarrow t = 1.469372, t = 3.597713$$

$$a = 1.469372$$

$$b = 3.597713$$

t	$N(t) = \int_a^t (A(x) - 400) dx$
a	0
b	71.254129
4	62.338346

The greatest number of vehicles in line is 71.

8-3b

80.  Oil is leaking from a pipeline on the surface of a lake and forms an oil slick whose volume increases at a constant rate of 2000 cubic centimeters per minute. The oil slick takes the form of a right circular cylinder with both its radius and height changing with time. (Note: The volume V of a right circular cylinder with radius r and height h is given by $V = \pi r^2 h$.)
- (a) At the instant when the radius of the oil slick is 100 centimeters and the height is 0.5 centimeter, the radius is increasing at the rate of 2.5 centimeters per minute. At this instant, what is the rate of change of the height of the oil slick with respect to time, in centimeters per minute?
- (b) A recovery device arrives on the scene and begins removing oil. The rate at which oil is removed is $R(t) = 400\sqrt{t}$ cubic centimeters per minute, where t is the time in minutes since the device began working. Oil continues to leak at the rate of 2000 cubic centimeters per minute. Find the time t when the oil slick reaches its maximum volume. Justify your answer.
- (c) By the time the recovery device began removing oil, 60,000 cubic centimeters of oil had already leaked. Write, but do not evaluate, an expression involving an integral that gives the volume of oil at the time found in part (b).

Part A

1 point earned for $\frac{dV}{dt} = 2000$ and $\frac{dr}{dt} = 2.5$

When $r = 100$ cm and $h = 0.5$ cm, $\frac{dV}{dt} = 2000$ cm³/min and $\frac{dr}{dt} = 2.5$ cm/min.

2 point earned for the correct expression of $\frac{dV}{dt}$

$$\frac{dV}{dt} = 2\pi r \frac{dr}{dt} h + \pi r^2 \frac{dh}{dt}$$

1 point earned for the correct answer

$$2000 = 2\pi(100)(2.5)(0.5) + \pi(100)^2 \frac{dh}{dt}$$

$$\frac{dh}{dt} = 0.038 \text{ or } 0.039 \text{ cm/min}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point earned for $\frac{dV}{dt} = 2000$ and $\frac{dr}{dt} = 2.5$

When $r = 100$ cm and $h = 0.5$ cm, $\frac{dV}{dt} = 2000$ cm³/min and $\frac{dr}{dt} = 2.5$ cm/min.

8-3b

2 point earned for the correct expression of $\frac{dV}{dt}$

$$\frac{dV}{dt} = 2\pi r \frac{dr}{dt} h + \pi r^2 \frac{dh}{dt}$$

1 point earned for the correct answer

$$2000 = 2\pi(100)(2.5)(0.5) + \pi(100)^2 \frac{dh}{dt}$$

$$\frac{dh}{dt} = 0.038 \text{ or } 0.039 \text{ cm/min}$$

Part B

1 point earned for $R(t) = 2000$

$$\frac{dV}{dt} = 2000 - R(t), \text{ so } \frac{dV}{dt} = 0 \text{ when } R(t) = 2000.$$

1 point earned for correct answer

This occurs when $t = 25$ minutes.

1 point earned for the justification

Since $\frac{dV}{dt} > 0$ for $0 < t < 25$ and $\frac{dV}{dt} < 0$ for $t > 25$, the oil slick reaches its maximum volume 25 minutes after the device begins working.



0	1	2	3
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The student response earns all of the following points:

1 point earned for $R(t) = 2000$

$$\frac{dV}{dt} = 2000 - R(t), \text{ so } \frac{dV}{dt} = 0 \text{ when } R(t) = 2000.$$

1 point earned for correct answer

This occurs when $t = 25$ minutes.

1 point earned for the justification

Since $\frac{dV}{dt} > 0$ for $0 < t < 25$ and $\frac{dV}{dt} < 0$ for $t > 25$, the oil slick reaches its maximum volume 25 minutes after the device begins working.

Part C

1 point earned for the correct limits and initial condition

8-3b

1 point earned for the integrand

The volume of oil, in cm^3 , in the slick at time $t = 25$ minutes is given by $60,000 + \int_0^{25} (2000 - R(t))dt$.



0	1	2
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The student response earns all of the following points:

1 point earned for the correct limits and initial condition

1 point earned for the integrand

The volume of oil, in cm^3 , in the slick at time $t = 25$ minutes is given by $60,000 + \int_0^{25} (2000 - R(t))dt$.

8-3b

81. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

A particle, P , is moving along the x -axis. The velocity of particle P at time t is given by $v_P(t) = \sin(t^{1.5})$ for $0 \leq t \leq \pi$. At time $t = 0$, particle P is at position $x = 5$.

A second particle, Q , also moves along the x -axis. The velocity of particle Q at time t is given by $v_Q(t) = (t - 1.8) \cdot 1.25^t$ for $0 \leq t \leq \pi$. At time $t = 0$, particle Q is at position $x = 10$.

- Find the positions of particles P and Q at time $t = 1$.
- Are particles P and Q moving toward each other or away from each other at time $t = 1$? Explain your reasoning.
- Find the acceleration of particle Q at time $t = 1$. Is the speed of particle Q increasing or decreasing at time $t = 1$? Explain your reasoning.
- Find the total distance traveled by particle P over the time interval $0 \leq t \leq \pi$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for the explicit presentation of at least one definite integral, either $\int_0^1 v_P(t) dt$ or $\int_0^1 v_Q(t) dt$.

The first point must be earned to be eligible for the second and third points.

The second point is earned for adding the initial condition to at least one of the definite integrals and finding the correct position.

Writing $\int_0^1 v_P(t) + 5 = 5.370660$ does not earn a position point, because the missing dt makes this statement unclear or false. However, $5 + \int_0^1 v_P(t) = 5.370660$ does earn the position point because it is not ambiguous. Similarly, for the position of Q .

Read unlabeled answers presented left to right, or top to bottom, as $x_P(1)$ and $x_Q(1)$, respectively.

8-3b

Special case 1: A response of $x_P(1) = 5 + \int_0^a v_P(t)dt = 5.370660$ AND $x_Q(1) = 10 + \int_0^a v_Q(t)dt = 8.564355$ for $a \neq 1$ earns one point.

Special case 2: A response of $x_P(1) = 5 + \int v_P(t)dt = 5.370660$ AND $x_Q(1) = 10 + \int v_Q(t)dt = 8.564355$ or the equivalent, never providing the definite integrals, earns one point.

Degree mode: A response that presents answers obtained by using a calculator in degree mode does not earn the first point it would have otherwise earned. The response is generally eligible for all subsequent points (unless no answer is possible in degree mode or the question is made simpler by using degree mode). In degree mode, $x_P(1)$ is 5.007 (or 5.006).



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ One definite integral
- ☐ One position
- ☐ The other position

Solution:

$$x_P(1) = 5 + \int_0^1 v_P(t)dt = 5.370660$$

At time $t = 1$, the position of particle P is $x = 5.371$ (or 5.370).

$$x_Q(1) = 10 + \int_0^1 v_Q(t)dt = 8.564355$$

At time $t = 1$, the position of particle Q is $x = 8.564$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for using the sign of $v_P(1)$ or $v_Q(1)$ to determine the direction of motion for one of the particles. This point cannot be earned without reference to the sign of $v_P(1)$ or $v_Q(1)$.

It is not necessary to present an explicit value for $v_P(1)$ or $v_Q(1)$, but if a value is presented, it must be correct as far as reported, up to three places after the decimal.

Read with imported incorrect position values from part (a).

If one or both position values were not found in part (a), but are found in part (b), the points for part (a) are not earned retroactively.

To earn the second point the explanation must be based on the signs of $v_P(1)$ and $v_Q(1)$ and the relative positions of particle P and particle Q at $t = 1$. References to other values of time, such as $t = 0$, are not sufficient.

8-3b

Degree mode: $v_P(1) = 0.017$. (See degree mode statement in part (a).)



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Direction of motion for one particle
- ☐ Answer with explanation

Solution:

$$v_P(1) = \sin(1^{1.5}) = 0.841471 > 0$$

At time $t = 1$, particle P is moving to the right.

$$v_Q(1) = (1 - 1.8) \cdot 1.25^1 = -1 < 0$$

At time $t = 1$, particle Q is moving to the left.

At time $t = 1$, $x_P(1) < x_Q(1)$, so particle P is to the left of particle Q .

Thus, at time $t = 1$, particles P and Q are moving toward each other.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point the acceleration must be explicitly connected to v'_Q (e.g., $v'_Q(1) = 1.026856$).

The first point is not earned for an unsupported value of 1.027 (or 1.026). The setup, $v'_Q(1)$, must be shown. Presenting only $a_Q(1) = 1.027$ (or 1.026) without indication that $v'_Q = a_Q$ is not enough to earn the first point.

A response does not need to present a value for $v_Q(1)$; the sign is sufficient.

To earn the second point a response must compare the signs of a_Q and v_Q at $t = 1$. Considering only one sign is not sufficient.

After the first point has been earned, a response declaring only “velocity and acceleration are of opposite signs at $t = 1$ so the particle is slowing down” (or equivalent) earns the second point.

The second point may be earned without the first, as long as the response does not present an incorrect value or sign for $v_Q(1)$ and concludes the particle is slowing down because velocity and acceleration have opposite signs at $t = 1$.



0	1	2
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The student response accurately includes **both** of the criteria below.

8-3b

- ☐ Setup and acceleration
- ☐ Speed decreasing with reason

Solution:

$$a_Q(1) = v'_Q(1) = 1.026856$$

The acceleration of particle Q is 1.027 (or 1.026) at time $t = 1$.

$$v_Q(1) = -1 < 0 \text{ and } a_Q(1) > 0$$

The speed of particle Q is decreasing at time $t = 1$ because the velocity and acceleration have opposite signs.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for $\int_0^{\pi} |v_P(t)| dt$.

The first point can also be earned for a sum (or difference) of definite integrals, such as a

$$\int_0^{2.145029} v_P(t) dt - \int_{2.145029}^{\pi} v_P(t) dt, \text{ provided the response has indicated } v_P(2.145029) = 0.$$

The second point can only be earned for the correct answer.

The unsupported value 1.931 earns no points.

A response reporting the distance traveled by particle Q as $\int_0^{\pi} |v_Q(t)| dt = 3.506$ earns the first point and is not eligible for the second point.

In degree mode, the total distance traveled is 0.122. (See degree mode statement in part (a).) In the degree mode case, the response must present $\int_0^{\pi} |v_P(t)| dt$ in order to earn the first point because $\int_0^{\pi} |v_P(t)| dt = \int_0^{\pi} v_P(t) dt$.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Definite integral
- ☐ Answer

Solution:

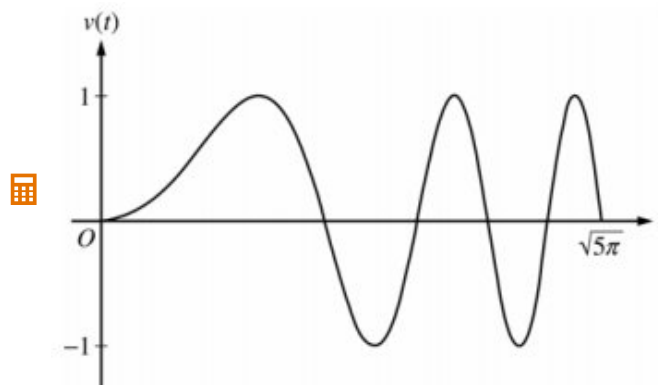
$$\int_0^{\pi} |v_P(t)| dt = 1.93148$$

8-3b

Over the time interval $0 \leq t \leq \pi$, the total distance traveled by particle P is 1.931.

8-3b

82.



A particle moves along the x -axis so that its velocity v at time $t \geq 0$ is given by $v(t) = \sin(t^2)$. The graph of v is shown above for $0 \leq t \leq \sqrt{5\pi}$. The position of the particle at time t is $x(t)$ and its position at time $t = 0$ is $x(0) = 5$.

- Find the acceleration of the particle at time $t = 3$.
- Find the total distance traveled by the particle from time $t = 0$ to $t = 3$.
- Find the position of the particle at time $t = 3$.
- For $0 \leq t \leq \sqrt{5\pi}$, find the time t at which the particle is farthest to the right. Explain your answer.

Part A

1 point is earned for $a(3)$

$$a(3) = v'(3) = 6 \cos 9 = -5.466 \text{ or } -5.467$$



0	1
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The student response earns all of the following points:

1 point is earned for $a(3)$

$$a(3) = v'(3) = 6 \cos 9 = -5.466 \text{ or } -5.467$$

Part B

1 point is earned for the setup

1 point is earned for the answer

$$\text{Distance} = \int_0^3 |v(t)| dt = 1.702$$

OR

For $0 < t < 3$, $v(t) = 0$ when $t = \sqrt{\pi} = 1.77245$ and $t = \sqrt{2\pi} = 2.50663$

8-3b

$$x(0) = 5$$

$$x(\sqrt{\pi}) = 5 + \int_0^{\sqrt{\pi}} v(t) dt = 5.89483$$

$$x(\sqrt{2\pi}) = 5 + \int_0^{\sqrt{2\pi}} v(t) dt = 5.43041$$

$$x(3) = 5 + \int_0^3 v(t) dt = 5.77356$$

$$|x(\sqrt{\pi}) - x(0)| + |x(\sqrt{2\pi}) - x(\sqrt{\pi})| + |x(3) - x(\sqrt{2\pi})| = 1.702$$



0	1	2
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The student response earns all of the following points:

1 point is earned for the setup

1 point is earned for the answer

$$\text{Distance} = \int_0^3 |v(t)| dt = 1.702$$

OR

For $0 < t < 3$, $v(t) = 0$ when $t = \sqrt{\pi} = 1.77245$ and $t = \sqrt{2\pi} = 2.50663$

$$x(0) = 5$$

$$x(\sqrt{\pi}) = 5 + \int_0^{\sqrt{\pi}} v(t) dt = 5.89483$$

$$x(\sqrt{2\pi}) = 5 + \int_0^{\sqrt{2\pi}} v(t) dt = 5.43041$$

$$x(3) = 5 + \int_0^3 v(t) dt = 5.77356$$

$$|x(\sqrt{\pi}) - x(0)| + |x(\sqrt{2\pi}) - x(\sqrt{\pi})| + |x(3) - x(\sqrt{2\pi})| = 1.702$$

Part C

1 point is earned for integrand

1 point is earned if uses $x(0) = 5$

8-3b

1 point is earned for the answer

$$x(3) = 5 + \int_0^3 v(t) dt = 5.773 \text{ or } 5.774$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for integrand

1 point is earned if uses $x(0) = 5$

1 point is earned for the answer

$$x(3) = 5 + \int_0^3 v(t) dt = 5.773 \text{ or } 5.774$$

Part D

1 point is earned if sets $v(t) = 0$

1 point is earned for the answer

1 point is earned for the reason

The particle's rightmost position occurs at time $t = \sqrt{\pi} = 1.772$.

The particle changes from moving right to moving left at those times t for which $v(t) = 0$ with $v(t)$ changing from positive to negative, namely at $t = \sqrt{\pi}, \sqrt{3\pi}, \sqrt{5\pi}$ ($t = 1.772, 3.070, 3.963$).

Using $x(T) = 5 + \int_0^T v(t) dt$, the particle's positions at the times it changes from rightward to leftward movement are:

$$\begin{array}{l} T : 0 \quad \sqrt{\pi} \quad \sqrt{3\pi} \quad \sqrt{5\pi} \\ x(T) : 5 \quad 5.895 \quad 5.788 \quad 5.752 \end{array}$$

The particle is farthest to the right when $T = \sqrt{\pi}$.



0	1	2	3
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8-3b

The student response earns all of the following points:

1 point is earned if sets $v(t) = 0$

1 point is earned for the answer

1 point is earned for the reason

The particle's rightmost position occurs at time $t = \sqrt{\pi} = 1.772$.

The particle changes from moving right to moving left at those times t for which $v(t) = 0$ with $v(t)$ changing from positive to negative, namely at $t = \sqrt{\pi}, \sqrt{3\pi}, \sqrt{5\pi}$ ($t = 1.772, 3.070, 3.963$).

Using $x(T) = 5 + \int_0^T v(t)dt$, the particle's positions at the times it changes from rightward to leftward movement are:

$$\begin{array}{l} T : 0 \quad \sqrt{\pi} \quad \sqrt{3\pi} \quad \sqrt{5\pi} \\ x(T) : 5 \quad 5.895 \quad 5.788 \quad 5.752 \end{array}$$

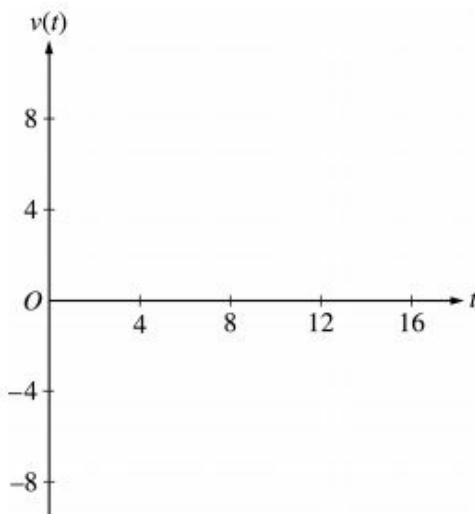
The particle is farthest to the right when $T = \sqrt{\pi}$.

8-3b

83. A particle moves along the x -axis so that its velocity v at any time t , for $0 \leq t \leq 16$, is given by $v(t) = e^{2 \sin t} - 1$. At time $t = 0$, the particle is at the origin.



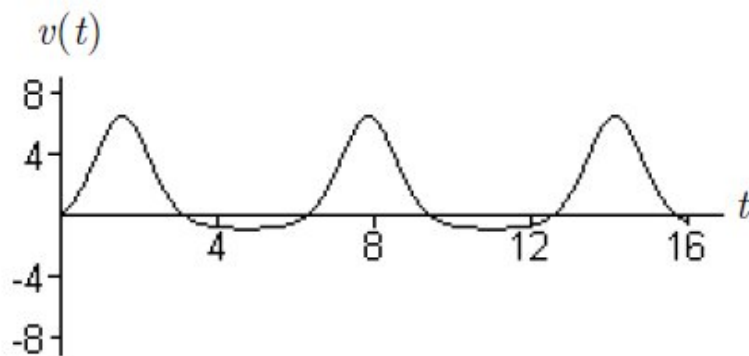
- (a) On the axes provided, sketch the graph of $v(t)$ for $0 \leq t \leq 16$.



- (b) During what intervals of time is the particle moving to the left? Give a reason for your answer.
 (c) Find the total distance traveled by the particle from $t = 0$ to $t = 4$.
 (d) Is there any time t , $0 < t \leq 16$, at which the particle returns to the origin? Justify your answer.

Part A

1 point is earned for the graph three "humps", periodic behavior, starts at origin, reasonable relative max and min values



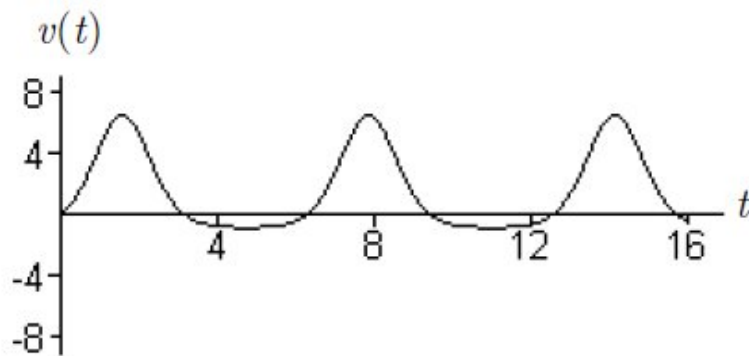
0

1

The student response earns all of the following points:

1 point is earned for the graph three "humps", periodic behavior, starts at origin, reasonable relative max and min values

8-3b

**Part B**

2 points are earned for the intervals

<-1> each missing or incorrect interval

1 point is earned for the reason

Particle is moving to the left when

$$v(t) < 0, \text{ i.e. } e^{2\sin t} < 1.$$

$$(\pi, 2\pi), (3\pi, 4\pi) \text{ and } (5\pi, 16]$$



0	1	2	3
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The student response earns all of the following points:

2 points are earned for the intervals

<-1> each missing or incorrect interval

1 point is earned for the reason

Particle is moving to the left when

$$v(t) < 0, \text{ i.e. } e^{2\sin t} < 1.$$

$$(\pi, 2\pi), (3\pi, 4\pi) \text{ and } (5\pi, 16]$$

Part C

1 point is earned for the limits of 0 and 4 on an integral of $v(t)$ or $|v(t)|$

or

8-3b

uses $x(0)$ and $x(4)$ to compute distance

1 point is earned for the handles change of direction at student's turning point

1 point is earned for the answer

$$\int_0^4 |v(t)| dt = 10.542$$

or

$$v(t) = e^{2 \sin t} - 1 = 0$$

$$t = 0 \text{ or } t = \pi$$

$$x(\pi) = \int_0^{\pi} v(t) dt = 10.10656$$

$$x(4) = \int_0^4 v(t) dt = 9.67066$$

$$|x(\pi) - x(0)| + |x(4) - x(\pi)| \\ = 10.542$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the limits of 0 and 4 on an integral of $v(t)$ or $|v(t)|$

or

uses $x(0)$ and $x(4)$ to compute distance

1 point is earned for the handles change of direction at student's turning point

1 point is earned for the answer

$$\int_0^4 |v(t)| dt = 10.542$$

or

8-3b

$$v(t) = e^{2 \sin t} - 1 = 0$$

$$t = 0 \text{ or } t = \pi$$

$$x(\pi) = \int_0^{\pi} v(t) dt = 10.10656$$

$$x(4) = \int_0^4 v(t) dt = 9.67066$$

$$|x(\pi) - x(0)| + |x(4) - x(\pi)| \\ = 10.542$$

Part D

1 point is earned for no such time

1 point is earned for the reason

There is no such time because $\int_0^T v(t) dt > 0$ for all $T > 0$.



0	1	2
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The student response earns all of the following points:

1 point is earned for no such time

1 point is earned for the reason

There is no such time because $\int_0^T v(t) dt > 0$ for all $T > 0$.

8-3b

84.  On a certain workday, the rate, in tons per hour, at which unprocessed gravel arrives at a gravel processing plant is modeled by $G(t) = 90 + 45\cos\left(\frac{t^2}{18}\right)$, where t is measured in hours and $0 \leq t \leq 8$. At the beginning of the workday ($t = 0$), the plant has 500 tons of unprocessed gravel. During the hours of operation, $0 \leq t \leq 8$, the plant processes gravel at a constant rate of 100 tons per hour.
- (a) Find $G'(5)$. Using correct units, interpret your answer in the context of the problem.
- (b) Find the total amount of unprocessed gravel that arrives at the plant during the hours of operation on this workday.
- (c) Is the amount of unprocessed gravel at the plant increasing or decreasing at time $t = 5$ hours? Show the work that leads to your answer.
- (d) What is the maximum amount of unprocessed gravel at the plant during the hours of operation on this workday? Justify your answer.

Part A

1 point is earned for $G'(5)$

1 point is earned for the interpretation with units

$$G'(5) = -24.588 \text{ (or } -24.587)$$

The rate at which gravel is arriving is decreasing by 24.588 (or 24.587) tons per hour per hour at time $t = 5$ hours.



0	1	2
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The student response earns all of the following points:

1 point is earned for $G'(5)$

1 point is earned for the interpretation with units

$$G'(5) = -24.588 \text{ (or } -24.587)$$

The rate at which gravel is arriving is decreasing by 24.588 (or 24.587) tons per hour per hour at time $t = 5$ hours.

Part B

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^8 G(t) dt = 825.551 \text{ tons}$$



0	1	2
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8-3b

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^8 G(t) dt = 825.551 \text{ tons}$$

Part C

1 point is earned for comparing $G(5)$ to 100

1 point is earned for the conclusion

$$G(5) = 98.140764 < 100$$

At time $t = 5$, the rate at which unprocessed gravel is arriving is less than the rate at which it is being processed.

Therefore, the amount of unprocessed gravel at the plant is decreasing at time $t = 5$.



0	1	2
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The student response earns all of the following points:

1 point is earned for comparing $G(5)$ to 100

1 point is earned for the conclusion

$$G(5) = 98.140764 < 100$$

At time $t = 5$, the rate at which unprocessed gravel is arriving is less than the rate at which it is being processed.

Therefore, the amount of unprocessed gravel at the plant is decreasing at time $t = 5$.

Part D

1 point is earned for considering $A'(t) = 0$

1 point is earned for the answer

1 point is earned for the justification

The amount of unprocessed gravel at time t is given by $A(t) = 500 + \int_0^t (G(s) - 100)ds$.

$$A'(t) = G(t) - 100 = 0 \Rightarrow t = 4.923480$$

8-3b

t	$A(t)$
0	500
4.92348	635.376123
8	525.551089

The maximum amount of unprocessed gravel at the plant during this workday is 635.376 tons.



0	1	2	3
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The student response earns all of the following points:

1 point is earned for considering $A'(t) = 0$

1 point is earned for the answer

1 point is earned for the justification

The amount of unprocessed gravel at time t is given by $A(t) = 500 + \int_0^t (G(s) - 100)ds$.

$$A'(t) = G(t) - 100 = 0 \Rightarrow t = 4.923480$$

t	$A(t)$
0	500
4.92348	635.376123
8	525.551089

The maximum amount of unprocessed gravel at the plant during this workday is 635.376 tons.

8-3b

85.  A 12,000-liter tank of water is filled to capacity. At time $t = 0$, water begins to drain out of the tank at a rate modeled by $r(t)$, measured in liters per hour, where r is given by the piecewise-defined function

$$r(t) = \begin{cases} \frac{600t}{t+3} & \text{for } 0 \leq t \leq 5 \\ 1000e^{-0.2t} & \text{for } t > 5 \end{cases}$$

- (a) Is r continuous at $t = 5$? Show the work that leads to your answer.
- (b) Find the average rate at which water is draining from the tank between time $t = 0$ and time $t = 8$ hours.
- (c) Find $r'(3)$. Using correct units, explain the meaning of that value in the context of this problem.
- (d) Write, but do not solve, an equation involving an integral to find the time A when the amount of water in the tank is 9000 liters.

Part A

2 points are earned for conclusion with analysis

$$\lim_{t \rightarrow 5^-} r(t) = \lim_{t \rightarrow 5^-} \left(\frac{600t}{t+3} \right) = 375 = r(5)$$

$$\lim_{t \rightarrow 5^+} r(t) = \lim_{t \rightarrow 5^+} (1000e^{-0.2t}) = 367.879$$

Because the left-hand and right-hand limits are not equal, r is not continuous at $t = 5$.



0	1	2
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The student response earns all of the following points:

2 points are earned for conclusion with analysis

$$\lim_{t \rightarrow 5^-} r(t) = \lim_{t \rightarrow 5^-} \left(\frac{600t}{t+3} \right) = 375 = r(5)$$

$$\lim_{t \rightarrow 5^+} r(t) = \lim_{t \rightarrow 5^+} (1000e^{-0.2t}) = 367.879$$

Because the left-hand and right-hand limits are not equal, r is not continuous at $t = 5$.

Part B

1 point is earned for the integrand

1 point is earned for the limits and constant

1 point is earned for the answer

$$\frac{1}{8} \int_0^8 r(t) dt = \frac{1}{8} \left(\int_0^5 \frac{600t}{t+3} dt + \int_5^8 1000e^{-0.2t} dt \right)$$

8-3b

$$= 258.052 \text{ or } 258.053$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits and constant

1 point is earned for the answer

$$\frac{1}{8} \int_0^8 r(t) dt = \frac{1}{8} \left(\int_0^5 \frac{600t}{t+3} dt + \int_5^8 1000e^{-0.2t} dt \right)$$

$$= 258.052 \text{ or } 258.053$$

Part C

1 point is earned for $r'(3)$

1 point is earned for the meaning of $r'(3)$

$$r'(3) = 50$$

The rate at which water is draining out of the tank at time $t = 3$ hours is increasing at 50 liters/hour².



0	1	2
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The student response earns all of the following points:

1 point is earned for $r'(3)$

1 point is earned for the meaning of $r'(3)$

$$r'(3) = 50$$

The rate at which water is draining out of the tank at time $t = 3$ hours is increasing at 50 liters/hour².

Part D

1 point is earned for the integral

8-3b

1 point is earned for the equation

$$12,000 - \int_0^A r(t) dt = 9000$$



0	1	2
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The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the equation

$$12,000 - \int_0^A r(t) dt = 9000$$

8-3b

86.  A storm washed away sand from a beach, causing the edge of the water to get closer to a nearby road. The rate at which the distance between the road and the edge of the water was changing during the storm is modeled by $f(t) = \sqrt{t} + \cos t - 3$ meters per hour, t hours after the storm began. The edge of the water was 35 meters from the road when the storm began, and the storm lasted 5 hours. The derivative of $f(t)$ is $f'(t) = \frac{1}{2\sqrt{t}} - \sin t$.
- (a) What was the distance between the road and the edge of the water at the end of the storm?
- (b) Using correct units, interpret the value $f'(4) = 1.007$ in terms of the distance between the road and the edge of the water.
- (c) At what time during the 5 hours of the storm was the distance between the road and the edge of the water decreasing most rapidly? Justify your answer.
- (d) After the storm, a machine pumped sand back onto the beach so that the distance between the road and the edge of the water was growing at a rate of $g(p)$ meters per day, where p is the number of days since pumping began. Write an equation involving an integral expression whose solution would give the number of days that sand must be pumped to restore the original distance between the road and the edge of the water.

Part A

1 point is earned for the integral

1 point is earned for the answer

$$35 + \int_0^5 f(t) dt = 26.494 \text{ or } 26.495 \text{ meters}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$35 + \int_0^5 f(t) dt = 26.494 \text{ or } 26.495 \text{ meters}$$

Part B

1 point is earned for interpretation of $f'(4)$

1 point is earned for units

Four hours after the storm began, the rate of change of the distance between the road and the edge of the water is increasing at a rate of 1.007 meters/hours².

8-3b



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for interpretation of $f'(4)$

1 point is earned for units

Four hours after the storm began, the rate of change of the distance between the road and the edge of the water is increasing at a rate of 1.007 meters/hours².

Part C

1 point is earned if considers $f'(t) = 0$

1 point is earned for the answer

1 point is earned for justification

$f'(t) = 0$ when $t = 0.66187$ and $t = 2.84038$

The minimum of f for $0 \leq t \leq 5$ may occur at 0, 0.66187, 2.84038, or 5.

$f(0) = -2$

$f(0.66187) = -1.39760$

$f(2.84038) = -2.26963$

$f(5) = -0.48027$

The distance between the road and the edge of the water was decreasing most rapidly at time $t = 2.840$ hours after the storm began.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if considers $f'(t) = 0$

1 point is earned for the answer

1 point is earned for justification

$f'(t) = 0$ when $t = 0.66187$ and $t = 2.84038$

8-3b

The minimum of f for $0 \leq t \leq 5$ may occur at 0, 0.66187, 2.84038, or 5.

$$f(0) = -2$$

$$f(0.66187) = -1.39760$$

$$f(2.84038) = -2.26963$$

$$f(5) = -0.48027$$

The distance between the road and the edge of the water was decreasing most rapidly at time $t = 2.840$ hours after the storm began.

Part D

1 point is earned for the integral of g

1 point is earned for the answer

$$-\int_0^5 f(t)dt = \int_0^x g(p)dp$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral of g

1 point is earned for the answer

$$-\int_0^5 f(t)dt = \int_0^x g(p)dp$$

8-3b

87.  The tide removes sand from Sandy Point Beach at a rate modeled by the function R , given by

$$R(t) = 2 + 5 \sin\left(\frac{4\pi t}{25}\right).$$

A pumping station adds sand to the beach at a rate modeled by the function S , given by

$$S(t) = \frac{15t}{1+3t}.$$

Both $R(t)$ and $S(t)$ have units of cubic yards per hour and t is measured in hours for $0 \leq t \leq 6$. At time $t = 0$, the beach contains 2500 cubic yards of sand.

- How much sand will the tide remove from the beach during this 6-hour period? Indicate units of measure.
- Write an expression for $Y(t)$, the total number of cubic yards of sand on the beach at time t .
- Find the rate at which the total amount of sand on the beach is changing at time $t = 4$.
- For $0 \leq t \leq 6$, at what time t is the amount of sand on the beach a minimum? What is the minimum value? Justify your answers.

Part A

1 point is earned for integral

1 point is earned for the answer with units

$$\int_0^6 R(t) dt = 31.815 \text{ or } 31.816 \text{ yd}^3$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer with units

$$\int_0^6 R(t) dt = 31.815 \text{ or } 31.816 \text{ yd}^3$$

Part B

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$Y(t) = 2500 + \int_0^t (S(x) - R(x)) dx$$

8-3b



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$Y(t) = 2500 + \int_0^t (S(x) - R(x))dx$$

Part C

1 point is earned for the answer

$$Y'(t) = S(t) - R(t)$$

$$Y'(4) = S(4) - R(4) = -1.908 \text{ or } -1.909 \text{ yd}^3/\text{hr}$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

$$Y'(t) = S(t) - R(t)$$

$$Y'(4) = S(4) - R(4) = -1.908 \text{ or } -1.909 \text{ yd}^3/\text{hr}$$

Part D

1 point is earned for setting $Y'(t) = 0$

1 point is earned for the critical t -value

1 point is earned for the answer with justification

$$Y'(t) = 0 \text{ when } S(t) - R(t) = 0.$$

The only value in $[0, 6]$ to satisfy $S(t) = R(t)$ is $a = 5.117865$.

8-3b

t	$Y(t)$
0	2500
a	2492.3694
6	2493.2766

The amount of sand is a minimum when $t = 5.117$ or 5.118 hours. The minimum value is 2492.369 cubic yards.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for setting $Y'(t) = 0$

1 point is earned for the critical t -value

1 point is earned for the answer with justification

$Y'(t) = 0$ when $S(t) - R(t) = 0$.

The only value in $[0, 6]$ to satisfy $S(t) = R(t)$ is $a = 5.117865$.

t	$Y(t)$
0	2500
a	2492.3694
6	2493.2766

The amount of sand is a minimum when $t = 5.117$ or 5.118 hours. The minimum value is 2492.369 cubic yards.

8-3b

88.  There is no snow on Janet's driveway when snow begins to fall at midnight. From midnight to 9 A.M., snow accumulates on the driveway at a rate modeled by $f(t) = 7te^{\cos t}$ cubic feet per hour, where t is measured in hours since midnight. Janet starts removing snow at 6 A.M. ($t = 6$). The rate $g(t)$, in cubic feet per hour, at which Janet removes snow from the driveway at time t hours after midnight is modeled by

$$g(t) = \begin{cases} 0 & \text{for } 0 \leq t < 6 \\ 125 & \text{for } 6 \leq t < 7 \\ 108 & \text{for } 7 \leq t \leq 9. \end{cases}$$

- (a) How many cubic feet of snow have accumulated on the driveway by 6 A.M.?
 (b) Find the rate of change of the volume of snow on the driveway at 8 A.M.
 (c) Let $h(t)$ represent the total amount of snow, in cubic feet, that Janet has removed from the driveway at time t hours after midnight. Express h as a piecewise-defined function with domain $0 \leq t \leq 9$.
 (d) How many cubic feet of snow are on the driveway at 9 A.M.?

Part A

1 point is earned for integral

1 point is earned for the answer

$$\text{Area} = \int_0^6 f(t) dt = 142.274 \text{ or } 142.275 \text{ cubic feet}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

$$\text{Area} = \int_0^6 f(t) dt = 142.274 \text{ or } 142.275 \text{ cubic feet}$$

Part B

1 point is earned for the answer

Rate of change is $f(8) - g(8) = -59.582$ or -59.583 cubic feet per hour.



0	1
---	---

8-3b

The student response earns all of the following points:

1 point is earned for the answer

Rate of change is $f(8) - g(8) = -59.582$ or -59.583 cubic feet per hour.

Part C

1 point is earned for $0 \leq t \leq 6$

1 point is earned for $6 < t \leq 7$

1 point is earned for $7 < t \leq 9$

$$h(0) = 0$$

$$\text{For } 0 < t \leq 6, h(t) = h(0) + \int_0^t g(s) ds = 0 + \int_0^t 0 ds = 0.$$

$$\text{For } 6 < t \leq 7, h(t) = h(6) + \int_6^t g(s) ds = 0 + \int_6^t 125 ds = 125(t - 6).$$

$$\text{For } 7 < t \leq 9, h(t) = h(7) + \int_7^t g(s) ds = 125 + \int_7^t 108 ds = 125 + 108(t - 7).$$

$$\text{Thus, } h(t) = \begin{cases} 0 & \text{for } 0 \leq t \leq 6 \\ 125(t - 6) & \text{for } 6 < t \leq 7 \\ 125 + 108(t - 7) & \text{for } 7 < t \leq 9 \end{cases}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $0 \leq t \leq 6$

1 point is earned for $6 < t \leq 7$

1 point is earned for $7 < t \leq 9$

8-3b

$$h(0) = 0$$

$$\text{For } 0 < t \leq 6, h(t) = h(0) + \int_0^t g(s) ds = 0 + \int_0^t 0 ds = 0.$$

$$\text{For } 6 < t \leq 7, h(t) = h(6) + \int_6^t g(s) ds = 0 + \int_6^t 125 ds = 125(t - 6).$$

$$\text{For } 7 < t \leq 9, h(t) = h(7) + \int_7^t g(s) ds = 125 + \int_7^t 108 ds = 125 + 108(t - 7).$$

$$\text{Thus, } h(t) = \begin{cases} 0 & \text{for } 0 \leq t \leq 6 \\ 125(t - 6) & \text{for } 6 < t \leq 7 \\ 125 + 108(t - 7) & \text{for } 7 < t \leq 9 \end{cases}$$

Part D

1 point is earned for integral

1 point is earned for $h(9)$

1 point is earned for the answer

Amount of snow is $\int_0^9 f(t) dt - h(9) = 26.334$ or 26.335 cubic feet.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integral

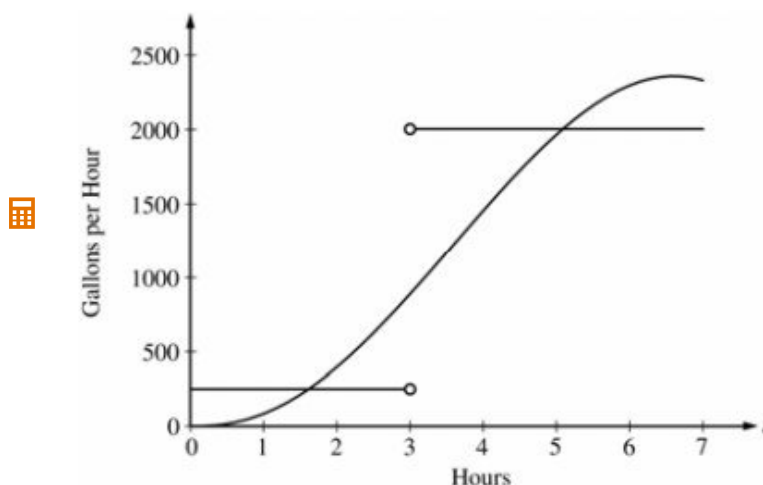
1 point is earned for $h(9)$

1 point is earned for the answer

Amount of snow is $\int_0^9 f(t) dt - h(9) = 26.334$ or 26.335 cubic feet.

8-3b

89.



The amount of water in a storage tank, in gallons, is modeled by a continuous function on the time interval $0 \leq t \leq 7$, where t is measured in hours. In this model, rates are given as follows:

- The rate at which water enters the tank is $f(t) = 100t^2 \sin(\sqrt{t})$ gallons per hour for $0 \leq t \leq 7$.
- The rate at which water leaves the tank is

$$g(t) = \begin{cases} 250 & \text{for } 0 \leq t < 3 \\ 2000 & \text{for } 3 < t \leq 7 \end{cases} \text{ gallons per hour.}$$

The graphs of f and g , which intersect at $t = 1.617$ and $t = 5.076$, are shown in the figure above. At time $t = 0$, the amount of water in the tank is 5000 gallons.

- How many gallons of water enter the tank during the time interval $0 \leq t \leq 7$? Round your answer to the nearest gallon.
- For $0 \leq t \leq 7$, find the time intervals during which the amount of water in the tank is decreasing. Give a reason for each answer.
- For $0 \leq t \leq 7$, at what time t is the amount of water in the tank greatest? To the nearest gallon, compute the amount of water at this time. Justify your answer.

Part A

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^7 f(t) dt \approx 8264 \text{ gallons}$$



0	1	2
---	---	---

The student response earns all of the following points:

8-3b

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^7 f(t)dt \approx 8264 \text{ gallons}$$

Part B

1 point is earned for the intervals

1 point is earned for the reason

The amount of water in the tank is decreasing on the intervals $0 \leq t \leq 1.617$ and $3 \leq t \leq 5.076$ because $f(t) < g(t)$ for $0 \leq t < 1.617$ and $3 < t < 5.076$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the intervals

1 point is earned for the reason

The amount of water in the tank is decreasing on the intervals $0 \leq t \leq 1.617$ and $3 \leq t \leq 5.076$ because $f(t) < g(t)$ for $0 \leq t < 1.617$ and $3 < t < 5.076$.

Part C

1 point is earned if identifies $t = 3$ as a candidate

1 point is earned for the integrand

1 point is earned for the amount of water at $t = 3$

1 point is earned for the amount of water at $t = 7$

1 point is earned for the conclusion

Since $f(t) - g(t)$ changes sign from positive to negative only at $t = 3$, the candidates for the absolute maximum are $t = 0$, 3 , and 7 .

8-3b

t (hours)	gallons of water
0	5000
3	$5000 + \int_0^3 f(t) dt - 250(3) = 5126.591$
7	$5126.591 + \int_3^7 f(t) dt - 2000(4) = 4513.807$

The amount of water in the tank is greatest at 3 hours. At that time, the amount of water in the tank, rounded to the nearest gallon, is 5127 gallons.



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned if identifies $t = 3$ as a candidate

1 point is earned for the integrand

1 point is earned for the amount of water at $t = 3$

1 point is earned for the amount of water at $t = 7$

1 point is earned for the conclusion

Since $f(t) - g(t)$ changes sign from positive to negative only at $t = 3$, the candidates for the absolute maximum are $t = 0, 3$, and 7.

t (hours)	gallons of water
0	5000
3	$5000 + \int_0^3 f(t) dt - 250(3) = 5126.591$
7	$5126.591 + \int_3^7 f(t) dt - 2000(4) = 4513.807$

The amount of water in the tank is greatest at 3 hours. At that time, the amount of water in the tank, rounded to the nearest gallon, is 5127 gallons.

8-3b

90.  Traffic flow is defined as the rate at which cars pass through an intersection, measured in cars per minute. The traffic flow at a particular intersection is modeled by the function F defined by

$$F(t) = 82 + 4 \sin\left(\frac{t}{2}\right) \text{ for } 0 \leq t \leq 30,$$

where $F(t)$ is measured in cars per minute and t is measured in minutes.

- (a) To the nearest whole number, how many cars pass through the intersection over the 30-minute period?
- (b) Is the traffic flow increasing or decreasing at $t = 7$? Give a reason for your answer.
- (c) What is the average value of the traffic flow over the time interval $10 \leq t \leq 15$? Indicate units of measure.
- (d) What is the average rate of change of the traffic flow over the time interval $10 \leq t \leq 15$? Indicate units of measure.

Part A

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\int_0^{30} F(t) dt = 2474 \text{ cars}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\int_0^{30} F(t) dt = 2474 \text{ cars}$$

Part B

1 point is earned for the answer with reason

$$F'(7) = -1.872 \text{ or } -1.873$$

Since $F'(7) < 0$, the traffic flow is decreasing at $t = 7$.

8-3b



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer with reason

$$F'(7) = -1.872 \text{ or } -1.873$$

Since $F'(7) < 0$, the traffic flow is decreasing at $t = 7$.

Part C

1 point is earned for limits

1 point is earned for the integrand

1 point is earned for the answer

$$\frac{1}{5} \int_{10}^{15} F(t) dt = 81.899 \text{ cars/min}$$

1 point is earned for units in (c) and (d)

Units of cars/min in (c) and cars/min² in (d)



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for limits

1 point is earned for the integrand

1 point is earned for the answer

$$\frac{1}{5} \int_{10}^{15} F(t) dt = 81.899 \text{ cars/min}$$

1 point is earned for units in (c) and (d)

Units of cars/min in (c) and cars/min² in (d)

8-3b

Part D

1 point is earned for the answer

$$\frac{F(15)-F(10)}{15-10} = 1.517 \text{ or } 1.518 \text{ cars/min}^2$$

1 point is earned for units in (c) and (d)

Units of cars/min in (c) and cars/min² in (d)



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

$$\frac{F(15)-F(10)}{15-10} = 1.517 \text{ or } 1.518 \text{ cars/min}^2$$

1 point is earned for units in (c) and (d)

Units of cars/min in (c) and cars/min² in (d)

8-3b

91.  At a certain height, a tree trunk has a circular cross section. The radius $R(t)$ of that cross section grows at a rate modeled by the function

$$\frac{dR}{dt} = \frac{1}{16}(3 + \sin(t^2)) \text{ centimeters per year}$$

for $0 \leq t \leq 3$, where time t is measured in years. At time $t = 0$, the radius is 6 centimeters. The area of the cross section at time t is denoted by $A(t)$.

(a) Write an expression, involving an integral, for the radius $R(t)$ for $0 \leq t \leq 3$. Use your expression to find $R(3)$.

(b) Find the rate at which the cross-sectional area $A(t)$ is increasing at time $t = 3$ years. Indicate units of measure.

(c) Evaluate $\int_0^3 A'(t) dt$. Using appropriate units, interpret the meaning of that integral in terms of cross-sectional area.

Part A

1 point is earned for the integral

1 point is earned for the expression for $R(t)$

1 point is earned for $R(3)$

$$R(t) = 6 + \int_0^t \frac{1}{16}(3 + \sin(x^2)) dx$$

$$R(3) = 6.610 \text{ or } 6.611$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the expression for $R(t)$

1 point is earned for $R(3)$

$$R(t) = 6 + \int_0^t \frac{1}{16}(3 + \sin(x^2)) dx$$

$$R(3) = 6.610 \text{ or } 6.611$$

Part B

1 point is earned for expression of $A(t)$

1 point is earned for expression of $A'(t)$

1 point is earned for answering with units

8-3b

$$A(t) = \pi(R(t))^2$$

$$A'(t) = 2\pi R(t)R'(t)$$

$$A'(3) = 8.858 \text{ cm}^2/\text{year}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for expression of $A(t)$

1 point is earned for expression of $A'(t)$

1 point is earned for answering with units

$$A(t) = \pi(R(t))^2$$

$$A'(t) = 2\pi R(t)R'(t)$$

$$A'(3) = 8.858 \text{ cm}^2/\text{year}$$

Part C

1 point is earned for using Fundamental Theorem of Calculus

1 point is earned for value of $\int_0^3 A'(t)dt$

1 point is earned for meaning of $\int_0^3 A'(t)dt$

$$\int_0^3 A'(t)dt = A(3) - A(0) = 24.200 \text{ or } 24.201$$

From time $t = 0$ to $t = 3$ years, the cross-sectional area grows by 24.201 square centimeters.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for using Fundamental Theorem of Calculus

8-3b

1 point is earned for value of $\int_0^3 A'(t)dt$

1 point is earned for meaning of $\int_0^3 A'(t)dt$

$$\int_0^3 A'(t)dt = A(3) - A(0) = 24.200 \text{ or } 24.201$$

From time $t = 0$ to $t = 3$ years, the cross-sectional area grows by 24.201 square centimeters.

8-3b

92.

t (hours)	$R(t)$ (gallons per hour)
0	9.6
3	10.4
6	10.8
9	11.2
12	11.4
15	11.3
18	10.7
21	10.2
24	9.6

The rate at which water flows out of a pipe, in gallons per hour, is given by a differentiable function R of time t . The table above shows the rate as measured every 3 hours for a 24-hour period.

- (a) Use a midpoint Riemann sum with 4 subdivisions of equal length to approximate $\int_0^{24} R(t)dt$. Using correct units, explain the meaning of your answer in terms of water flow.
- (b) Is there some time t , $0 < t < 24$, such that $R'(t) = 0$? Justify your answer.
- (c) The rate of water flow $R(t)$ can be approximated by $Q(t) = \frac{1}{79}(768 + 23t - t^2)$. Use $Q(t)$ to approximate the average rate of water flow during the 24-hour time period. Indicate units of measure.

Part A

1 point is earned for: $R(3) + R(9) + R(15) + R(21)$

1 point is earned for the answer

1 point is earned for the explanation

$$\begin{aligned}
 \int_0^{24} R(t)dt &\approx 6[R(3) + R(9) + R(15) + R(21)] \\
 &= 6[10.4 + 11.2 + 11.3 + 10.2] \\
 &= 258.6 \text{ gallons}
 \end{aligned}$$

This is an approximation to the total flow in gallons of water from the pipe in the 24-hour period.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $R(3) + R(9) + R(15) + R(21)$

1 point is earned for the answer

1 point is earned for the explanation

8-3b

$$\begin{aligned}
 \int_0^{24} R(t) dt &\approx 6 [R(3) + R(9) + R(15) + R(21)] \\
 &= 6 [10.4 + 11.2 + 11.3 + 10.2] \\
 &= 258.6 \text{ gallons}
 \end{aligned}$$

This is an approximation to the total flow in gallons of water from the pipe in the 24-hour period.

Part B

1 point is earned for the answer

1 point is earned for MVT or equivalent

Yes;

Since $R(0) = R(24) = 9.6$, the Mean Value Theorem guarantees that there is a t , $0 < t < 24$, such that $R'(t) = 0$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for MVT or equivalent

Yes;

Since $R(0) = R(24) = 9.6$, the Mean Value Theorem guarantees that there is a t , $0 < t < 24$, such that $R'(t) = 0$.

Part C

1 point is earned for the limits and average value constant

1 point is earned for: $Q(t)$ as integrand

1 point is earned for the answer

Average rate of flow

$\approx$ average value of $Q(t)$

$$\begin{aligned}
 &= \frac{1}{24} \int_0^{24} \frac{1}{79} (768 + 23t - t^2) dt \\
 &= 10.785 \text{ gal/hr or } 10.784 \text{ gal/hr}
 \end{aligned}$$

1 point is earned for units

(units) Gallons in part (a) and gallons/hr in part (c), or equivalent.

8-3b



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits and average value constant

1 point is earned for: $Q(t)$ as integrand

1 point is earned for the answer

Average rate of flow

$\approx$ average value of $Q(t)$

$$\begin{aligned} &= \frac{1}{24} \int_0^{24} \frac{1}{79} (768 + 23t - t^2) dt \\ &= 10.785 \text{ gal/hr or } 10.784 \text{ gal/hr} \end{aligned}$$

1 point is earned for units

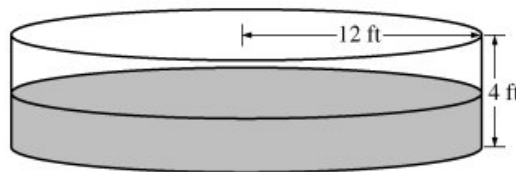
(units) Gallons in part (a) and gallons/hr in part (c), or equivalent.

8-3b

93.



t	0	2	4	6	8	10	12
$P(t)$	0	46	53	57	60	62	63



The figure above shows an aboveground swimming pool in the shape of a cylinder with a radius of 12 feet and a height of 4 feet. The pool contains 1000 cubic feet of water at time $t = 0$. During the time interval $0 \leq t \leq 12$ hours, water is pumped into the pool at the rate $P(t)$ cubic feet per hour. The table above gives values of $P(t)$ for selected values of t . During the same time interval, water is leaking from the pool at the rate $R(t)$ cubic feet per hour, where $R(t) = 25e^{-0.05t}$. (Note: The volume V of a cylinder with radius r and height h is given by $V = \pi r^2 h$.)

- Use a midpoint Riemann sum with three subintervals of equal length to approximate the total amount of water that was pumped into the pool during the time interval $0 \leq t \leq 12$ hours. Show the computations that lead to your answer.
- Calculate the total amount of water that leaked out of the pool during the time interval $0 \leq t \leq 12$ hours.
- Use the results from parts (a) and (b) to approximate the volume of water in the pool at time $t = 12$ hours. Round your answer to the nearest cubic foot.
- Find the rate at which the volume of water in the pool is increasing at time $t = 8$ hours. How fast is the water level in the pool rising at $t = 8$ hours? Indicate units of measure in both answers.

Part A

1 point is earned for midpoint sum

1 point is earned for the answer

$$\int_0^{12} P(t) dt \approx 46 \cdot 4 + 57 \cdot 4 + 62 \cdot 4 = 660 \text{ ft}^3$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for midpoint sum

1 point is earned for the answer

$$\int_0^{12} P(t) dt \approx 46 \cdot 4 + 57 \cdot 4 + 62 \cdot 4 = 660 \text{ ft}^3$$

Part B

1 point is earned for the integral

1 point is earned for the answer

8-3b

$$\int_0^{12} R(t) dt = 225.594 \text{ ft}^3$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^{12} R(t) dt = 225.594 \text{ ft}^3$$

Part C

1 point is earned for the answer

$$1000 + \int_0^{12} P(t) dt - \int_0^{12} R(t) dt = 1434.406$$

At time $t = 12$ hours, the volume of water in the pool is approximately 1434 ft^3 .



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

$$1000 + \int_0^{12} P(t) dt - \int_0^{12} R(t) dt = 1434.406$$

At time $t = 12$ hours, the volume of water in the pool is approximately 1434 ft^3 .

Part D

1 point is earned for $V'(8)$

8-3b

1 point is earned for equation relating $\frac{dV}{dt}$ and $\frac{dh}{dt}$

1 point is earned for $\left. \frac{dh}{dt} \right|_{t=8}$

1 point is earned for units of ft^3/hr and ft/hr

$$V'(t) = P(t) - R(t)$$

$$V'(8) = P(8) - R(8) = 60 - 25e^{-0.4} = 43.241 \text{ or } 43.242 \text{ ft}^3/\text{hr}$$

$$V = \pi(12)^2 h$$

$$\frac{dV}{dt} = 144\pi \frac{dh}{dt}$$

$$\left. \frac{dh}{dt} \right|_{t=8} = \frac{1}{144\pi} \cdot \left. \frac{dV}{dt} \right|_{t=8} = 0.095 \text{ or } 0.096 \text{ ft}/\text{hr}$$



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for $V'(8)$

1 point is earned for equation relating $\frac{dV}{dt}$ and $\frac{dh}{dt}$

1 point is earned for $\left. \frac{dh}{dt} \right|_{t=8}$

1 point is earned for units of ft^3/hr and ft/hr

$$V'(t) = P(t) - R(t)$$

$$V'(8) = P(8) - R(8) = 60 - 25e^{-0.4} = 43.241 \text{ or } 43.242 \text{ ft}^3/\text{hr}$$

8-3b

$$V = \pi(12)^2 h$$

$$\frac{dV}{dt} = 144\pi \frac{dh}{dt}$$

$$\left. \frac{dh}{dt} \right|_{t=8} = \frac{1}{144\pi} \cdot \left. \frac{dV}{dt} \right|_{t=8} = 0.095 \text{ or } 0.096 \text{ ft/hr}$$

8-3b

94. Water is pumped into an underground tank at a constant rate of 8 gallons per minute. Water leaks out of the tank at the rate of $\sqrt{t+1}$ gallons per minute, for $0 \leq t \leq 120$ minutes. At time $t = 0$, the tank contains 30 gallons of water.
- How many gallons of water leak out of the tank from time $t = 0$ to $t = 3$ minutes?
 - How many gallons of water are in the tank at time $t = 3$ minutes?
 - Write an expression for $A(t)$, the total number of gallons of water in the tank at time t .
 - At what time t , for $0 \leq t \leq 120$, is the amount of water in the tank a maximum? Justify your answer.

Part A

Method 1:

2 points are earned for the definite integral

1: limits

1: integrand

1 point is earned for the answer

- or -

Method 2:

1 point is earned for antiderivative with C

1 point is earned for solves for C using $L(0) = 0$

1 point is earned for the answer

$$\text{Method 1: } \int_0^3 \sqrt{t+1} \, dt = \frac{2}{3}(t+1)^{3/2} \Big|_0^3 = \frac{14}{3}$$

- or -

Method 2: $L(t)$ = gallons leaked in first t minutes

$$\frac{dL}{dt} = \sqrt{t+1}; \quad L(t) = \frac{2}{3}(t+1)^{3/2} + C$$

$$L(0) = 0; \quad C = -\frac{2}{3}$$

$$L(t) = \frac{2}{3}(t+1)^{3/2} - \frac{2}{3}; \quad L(3) = \frac{14}{3}$$



0	1	2	3	4	5	6
---	---	---	---	---	---	---

The student response earns all of the following points:

8-3bMethod 1:

2 points are earned for the definite integral

1: limits

1: integrand

1 point is earned for the answer

- or -

Method 2:

1 point is earned for antiderivative with C

1 point is earned for solves for C using $L(0) = 0$

1 point is earned for the answer

Method 1: $\int_0^3 \sqrt{t+1} \, dt = \frac{2}{3}(t+1)^{3/2} \Big|_0^3 = \frac{14}{3}$

- or -

Method 2: $L(t)$ = gallons leaked in first t minutes

$$\frac{dL}{dt} = \sqrt{t+1}; \quad L(t) = \frac{2}{3}(t+1)^{3/2} + C$$

$$L(0) = 0; \quad C = -\frac{2}{3}$$

$$L(t) = \frac{2}{3}(t+1)^{3/2} - \frac{2}{3}; \quad L(3) = \frac{14}{3}$$

Part B

1 point is earned for the answer

$$30 + 8 \cdot 3 - \frac{14}{3} = \frac{148}{3}$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

8-3b

$$30 + 8 \cdot 3 - \frac{14}{3} = \frac{148}{3}$$

Part C

Method 1:

1 point is earned for: $30 + 8t$

1 point is earned for: $-\int_0^t \sqrt{x+1} \, dx$

- or -

Method 2:

1 point earned for antiderivative with C

1 point is earned for the answer

Method 1:

$$\begin{aligned} A(t) &= 30 + \int_0^t (8 - \sqrt{x+1}) \, dx \\ &= 30 + 8t - \int_0^t \sqrt{x+1} \, dx \end{aligned}$$

- or -

Method 2:

$$\begin{aligned} \frac{dA}{dt} &= 8 - \sqrt{t+1} \\ A(t) &= 8t - \frac{2}{3}(t+1)^{3/2} + C \\ 30 &= 8(0) - \frac{2}{3}(0+1)^{3/2} + C; \quad C = \frac{92}{3} \\ A(t) &= 8t - \frac{2}{3}(t+1)^{3/2} + \frac{92}{3} \end{aligned}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

Method 1:

8-3b

1 point is earned for: $30 + 8t$

1 point is earned for: $-\int_0^t \sqrt{x+1} \, dx$

- or -

Method 2:

1 point earned for antiderivative with C

1 point is earned for the answer

Method 1:

$$\begin{aligned} A(t) &= 30 + \int_0^t (8 - \sqrt{x+1}) \, dx \\ &= 30 + 8t - \int_0^t \sqrt{x+1} \, dx \end{aligned}$$

- or -

Method 2:

$$\begin{aligned} \frac{dA}{dt} &= 8 - \sqrt{t+1} \\ A(t) &= 8t - \frac{2}{3}(t+1)^{3/2} + C \\ 30 &= 8(0) - \frac{2}{3}(0+1)^{3/2} + C; \quad C = \frac{92}{3} \\ A(t) &= 8t - \frac{2}{3}(t+1)^{3/2} + \frac{92}{3} \end{aligned}$$

Part D

1 point is earned for set $A'(t) = 0$

1 point is earned for solves for t

1 point is earned for the justification

$$A'(t) = 8 - \sqrt{t+1} = 0 \text{ when } t = 63$$

$A'(t)$ is positive for $0 < t < 63$ and negative for $63 < t < 120$. Therefore there is a maximum at $t = 63$.

8-3b



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for set $A'(t) = 0$

1 point is earned for solves for t

1 point is earned for the justification

$$A'(t) = 8 - \sqrt{t+1} = 0 \text{ when } t = 63$$

$A'(t)$ is positive for $0 < t < 63$ and negative for $63 < t < 120$. Therefore there is a maximum at $t = 63$.

8-3b

95.

t (minutes)	0	4	9	15	20
$W(t)$ (degrees Fahrenheit)	55.0	57.1	61.8	67.9	71.0

 The temperature of water in a tub at time t is modeled by a strictly increasing, twice-differentiable function W , where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. At time $t = 0$, the temperature of the water is 55°F . The water is heated for 30 minutes, beginning at time $t = 0$. Values of $W(t)$ at selected times t for the first 20 minutes are given in the table above.

(a) Use the data in the table to estimate $W'(12)$. Show the computations that lead to your answer. Using correct units, interpret the meaning of your answer in the context of this problem.

(b) Use the data in the table to evaluate $\int_0^{20} W'(t) dt$. Using correct units, interpret the meaning of $\int_0^{20} W'(t) dt$ in the context of this problem.

(c) For $0 \leq t \leq 20$, the average temperature of the water in the tub is $\frac{1}{20} \int_0^{20} W(t) dt$. Use a left Riemann sum with the four subintervals indicated by the data in the table to approximate $\frac{1}{20} \int_0^{20} W(t) dt$. Does this approximation overestimate or underestimate the average temperature of the water over these 20 minutes? Explain your reasoning.

(d) For $20 \leq t \leq 25$, the function W that models the water temperature has first derivative given by $W'(t) = 0.4\sqrt{t} \cos(0.06t)$. Based on the model, what is the temperature of the water at time $t = 25$?

Part A

1 point is earned for the estimation

1 point is earned for the interpretation with units

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6} \\ = 1.017 \text{ (or } 1.016\text{)}$$

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t = 12$ minutes.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the estimation

1 point is earned for the interpretation with units

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6} \\ = 1.017 \text{ (or } 1.016\text{)}$$

8-3b

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t = 12$ minutes.

Part B

1 point is earned for the value

$$\int_0^{20} W'(t) dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

1 point is earned for the interpretation with units

The water has warmed by 16°F over the interval from $t = 0$ to $t = 20$ minutes.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the value

$$\int_0^{20} W'(t) dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

1 point is earned for the interpretation with units

The water has warmed by 16°F over the interval from $t = 0$ to $t = 20$ minutes.

Part C

1 point is earned for the left Riemann sum

1 points are earned the approximation

1 point is earned for the underestimate with reason

$$\begin{aligned} \frac{1}{20} \int_0^{20} W(t) dt &\approx \frac{1}{20} (4 \cdot W(0) + 5 \cdot W(4) + 6 \cdot W(9) + 5 \cdot W(15)) \\ &= \frac{1}{20} (4 \cdot 55.0 + 5 \cdot 57.1 + 6 \cdot 61.8 + 5 \cdot 67.9) \\ &= \frac{1}{20} \cdot 1215.8 = 60.79 \end{aligned}$$

This approximation is an underestimate, because a left Riemann sum is used and the function W is strictly increasing.

8-3b



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the left Riemann sum

1 points are earned the approximation

1 point is earned for the underestimate with reason

$$\begin{aligned}
 \frac{1}{20} \int_0^{20} W(t) dt &\approx \frac{1}{20} (4 \cdot W(0) + 5 \cdot W(4) + 6 \cdot W(9) + 5 \cdot W(15)) \\
 &= \frac{1}{20} (4 \cdot 55.0 + 5 \cdot 57.1 + 6 \cdot 61.8 + 5 \cdot 67.9) \\
 &= \frac{1}{20} \cdot 1215.8 = 60.79
 \end{aligned}$$

This approximation is an underestimate, because a left Riemann sum is used and the function W is strictly increasing.

Part D

1 point is earned for the integral

1 point is earned for the answer

$$\begin{aligned}
 W(25) &= 71.0 + \int_{20}^{25} W'(t) dt \\
 &= 71.0 + 2.043155 = 73.043
 \end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\begin{aligned}
 W(25) &= 71.0 + \int_{20}^{25} W'(t) dt \\
 &= 71.0 + 2.043155 = 73.043
 \end{aligned}$$